# CHAPTER 14
# LIGHT

## 3.2 Light

### 3.2.1 Reflection of light

1 Define and use the terms normal, angle of incidence and angle of reflection

2 Describe an experiment to illustrate the law of reflection

3 Describe an experiment to find the position and characteristics of an optical image formed by a plane mirror (same size, same distance from mirror as object and virtual)

4 State that for reflection, the angle of incidence is equal to the angle of reflection and use this in constructions, measurements and calculations

### 3.2.2 Refraction of light

1 Define and use the terms normal, angle of incidence and angle of refraction

2 Define refractive index $n$ as $n = \frac{\sin i}{\sin r}$; recall and use this equation

3 Describe an experiment to show refraction of light by transparent blocks of different shapes

4 Define the terms critical angle and total internal reflection; recall and use the equation

$$n = \frac{1}{\sin c}$$

5 Describe experiments to show internal reflection and total internal reflection

6 Describe the use of optical fibres, particularly in telecommunications, stating the advantages of their use in each context

### 3.2.3 Thin lenses

1 Describe the action of thin converging and thin diverging lenses on a parallel beam of light

2 Define and use the terms focal length, principal axis and principal focus (focal point)

3 Draw ray diagrams to illustrate the formation of real and virtual images of an object by a converging lens and know that a real image is formed by converging rays and a virtual image is formed by diverging rays

4 Define linear magnification as the ratio of image length to object length; recall and use the equation

$$\text{linear magnification} = \frac{\text{image length}}{\text{object length}}$$

5 Describe the use of a single lens as a magnifying glass

6 Draw ray diagrams to show the formation of images in the normal eye, a short-sighted eye and a long-sighted eye

7 Describe the use of converging and diverging lenses to correct long-sightedness and short-sightedness

### 3.2.4 Dispersion of light

1 Describe the dispersion of light as illustrated by the refraction of white light by a glass prism

2 Know the traditional seven colours of the visible spectrum in order of frequency and in order of wavelength

**1. O/N/23/11/Q18**

A plane mirror on a vertical wall forms an image of an object placed in front of it.

Which characteristics describe the image?

**A** real and inverted

**B** real and upright

**C** virtual and inverted

**D** virtual and upright

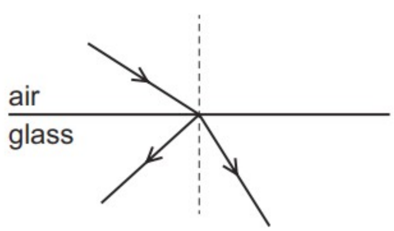
**2. O/N/23/11/Q19**

The diagrams show light travelling in air incident on the surface of a glass block.

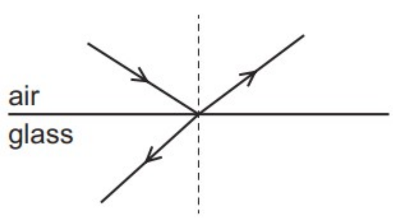
Some light is reflected and some light is refracted.

Which diagram shows the reflection and refraction of the light?

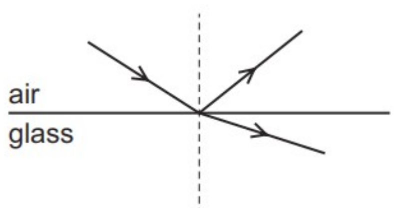
**A**



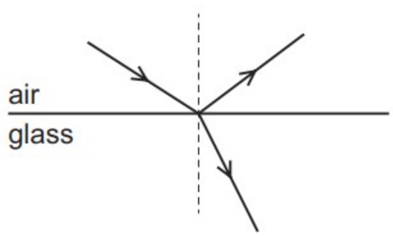

**B**




**C**




**D**




**3. O/N/23/11/20**

Which statement is correct?

**A** Total internal reflection only occurs when light travels from air into glass.

**B** The larger the refractive index of glass, the larger is the critical angle.

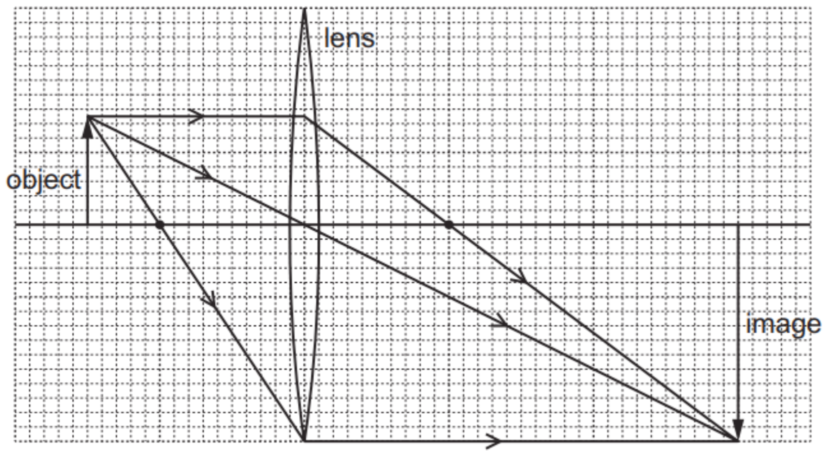
**C** When total internal reflection occurs, the angle of incidence is equal to the angle of reflection.

**D** When total internal reflection occurs, the angle of incidence is less than the critical angle.

**4. O/N/23/11/Q21**

An object is placed in front of a converging lens of focal length 4.0 cm. The height of the image is 6.0 cm.

The arrangement is shown on the scale diagram.



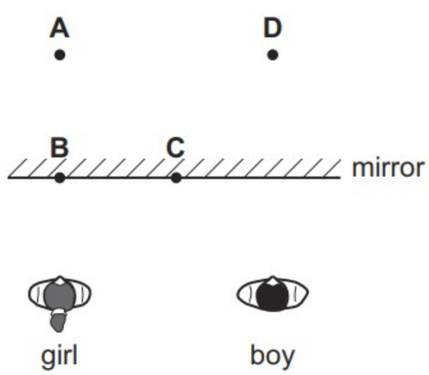

What is the linear magnification produced by the lens?

**A** 0.50 **B** 1.5 **C** 2.0 **D** 6.0

**5. O/N/23/12/Q26**

A boy stands beside a girl in front of a large vertical plane mirror. They are equal distances from the mirror, as shown. The boy sees an image of the girl.

Where is the girl's image?




**6. O/N/23/12/Q27**

Light travelling in air is incident on a water surface.

The refractive index of water is 1.3 and the angle of refraction in the water is 40°.

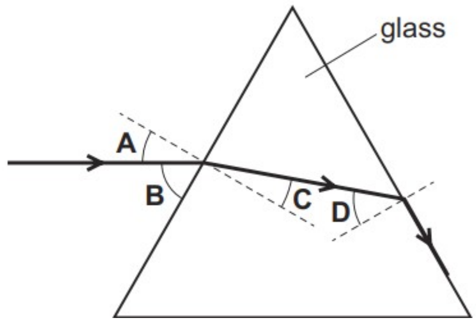
What is the angle of incidence in the air?

**A** 30° **B** 50° **C** 52° **D** 57°

**7. O/N/23/12/Q28**

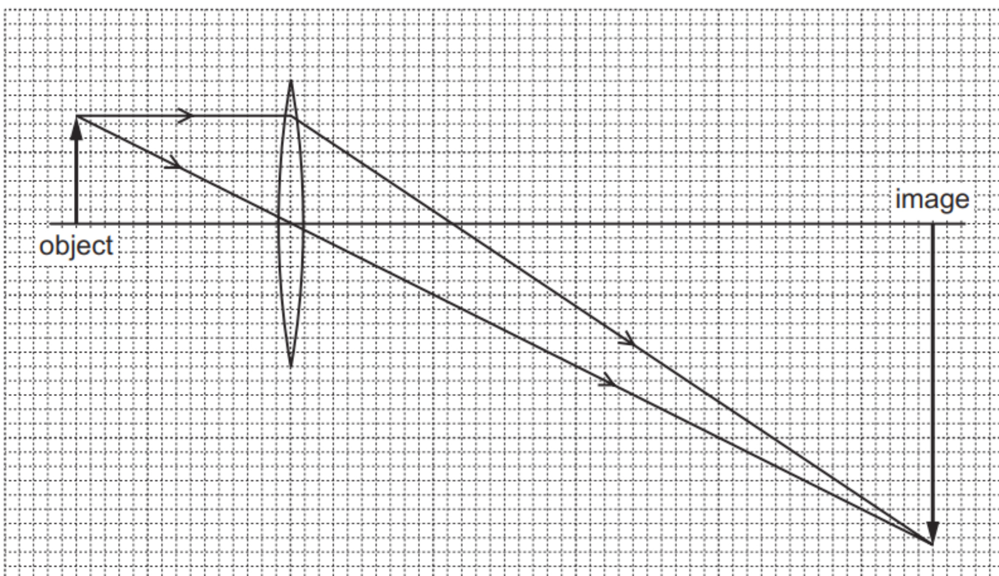
The diagram shows a ray of light passing through a triangular glass block.

Which angle is the critical angle of light in glass?




**8. O/N/23/12/Q29**

A lens is used to produce a magnified image, as shown in the scale diagram.

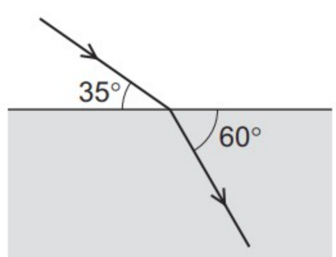



What is the linear magnification produced by the lens?

**A** 0.33 **B** 3.0 **C** 4.0 **D** 6.0

**9. M/J/23/12/Q20**

A ray of light passes into a block of transparent material as shown.



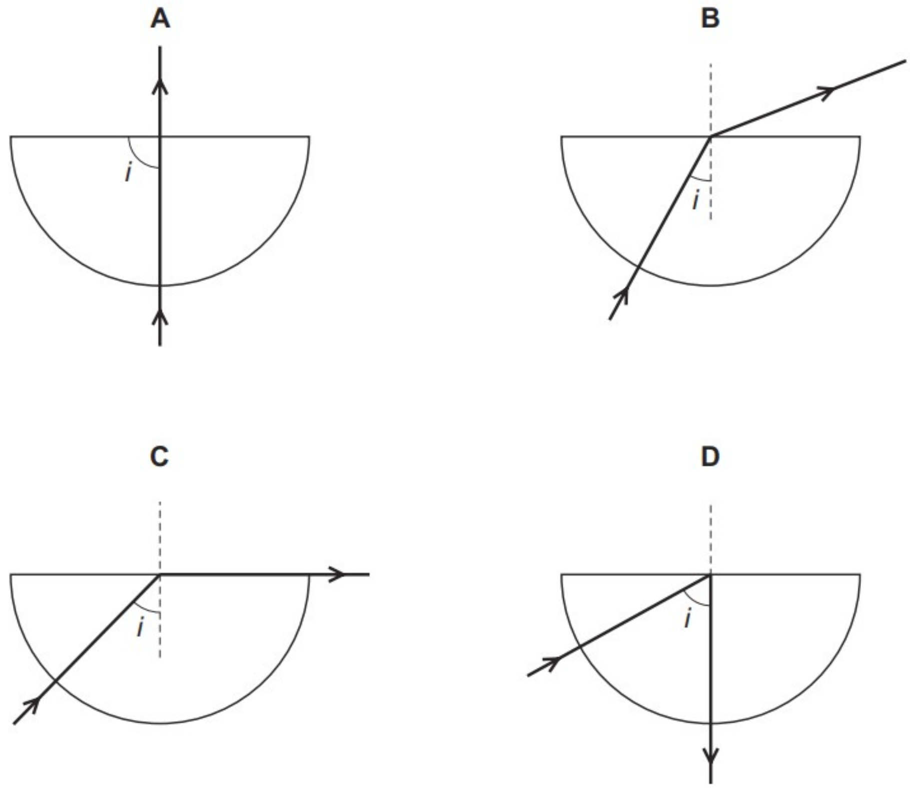

What is the refractive index of the transparent material?

**A** 0.66 **B** 1.15 **C** 1.64 **D** 1.83

**10. M/J/23/12/Q21**

The diagrams show light incident on the straight edge of a semi-circular glass block after passing through it.

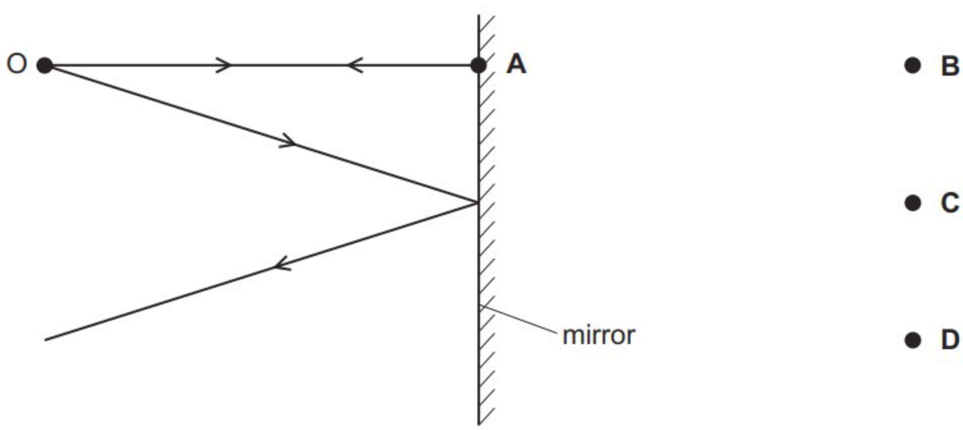
Which diagram shows refraction and an angle *i* equal to the critical angle of the glass?




**11. M/J/23/11/Q19**

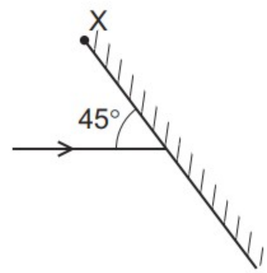
The diagram shows two divergent rays of light from an object O being reflected from a plane mirror.

At which position is the image formed?



**12. M/J/23/11/Q20**

Light is incident on a mirror at an angle of 45° as shown. The mirror can be rotated about an axis into the page through point X.




The mirror is rotated until the light is reflected back along its original path.

Through which angle is the mirror rotated?

**A** 22.5° clockwise

**B** 22.5° anticlockwise

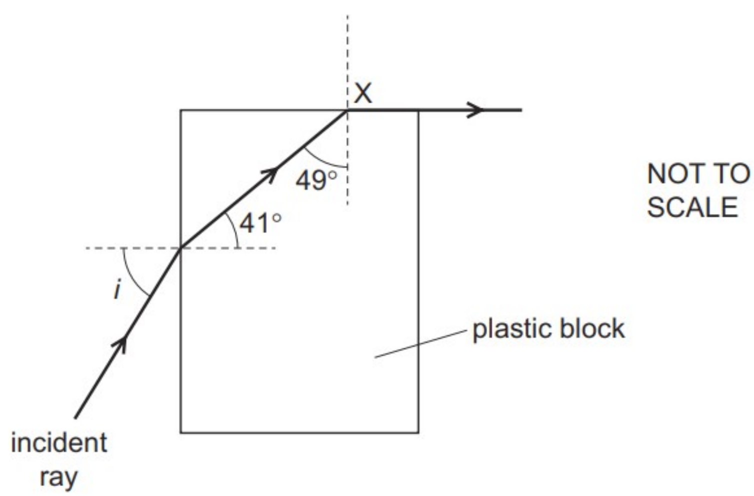
**C** 45° clockwise

**D** 45° anticlockwise

**13. M/J/23/11/Q21**

A ray of light is incident on a plastic block in air, at an angle of incidence *i*. The refractive index of the plastic is *n*.

The light ray refracts along the plastic-air boundary when it reaches X.




Which equation is correct?

**A** $\sin i = \frac{\sin 41°}{n}$

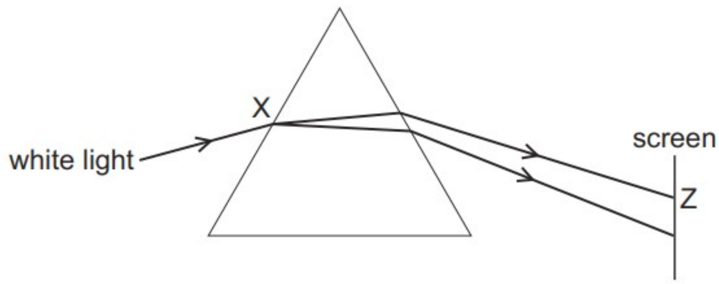
**B** $\sin i = \frac{n}{\sin 49°}$

**C** $\sin i = n \times \sin 41°$

**D** $\sin i = n \times \sin 49°$

**14. M/J/23/11/Q22**

The diagram shows white light entering a prism.

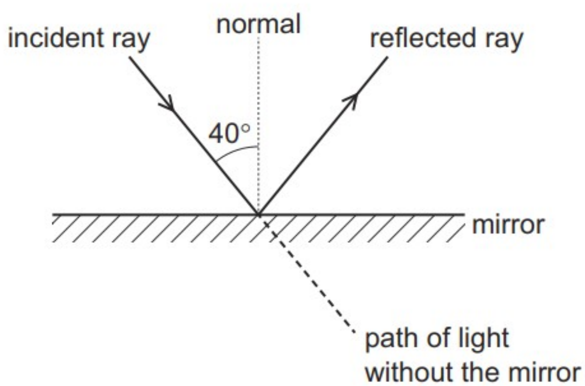



What happens at point X and what is the colour of the light that strikes the screen at point Z?

| | at point X | at point Z |
|---|---|---|
| **A** | dispersion only | blue light |
| **B** | dispersion and refraction | blue light |
| **C** | dispersion and refraction | red light |
| **D** | refraction only | red light |

**15. O/N/22/11/Q24**

A mirror is placed in the path of a ray of light.

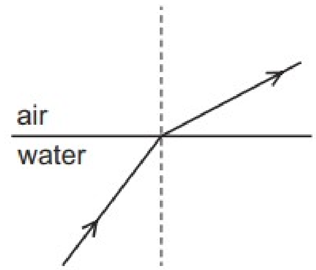



Through which angle does the direction of the ray of light change?

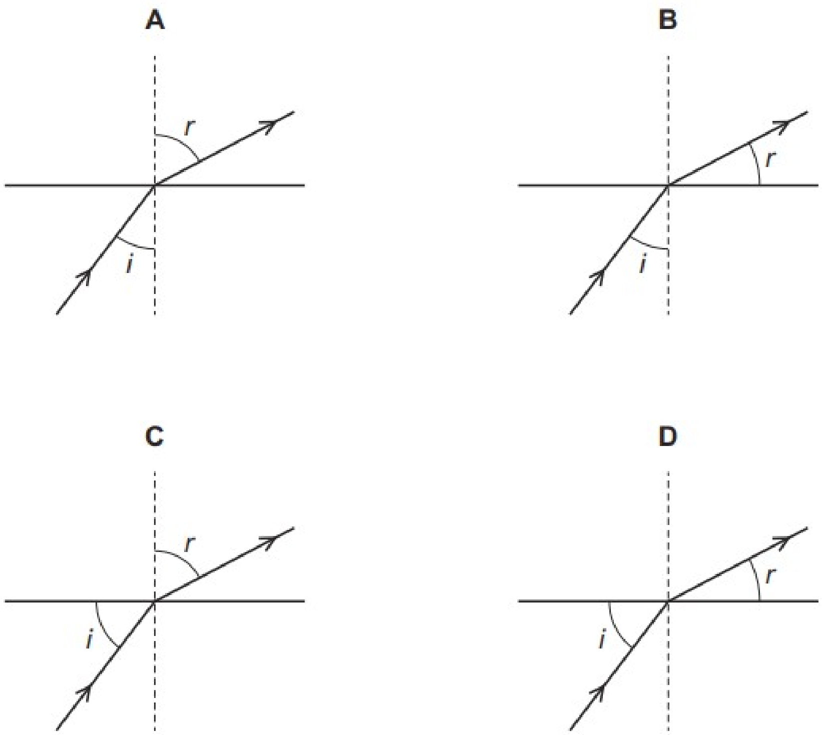
**A** 40° **B** 90° **C** 100° **D** 140°

**16. O/N/22/12/Q23**

A ray of light in water is refracted at the surface into air.

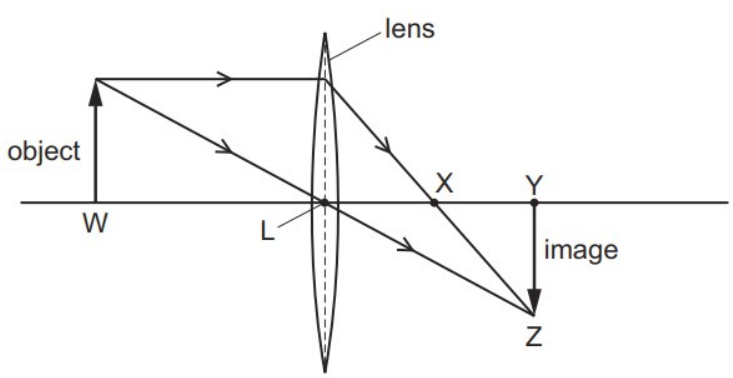



Which diagram shows the angle of incidence *i* and the angle of refraction *r*?




**17. O/N/22/12/Q24**

A thin converging lens forms a real, focused image of an object, as shown.

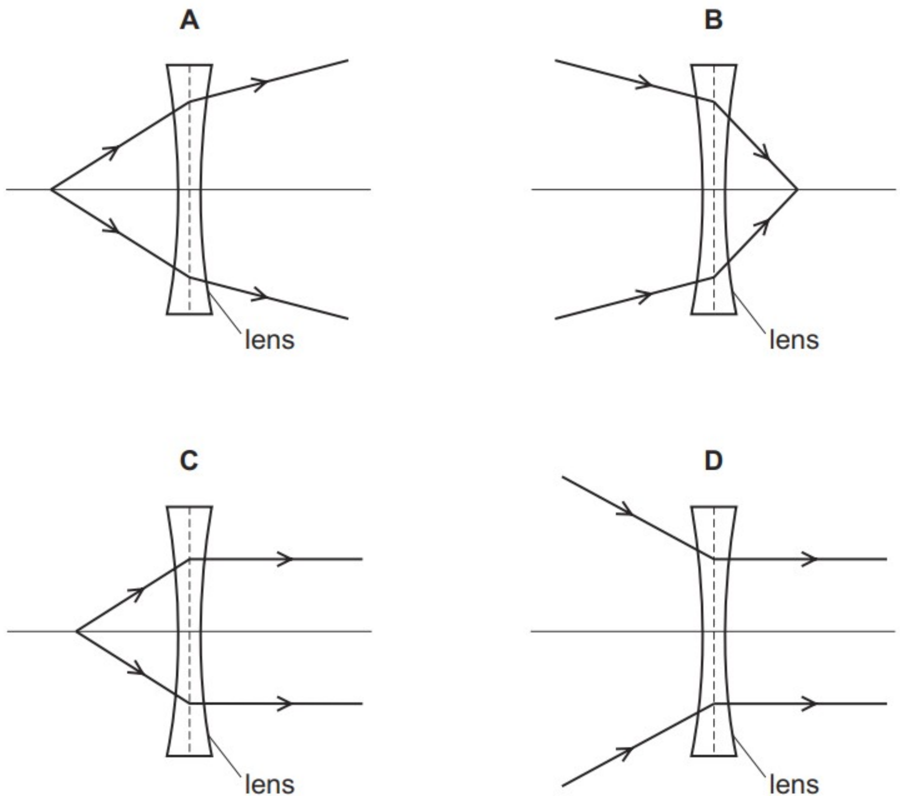



Which distance is equal to the focal length of the lens?

**A** LW **B** LX **C** LY **D** LZ

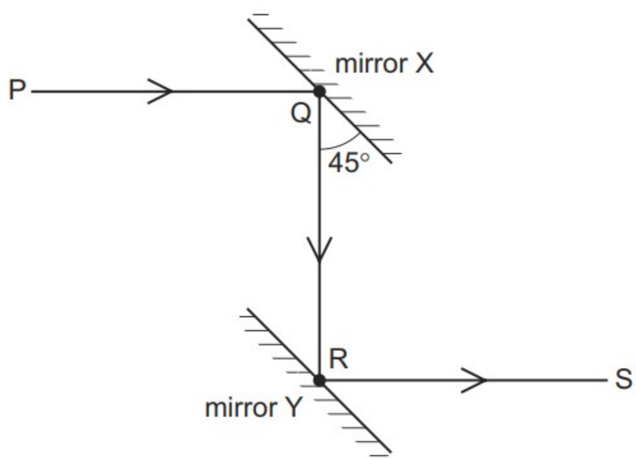
**18. O/N/22/12/Q25**

Which ray diagram shows the action of a diverging lens?




**19. M/J/22/12/Q24**

The diagram shows a ray PQ reflected by mirror X to a parallel mirror Y. The reflected ray along RS is parallel to PQ.




Which statement is correct?

**A** The angle between PQ and QR is 45°.

**B** The angle between QR and RS is 180°.

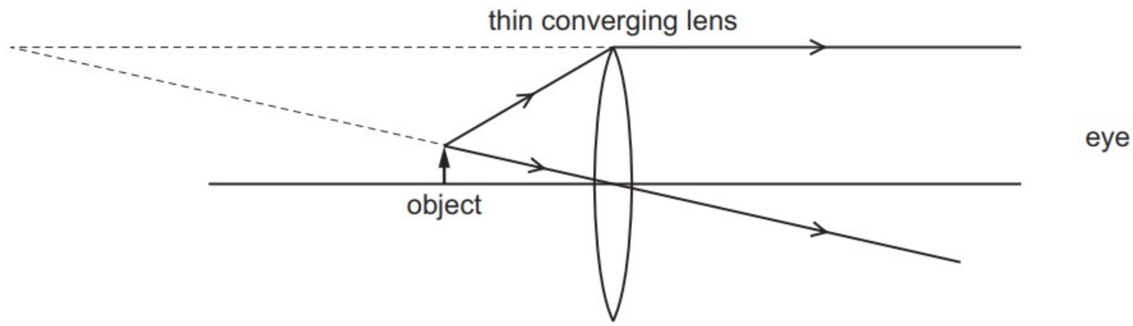
**C** The angle of incidence of PQ on mirror X is 60°.

**D** The angle of incidence of QR on mirror Y is 45°.

## 20. M/J/22/12/Q25

An object is viewed through a thin converging lens.

The diagram shows the paths of two rays from the top of the object to an eye.




How does the image compare with the object?

**A** It is larger and inverted.

**B** It is larger and upright.

**C** It is smaller and inverted.

**D** It is smaller and upright.

## 21. M/J/22/12/Q26

Violet and indigo light have the shortest wavelengths in the spectrum of visible light.

Which three colours, in order of increasing wavelength, immediately follow indigo?

**A** blue → green → orange

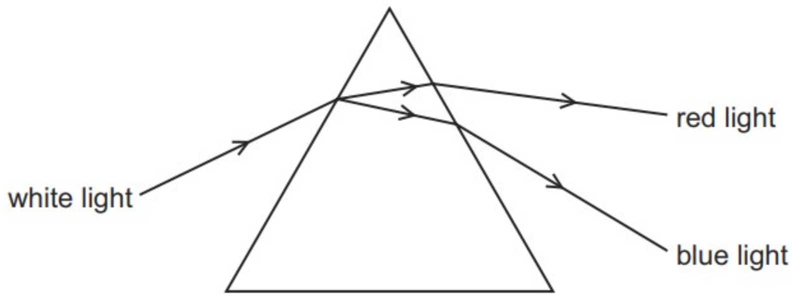
**B** blue → green → yellow

**C** green → blue → yellow

**D** yellow → green → orange

**22. M/J/22/11/Q22**

A ray of white light enters a prism as shown.



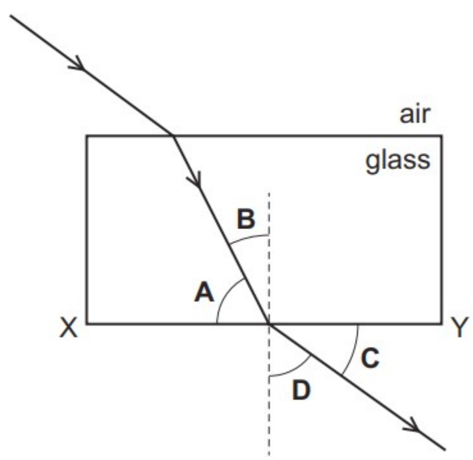

Which row is correct?

| | wave properties observed | frequency of red light compared with frequency of blue light |
|---|---|---|
| **A** | dispersion only | smaller |
| **B** | refraction only | greater |
| **C** | dispersion and refraction | smaller |
| **D** | dispersion and refraction | greater |

**23. O/N/21/11/Q20**

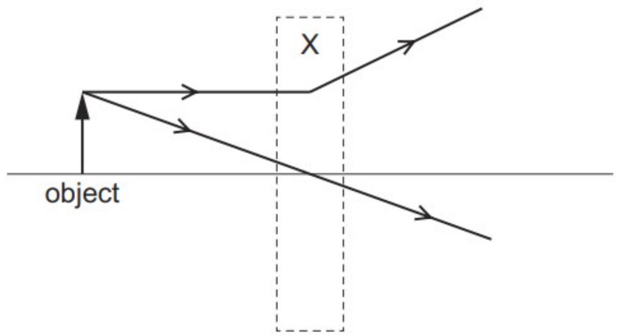
A ray of light passes into a glass block. It travels through the glass block and then emerges into the air.

Which angle is the angle of refraction at the surface XY?



## 24. O/N/21/11/Q21

Two rays of light pass through a lens in region X.




Which type of lens is in region X and which type of image is formed?

| | type of lens | type of image |
|---|---|---|
| **A** | converging | real |
| **B** | converging | virtual |
| **C** | diverging | real |
| **D** | diverging | virtual |

## 25. O/N/21/11/Q22

A girl is long-sighted.

Which statement is correct?

**A** She sees close objects less clearly than a person with normal vision.

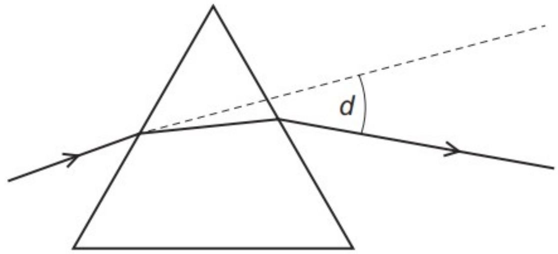
**B** She sees distant objects more clearly than a person with normal vision.

**C** The fault is corrected with a diverging lens.

**D** The image of a close object is formed in front of her retina.

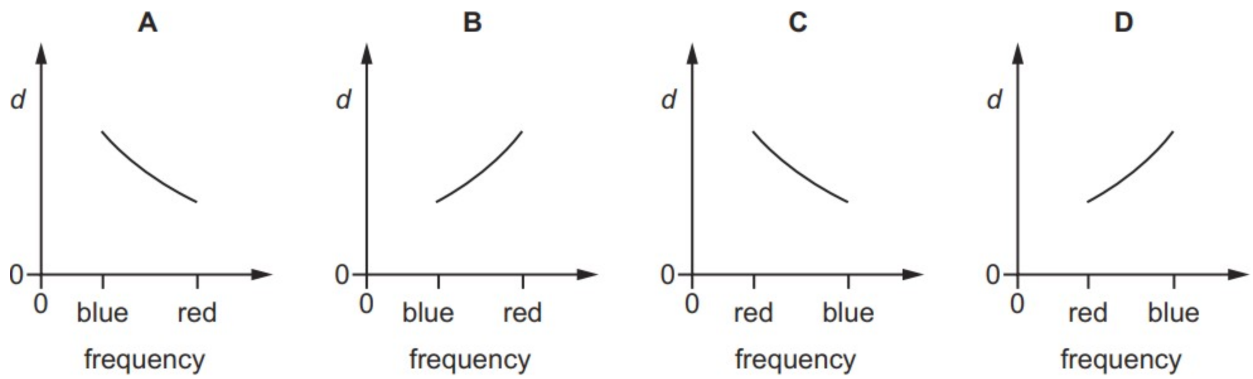
## 26. O/N/21/11/Q23

Light rays are deviated by a prism.



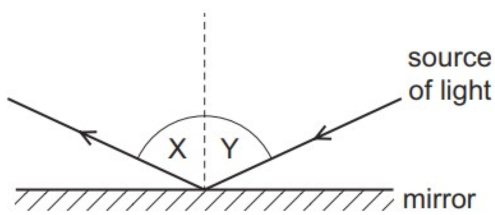

The deviation angle *d* is measured for light rays of different frequency, including blue light and red light.

Which graph of *d* against frequency is correct?




## 27. O/N/21/12/Q23

Light reflects from a plane mirror as shown.

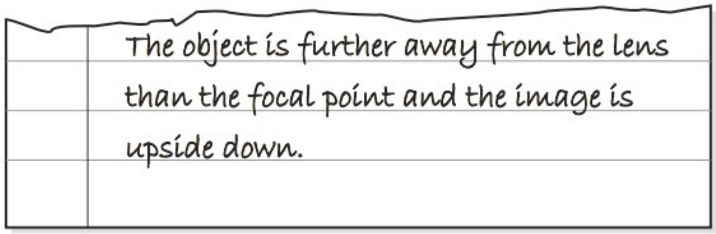



Which row is always correct?

| | symbol of angle X | relationship between X and Y |
|---|---|---|
| **A** | *i* | X = Y |
| **B** | *i* | X + Y = 90° |
| **C** | *r* | X = Y |
| **D** | *r* | X + Y = 90° |

## 28. O/N/21/12/Q24

A piece of paper torn from an exercise book is shown.




Which process is being described?

**A** the formation of a virtual image by a diverging lens

**B** the formation of a virtual image by a converging lens

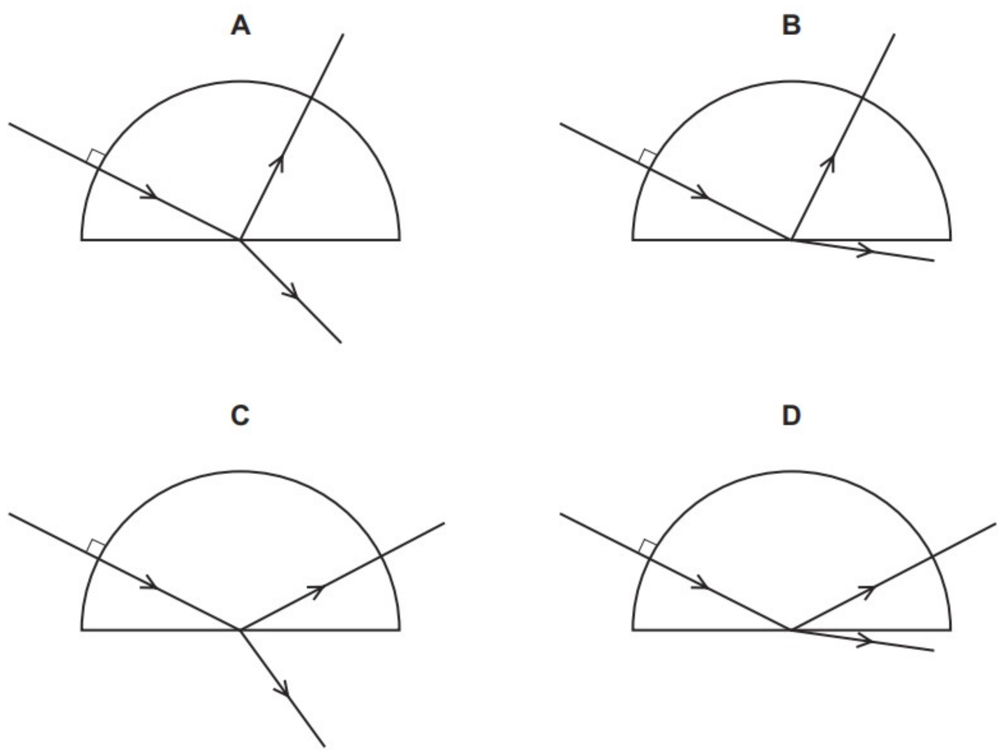
**C** the formation of a real image by a diverging lens

**D** the formation of a real image by a converging lens

## 29. M/J/21/12/Q24

A ray of red light in air enters a semi-circular block.

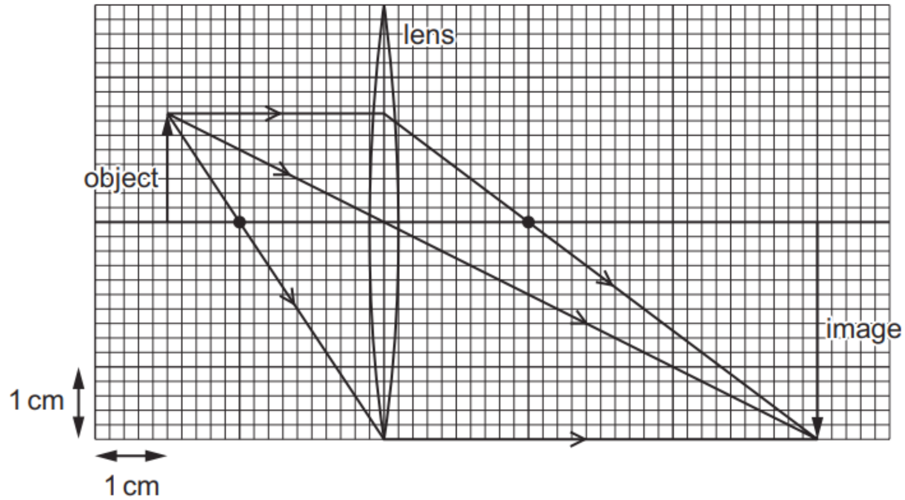
Which diagram shows the partial reflection and the refraction of the ray?



**30. M/J/21/12/Q25**

An object of height 1.5 cm is placed in front of a converging lens of focal length 2.0 cm.

The arrangement is shown on the full-scale ray diagram.




What is the linear magnification produced by the lens?

**A** 2.0 **B** 3.0 **C** 4.0 **D** 6.0

**31. M/J/21/12/Q26**

In which optical instrument is the distance between the object and the lens less than the focal length of the lens?

**A** camera

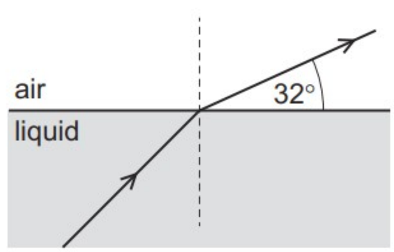
**B** magnifying glass

**C** photographic enlarger

**D** projector

**32. M/J/21/11/Q21**

Light refracts from a liquid into air as shown.




not to scale

The refractive index for light moving from air to the liquid is 1.4.

What is the angle of incidence in the liquid?

**A** 22° **B** 37° **C** 41° **D** 45°

**33. M/J/21/11/Q23**

Which statement about human vision is correct?

**A** In a normal eye, the image on the retina is magnified and upright.

**B** In a long-sighted eye, distant objects form images in front of the retina.

**C** Short-sighted eyes produce only virtual images.

**D** Short-sight is corrected by the use of a diverging lens.

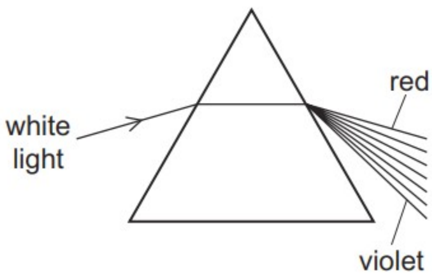
**34. M/J/21/11/Q24**

White light enters a prism and forms a spectrum.

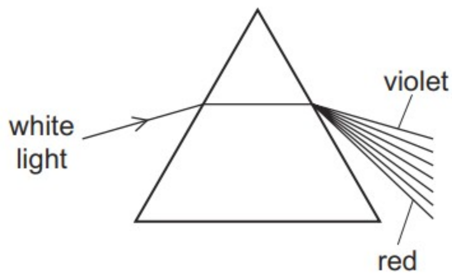
The rays in the air are labelled.

Which diagram shows how the white light is dispersed by the prism?

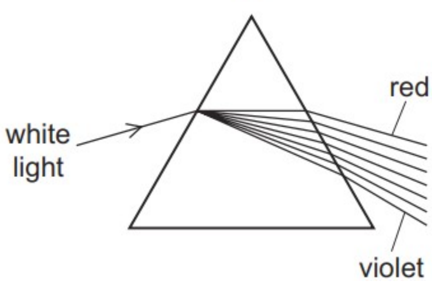
**A**



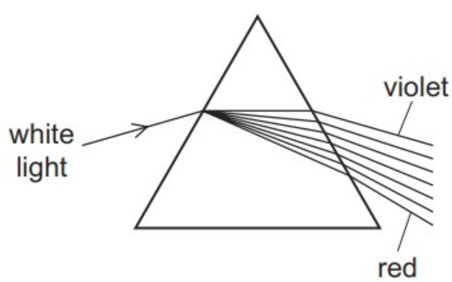

**B**




**C**

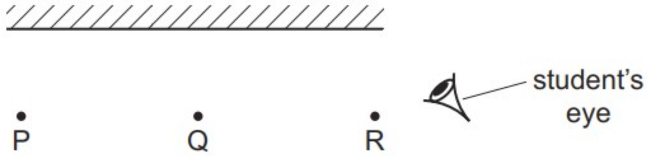



**D**



## 35. O/N/20/11/Q27

Three objects P, Q and R are placed in front of a plane mirror.




The student's eye is positioned as shown.

Which of the images of P, Q and R can the student see in the mirror?

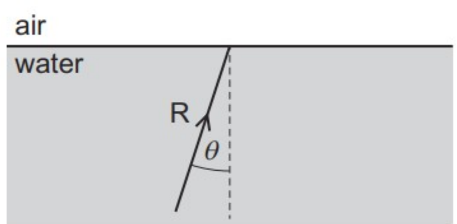
| | P | Q | R |
|---|---|---|---|
| **A** | ✓ | ✓ | ✓ |
| **B** | ✓ | ✓ | ✗ |
| **C** | ✓ | ✗ | ✗ |
| **D** | ✗ | ✗ | ✗ |

key
✓ = can see
✗ = cannot see

## 36. O/N/20/11/Q28

A ray of light R is incident on a water-to-air surface with an angle of incidence $\theta$.

The angle $\theta$ is less than the critical angle *c*.




Which statement describes the subsequent path of R?

**A** It travels back into the water with an angle of reflection equal to *c*.

**B** It travels back into the water with an angle of reflection greater than *c*.

**C** It travels into air with an angle of refraction greater than $\theta$.

**D** It travels into air with an angle of refraction smaller than $\theta$.

## 37. O/N/20/12/Q24

Which statement about light passing from air to glass is correct?

**A** The frequency of the light waves decreases.

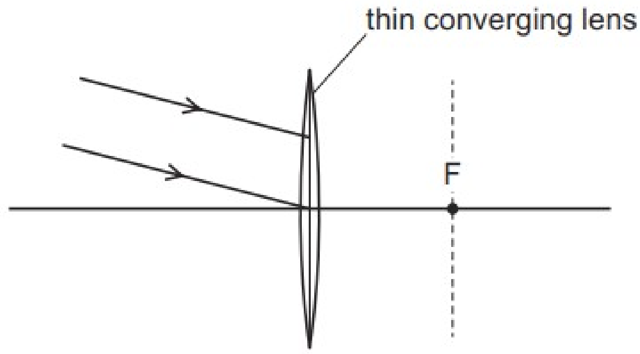
**B** The speed of the light waves decreases.

**C** The wavelength of the light waves increases.

**D** The wavelength of the light waves remains unchanged.

**38. O/N/20/11/Q29**

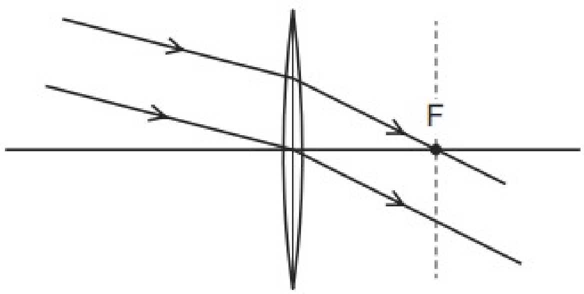
A parallel beam of light is incident on a thin converging lens.



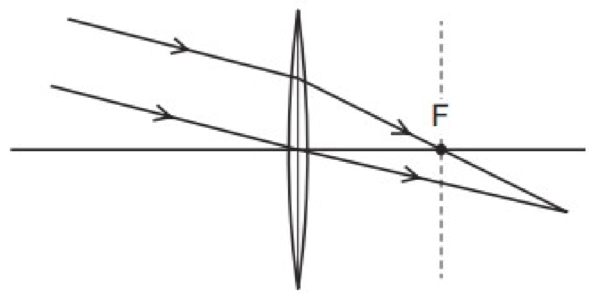

F is one focal point of the lens.

Which ray diagram shows the light after it has passed through the lens?

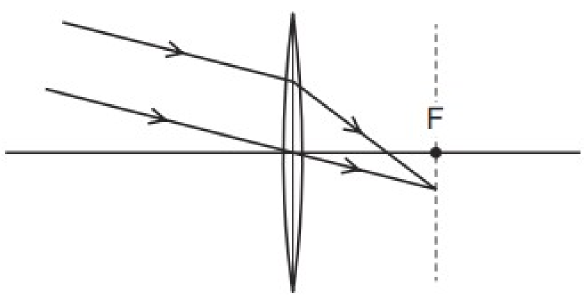
A



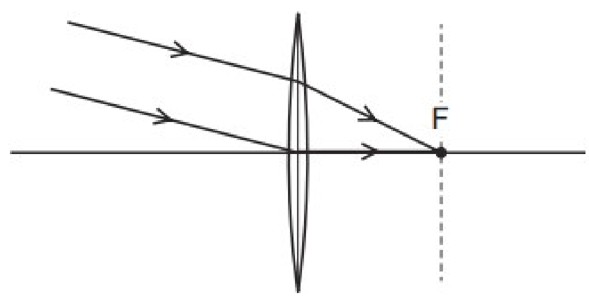

B




C




D




**39. O/N/20/12/Q23**

A plane mirror on a vertical wall forms an image of an object placed in front of it.

Which characteristics describe the image?

**A** real, inverted and smaller than the object

**B** real, upright and the same size as the object

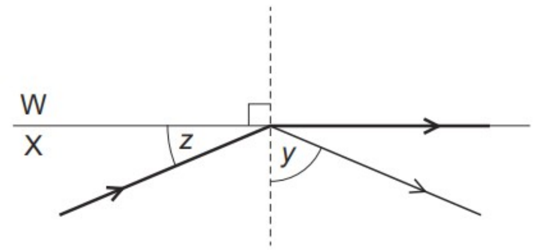
**C** virtual, upright and smaller than the object

**D** virtual, upright and the same size as the object

**40. M/J/20/12/Q31**

The diagram shows a ray of light incident on the boundary between two mediums W and X.

The mediums have different refractive indexes.




Some light is reflected and some passes along the surface between the two mediums. Angle *y* is greater than angle *z*.

Which statement is correct?

**A** W has a greater refractive index than X and angle *y* is equal to the critical angle.

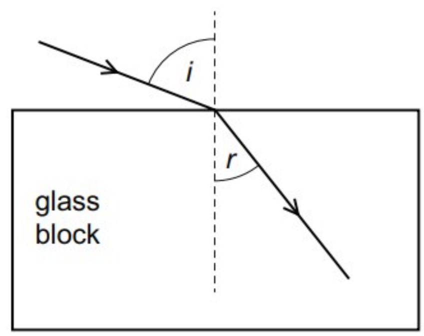
**B** W has a greater refractive index than X and angle *z* is equal to the critical angle.

**C** X has a greater refractive index than W and angle *y* is equal to the critical angle.

**D** X has a greater refractive index than W and angle *z* is equal to the critical angle.

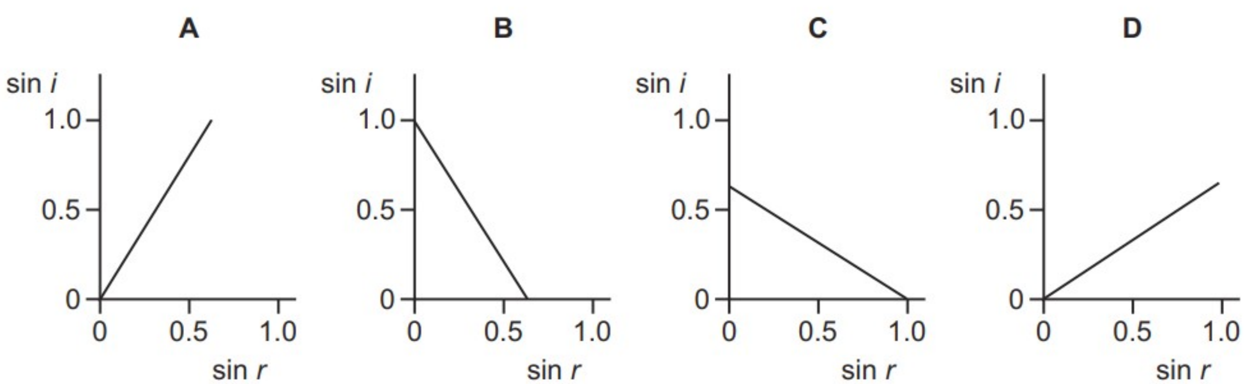
**41. M/J/20/12/Q32**

Light enters a glass block at an angle of incidence *i* and it produces an angle of refraction *r* in the glass.




Several different values of *i* and *r* are measured, and a graph is drawn of sin *i* against sin *r*.

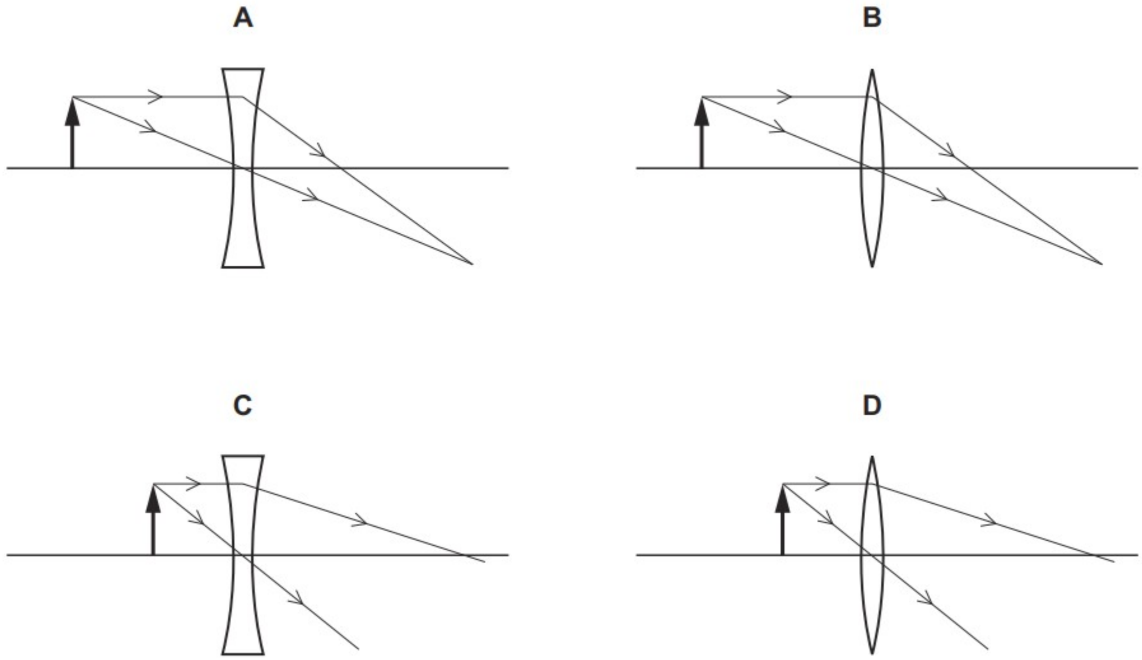
Which graph is correct?



**42. M/J/20/12/Q33**

A converging glass lens is used to produce a virtual, magnified image.

Which ray diagram shows the rays passing through the converging lens?

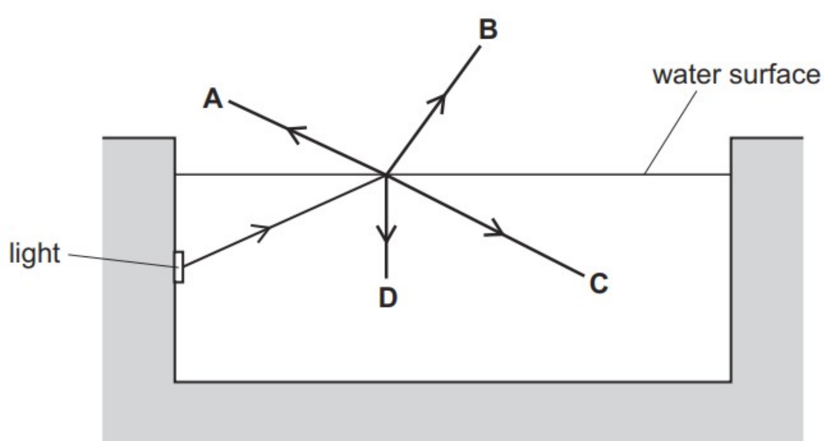



**43. M/J/20/11/Q27**

A swimming pool is lit by an underwater light.

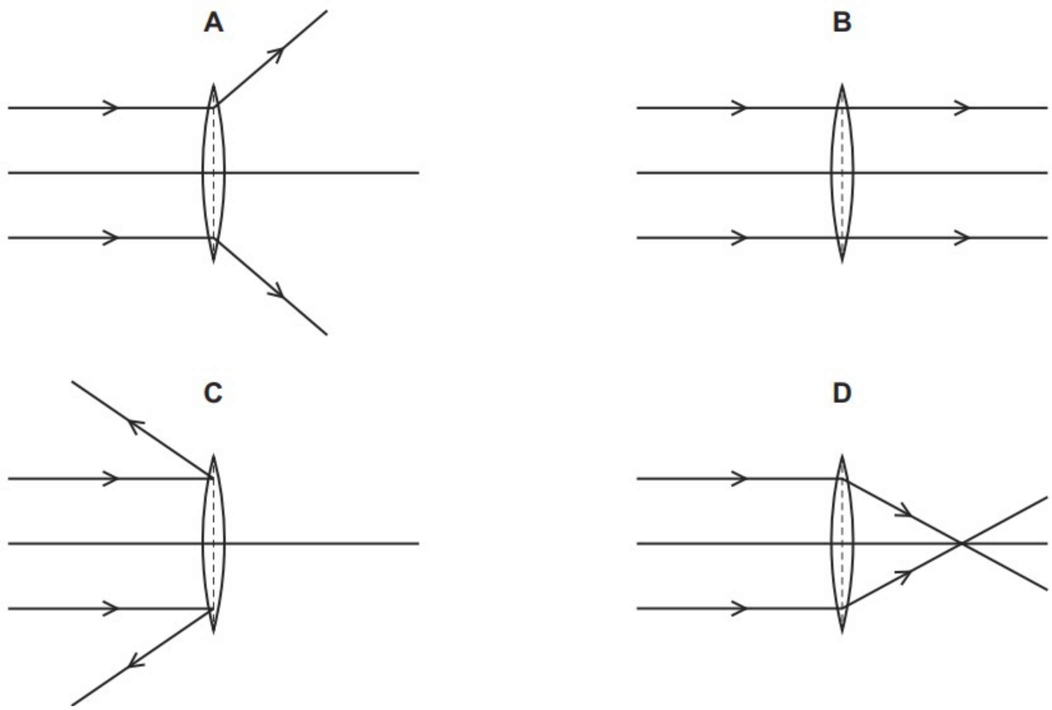
A ray of light is incident on the water surface.

What is the correct path for the ray of light?



**44. M/J/20/11/Q28**

Which diagram shows the action of a converging lens on a parallel beam of light?




**45. M/J/20/11/Q29**

What is a feature of red light compared with that of violet light?

**A** A prism deviates red light more.

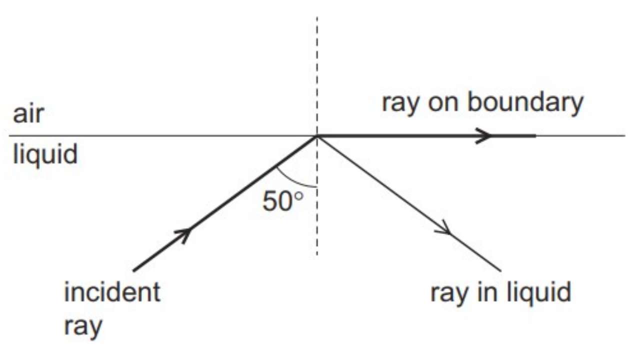
**B** Red light has a lower frequency.

**C** Red light has a shorter wavelength.

**D** The speed of red light in a vacuum is smaller.

**46. O/N/19/11/Q26**

The diagram shows a ray of light in liquid incident on the boundary with air. Two other rays are observed. One is in the liquid and the other is in the air on the boundary.

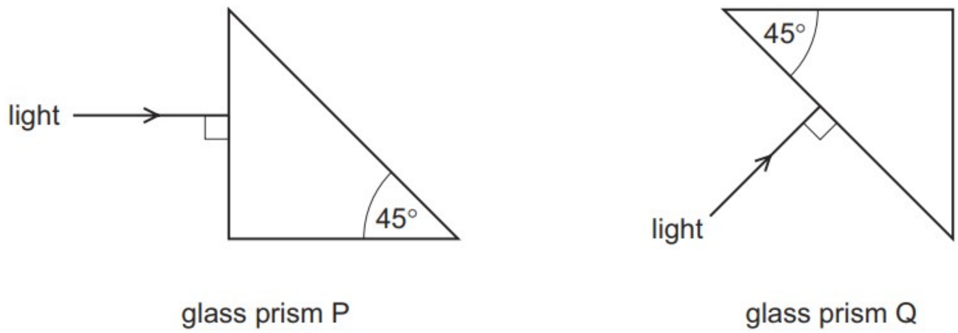



What is the angle of refraction?

**A** 0° **B** 40° **C** 50° **D** 90°

**47. O/N/19/11/Q27**

Light is incident at 90° on the surfaces of two glass prisms P and Q.




The critical angle for light travelling from glass into air is 42°.

Where does total internal reflection occur?

**A** in P and in Q

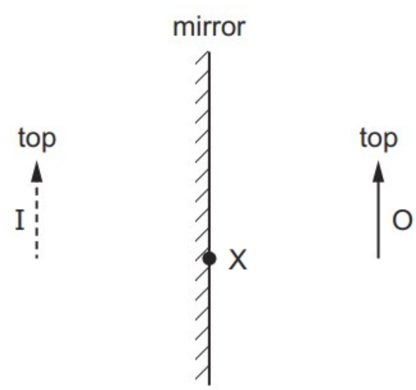
**B** in P only

**C** in Q only

**D** in neither P nor Q

**48. O/N/19/12/Q24**

An object O is placed in front of a plane mirror. I is the image formed.




A ray from the top of the object is incident on the mirror at X.

What happens to this ray?

**A** It reflects and passes through the bottom of O.

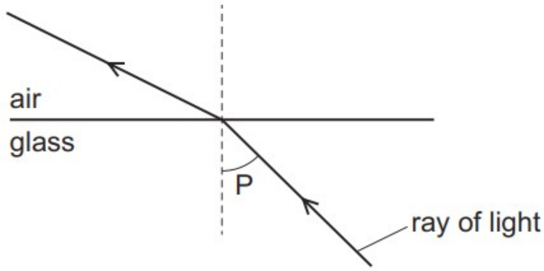
**B** It reflects and passes through the top of O.

**C** It reflects as though it came from the bottom of I.

**D** It reflects as though it came from the top of I.

**49. O/N/19/12/Q25**

The diagram shows light passing from glass into air.




What is the name of angle P?

**A** the angle of incidence

**B** the angle of reflection

**C** the angle of refraction

**D** the critical angle

**50. M/J/19/12/Q27**

An object is placed at a distance from a converging lens that is equal to twice the focal length of the lens.

Which statement about the image is correct?

**A** It is enlarged.

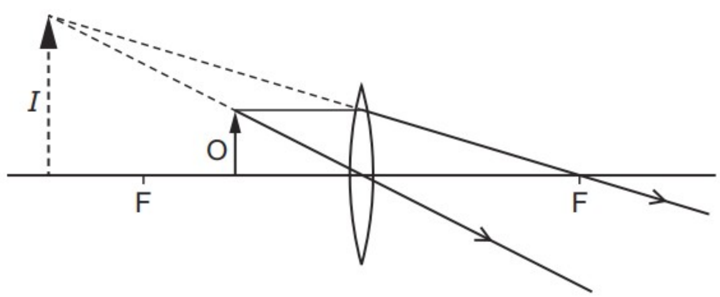
**B** It is inverted.

**C** It is on the same side of the lens as the object.

**D** It is virtual.

**51. M/J/19/12/Q28**

The lens in the diagram produces an image $I$ of the object O.




Why is this **not** the ray diagram for a photographic enlarger?

**A** The image is magnified.

**B** The image is virtual.

**C** The lens is a converging lens.

**D** The lens is too thin.

## 52. M/J/19/11/Q24

A ray of light strikes a plane mirror at an angle of incidence of 20°.

The angle of incidence is then increased by 5°.

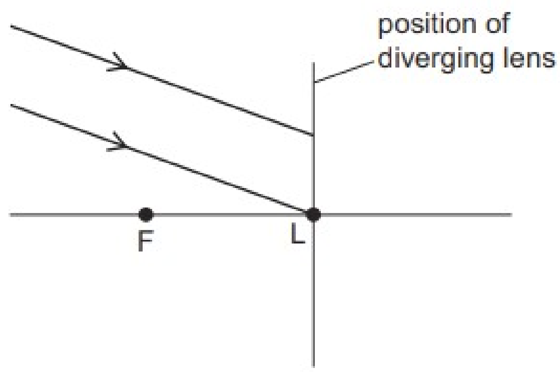
What is the new angle between the incident ray and the reflected ray?

**A** 10° **B** 25° **C** 45° **D** 50°

## 53. M/J/19/11/Q25

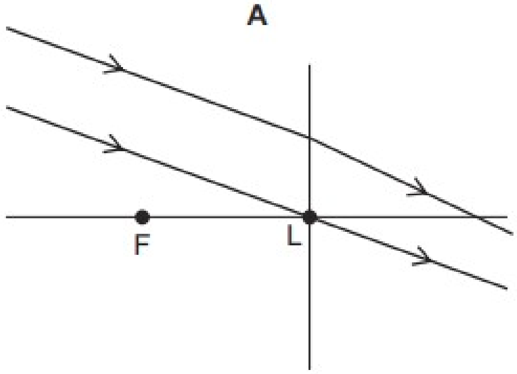
A parallel beam of light is incident on a thin diverging lens.

The focal length of the lens is FL, as shown in the diagram.

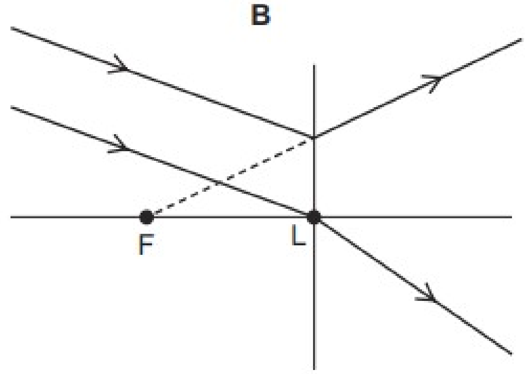



Which ray diagram shows the beam after it has passed through the lens?

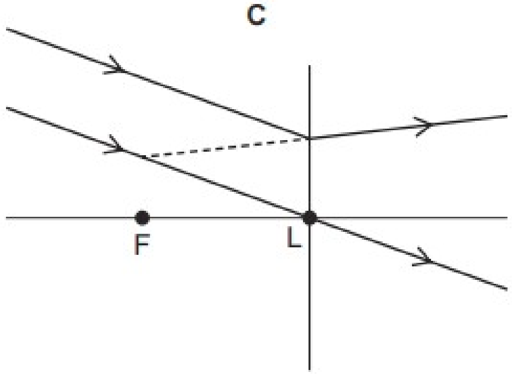
**A**



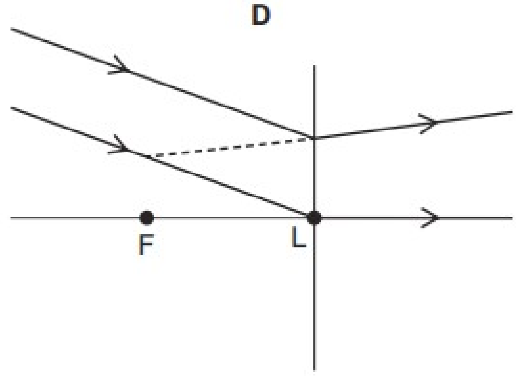

**B**




**C**




**D**



**54. M/J/19/11/Q26**

The following lists show colours of the spectrum.

Which list shows these colours in order of increasing frequency?

**A** blue, violet, red, orange, yellow, green

**B** green, blue, violet, red, orange, yellow

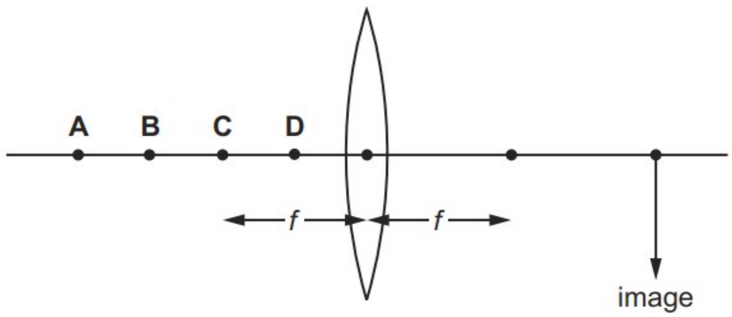
**C** red, orange, violet, yellow, green, blue

**D** red, orange, yellow, green, blue, violet

**55. O/N/18/11/Q19**

The diagram shows a thin converging lens of focal length *f*.

Where must an object be placed to produce a real image in the position shown?




**56. O/N/18/11/Q20**

White light is dispersed by a prism. Compared with blue light, the **red** light is

**A** slowed down less and refracted less.

**B** slowed down less and refracted more.

**C** slowed down more and refracted less.

**D** slowed down more and refracted more.

**57. ON/18/12/Q25**

A student reads the following in her physics book.

'The incident angle is greater than 42° which is the critical angle for glass in air.'

What is the student reading about?

**A** focal length of a glass lens

**B** reflection in a plane mirror

**C** refraction as light enters glass

**D** total internal reflection

**58. M/J/18/12/Q28**

Which statement about blue light is correct?

**A** Blue light has a smaller frequency than red light.

**B** Blue light has a longer wavelength than red light.

**C** Blue light has the same speed in glass as red light.

**D** Blue light is refracted more by a glass prism than red light.

**59. M/J/18/12/Q29**

Which device uses total internal reflection?

**A** magnifying glass

**B** optical fibre

**C** photographic enlarger

**D** projector

**60. M/J/18/11/Q23**

Which statement is correct?

**A** Total internal reflection only occurs when light travels from air into glass.

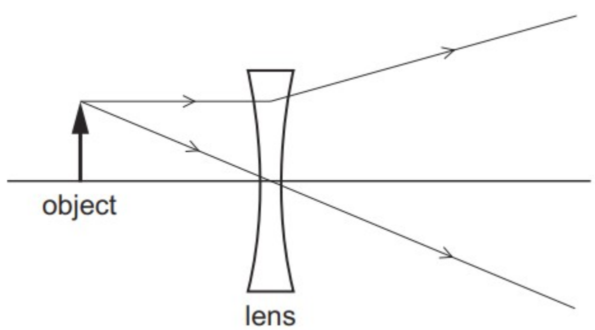
**B** The larger the refractive index of glass, the larger is the critical angle.

**C** When total internal reflection occurs, the angle of incidence is equal to the angle of reflection.

**D** When total internal reflection occurs, the angle of incidence is less than the critical angle.

**61. M/J/18/11/Q24**

The diagram shows the paths of two rays from the top of an object through a lens.




The object is viewed from the opposite side of the lens to the object.

How does the image compare with the object?

**A** larger and inverted

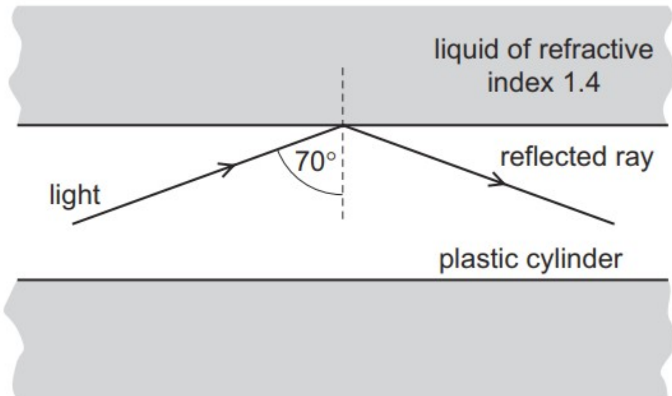
**B** larger and the same way up

**C** smaller and inverted

**D** smaller and the same way up

**62. ON/17/11/Q20**

A solid plastic cylinder is immersed in a liquid of refractive index 1.4. Light travelling in the plastic cylinder strikes the inside surface at an angle of incidence of 70°. The light undergoes total internal reflection.




What are the values of the critical angle in the plastic and the refractive index of the plastic?

| | critical angle in plastic | refractive index of plastic |
|---|---|---|
| **A** | greater than 70° | greater than 1.4 |
| **B** | greater than 70° | less than 1.4 |
| **C** | less than 70° | greater than 1.4 |
| **D** | less than 70° | less than 1.4 |

**63. O/N/17/11/Q21**

What is the name and shape of the lens used to correct short sight?

| | name of lens | shape of lens |
|---|---|---|
| **A** | converging |  |
| **B** | converging |  |
| **C** | diverging |  |
| **D** | diverging |  |

## 64. O/N/17/11/Q22

An object is placed in front of a converging lens. The lens forms a magnified image of the object on a screen.

Which statement is correct?

**A** The distance between the object and the lens is greater than the focal length.

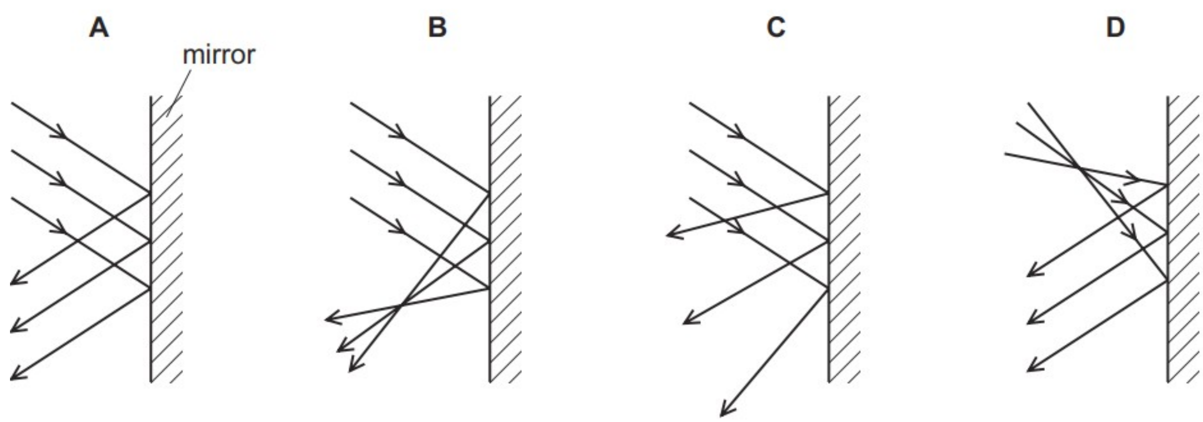
**B** The image formed is a virtual image.

**C** The image is the right way up.

**D** The lens is acting as a magnifying glass.

## 65. O/N/17/12/Q20

Which diagram shows reflection by a plane mirror?




## 66. O/N/17/12/Q21

Light is incident on a plastic block of refractive index 1.5.

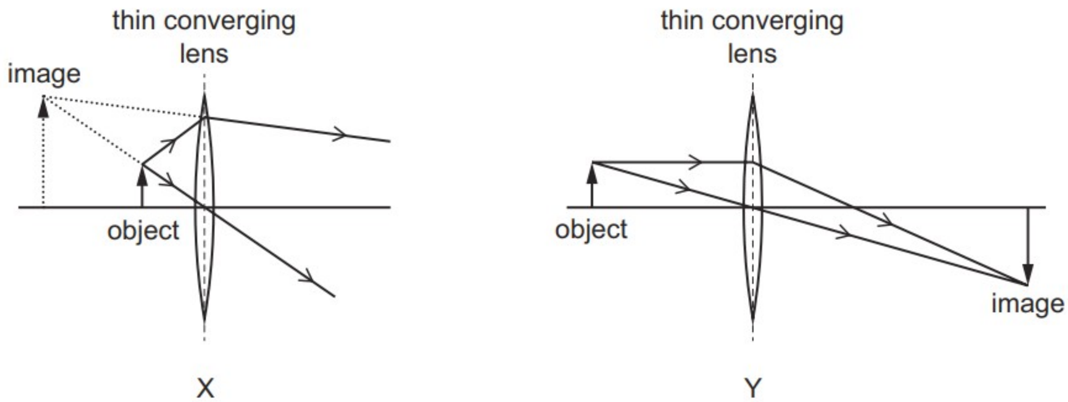
The angle of incidence is 50°.

What is the angle of refraction?

**A** 31° **B** 33° **C** 40° **D** 75°

**67. O/N/17/12/Q23**

The ray diagrams, X and Y, show two ways in which a thin converging lens produces an image that is larger than the object.



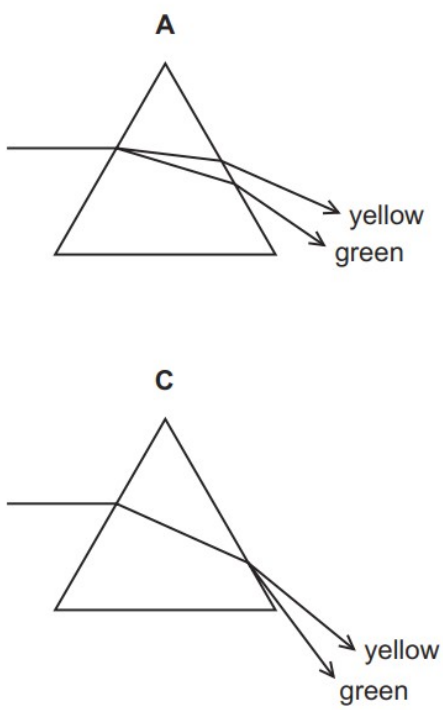

Which devices use a lens as shown in diagram X and in diagram Y?

| | X | Y |
|---|---|---|
| **A** | camera | magnifying glass |
| **B** | magnifying glass | projector |
| **C** | photographic enlarger | camera |
| **D** | photographic enlarger | projector |

**68. M/J/17/12/Q28**

A narrow beam of yellow and green light is separated as it passes through a prism.

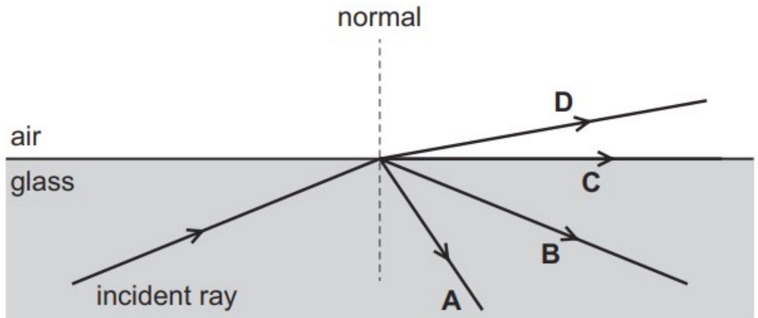
Which ray diagram is correct?



**69. M/J/17/11/Q27**

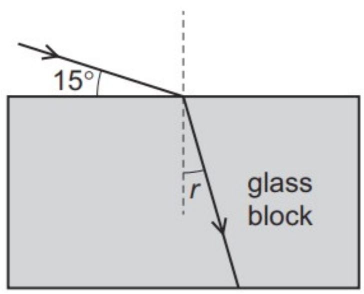
Light travelling in glass is incident on a glass-air boundary. The angle of incidence of the light is greater than the critical angle.

Which arrow shows the direction of the light after it is incident on the boundary?




**70. M/J/17/11/Q28**

Light strikes the top surface of a glass block at an angle of 15° as shown.




(not to scale)

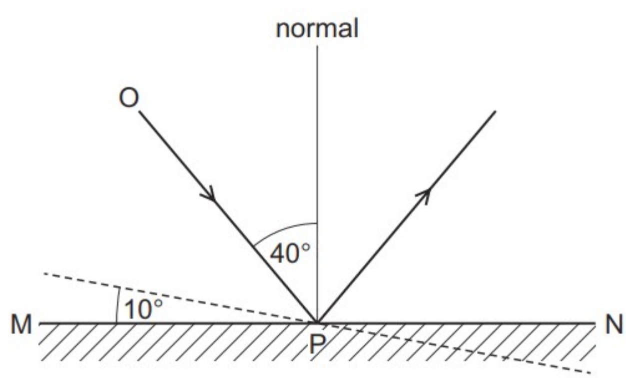
The refractive index of glass is 1.5.

What is the angle of refraction *r*?

**A** 10° **B** 23° **C** 40° **D** 50°

**71. O/N/16/11/Q23**

The angle of incidence of ray OP on the plane mirror MN is 40°.




The mirror is rotated through 10°, as shown by the dashed line. The direction of the incident ray OP does not change.

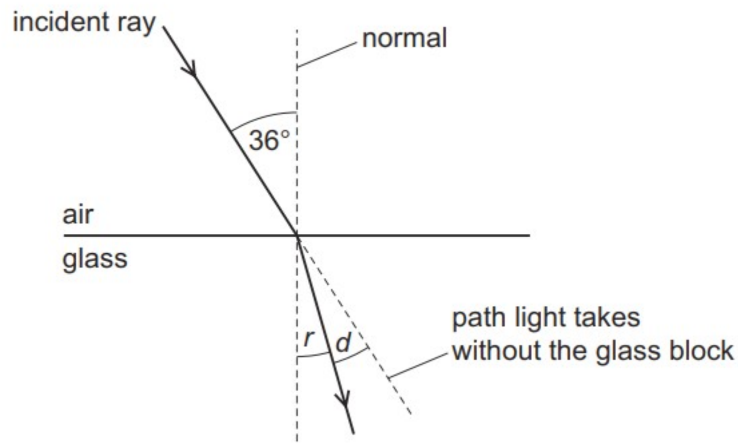
What is the new angle of incidence?

**A** 30° **B** 40° **C** 50° **D** 60°

**72. O/N/16/11/Q24**

A ray of light is incident on the surface of a glass block. The diagram is not drawn to scale.

The refractive index of the glass is 1.5.



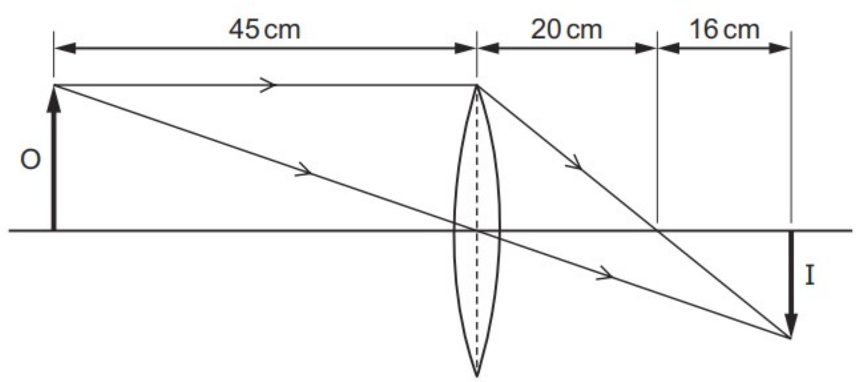

The angle of refraction is *r*. The angle between the refracted ray and the path the light takes without the glass block is *d*.

What are *r* and *d*?

| | *r* / ° | *d* / ° |
|---|---|---|
| **A** | 23 | 12 |
| **B** | 24 | 12 |
| **C** | 23 | 13 |
| **D** | 24 | 13 |

**73. O/N/16/11/Q25**

In the diagram, a convex lens forms an image I of an object O. The diagram is not drawn to scale.

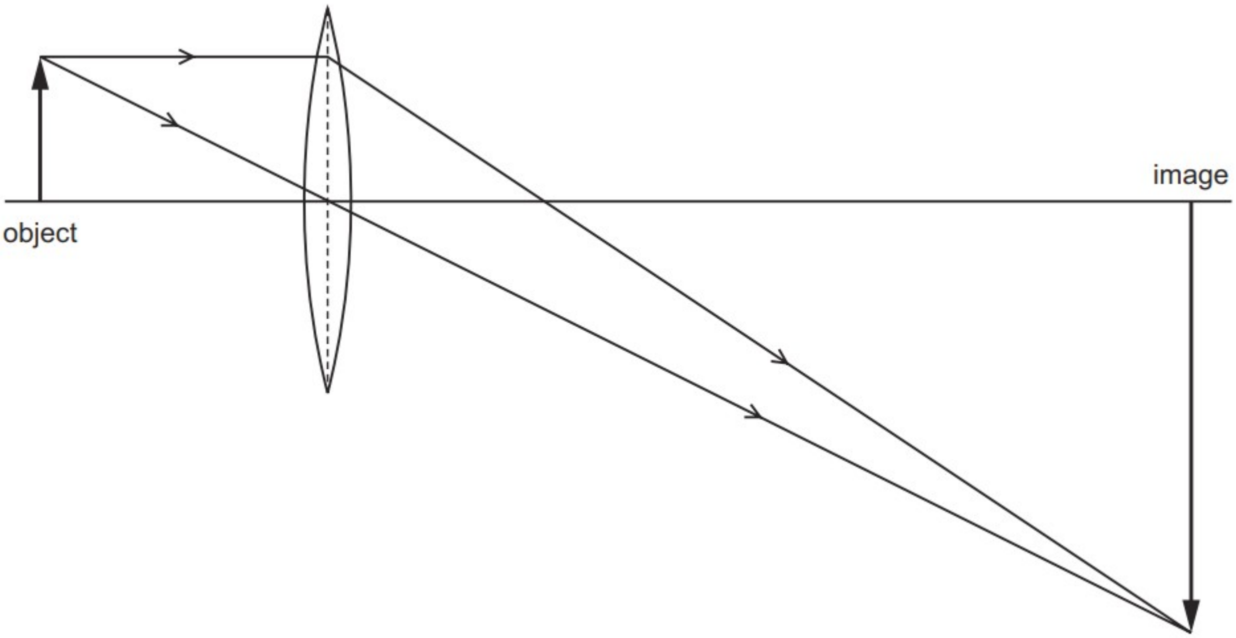



What is the focal length of the lens?

**A** 16 cm **B** 20 cm **C** 36 cm **D** 45 cm

**74. O/N/16/11/Q26**

A lens is used to produce a magnified image, as shown in the scale diagram.

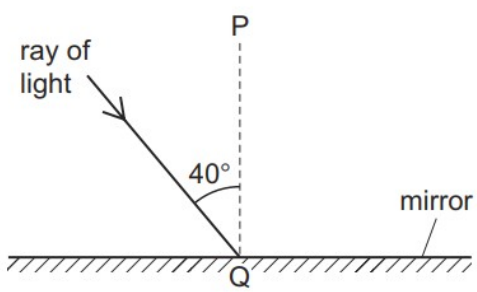



What is the linear magnification produced by the lens?

**A** 0.33 **B** 3.0 **C** 4.0 **D** 6.0

**75. M/J/16/12/Q18**

The diagram shows light incident on a plane mirror.

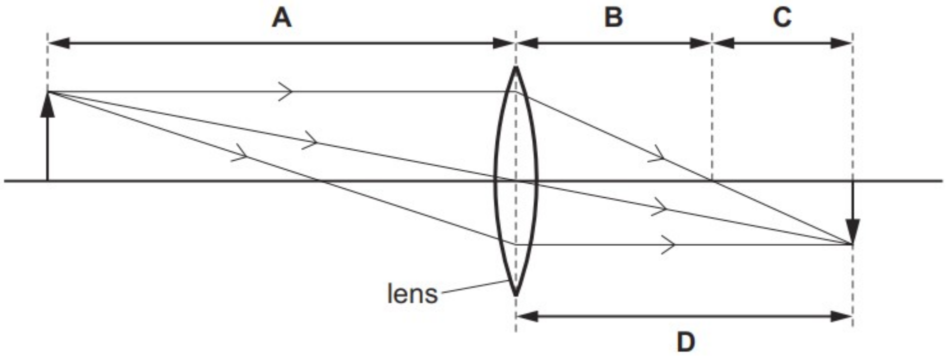



Which row gives the angle of reflection and the name of line PQ?

| | angle of reflection | the line PQ is called the |
|---|---|---|
| **A** | 40° | normal |
| **B** | 40° | reflected ray |
| **C** | 50° | normal |
| **D** | 50° | reflected ray |

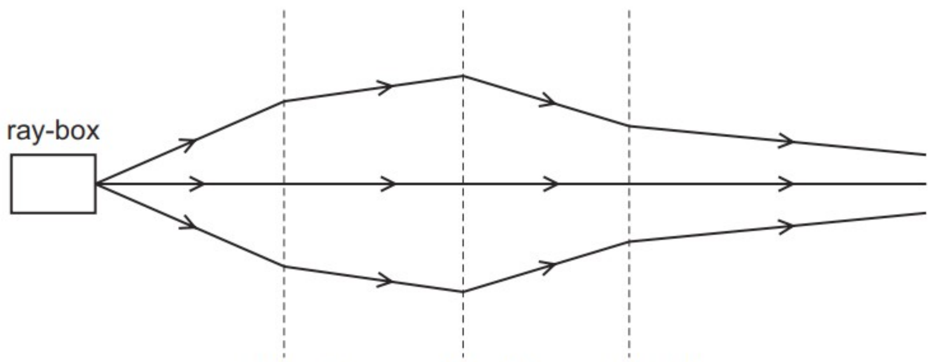
**76. M/J/16/12/Q19**

Which length is the focal length of the lens shown in the diagram?




**77. M/J/16/12/Q20**

The rays of light from a ray-box pass through three lenses placed at positions 1, 2 and 3.

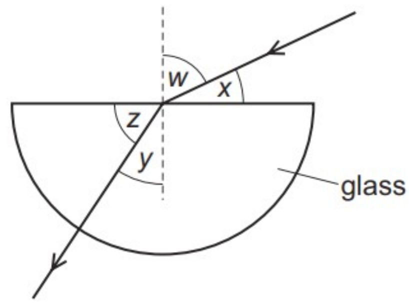



What type of lens is used at each position?

| | position 1 | position 2 | position 3 |
|---|---|---|---|
| **A** | converging | converging | converging |
| **B** | converging | converging | diverging |
| **C** | diverging | converging | diverging |
| **D** | diverging | diverging | converging |

## 78. M/J/16/12/Q21

Light passes from air into a block of glass, as shown.




Which expression is equal to the refractive index of glass?

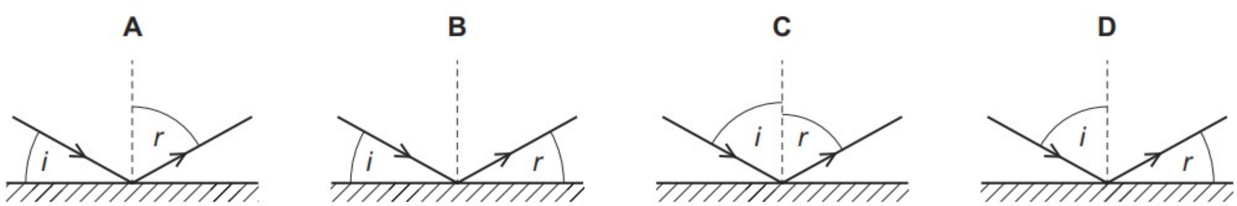
**A** $\dfrac{\sin w}{\sin y}$ **B** $\dfrac{\sin w}{\sin z}$ **C** $\dfrac{\sin y}{\sin w}$ **D** $\dfrac{\sin z}{\sin x}$

## 79. M/J/16/11/Q21

Light is incident on a mirror. The light is reflected from the mirror.

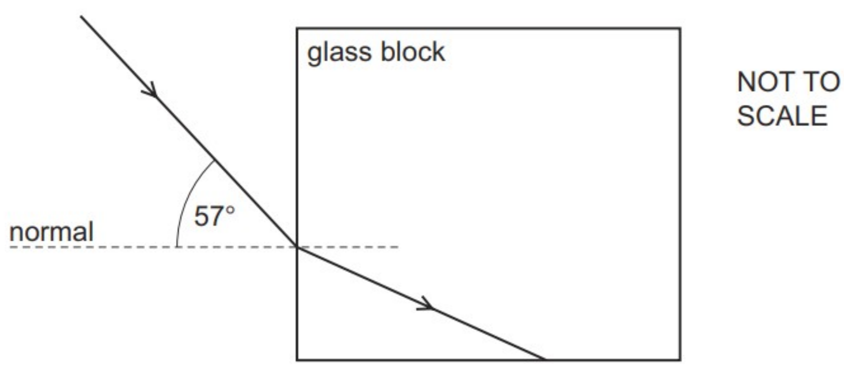
The angle of incidence is $i$ and the angle of reflection is $r$.

Which diagram correctly shows $i$ and $r$?




## 80. M/J/16/11/Q23

Light passes from air into a glass block of refractive index 1.5, as shown.




What is the angle of refraction in the glass and what is the critical angle?

| | angle of refraction | critical angle |
|---|---|---|
| **A** | 34° | 42° |
| **B** | 34° | 60° |
| **C** | 38° | 42° |
| **D** | 38° | 60° |

## 81. O/N/15/11/Q20

Which statement about red light and blue light is correct?

**A** Red light has a higher frequency than blue light.

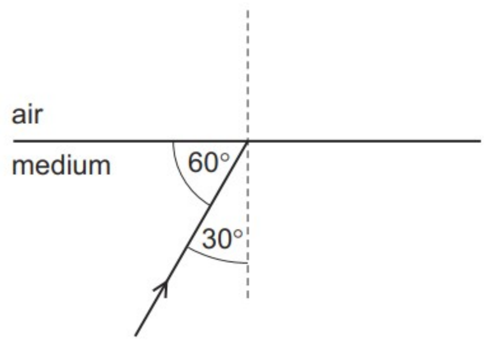
**B** Red light has a longer wavelength than blue light.

**C** Red light has the same speed in glass as blue light.

**D** Red light is refracted by a glass prism more than blue light.

## 82. O/N/15/11/Q21

A ray of light in a transparent medium of refractive index 1.8 is incident on the surface as shown. The light enters air.



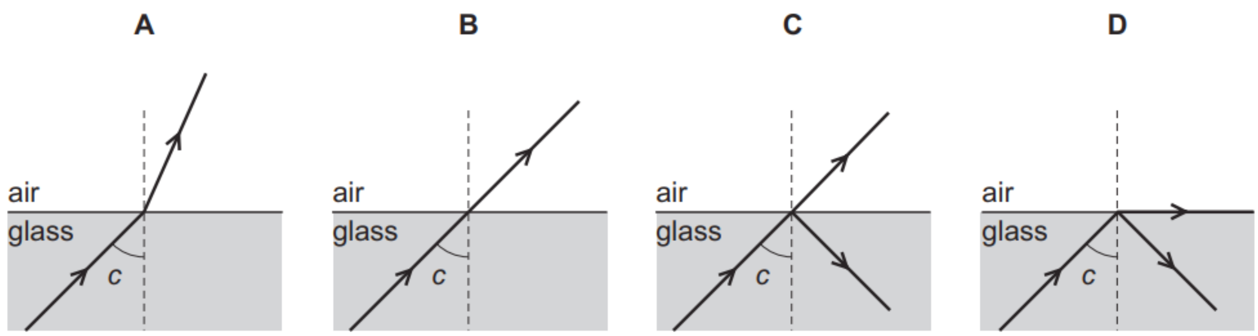

What is the angle between the refracted ray and the normal in air?

**A** 29° **B** 33° **C** 54° **D** 64°

## 83. M/J/15/12/Q25

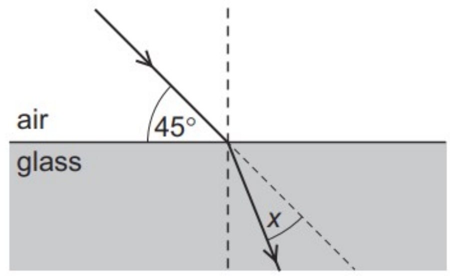
A ray of light in glass is incident on the surface at an angle *c*. The angle *c* is the critical angle.

Which diagram shows what happens to the light?



**84. M/J/15/12/Q26**

A ray of light is incident on the surface of a glass block, as shown in the diagram below.




The refractive index of the glass is 1.5.

The light ray changes direction when entering the glass.

What is the angle *x* through which the ray moves?

**A** 30° **B** 28° **C** 17° **D** 15°

**85. M/J/15/12/Q27**

An image is formed by a thin converging lens when it is used as a magnifying glass.

What is the correct description of the image?

**A** real and erect

**B** real and inverted

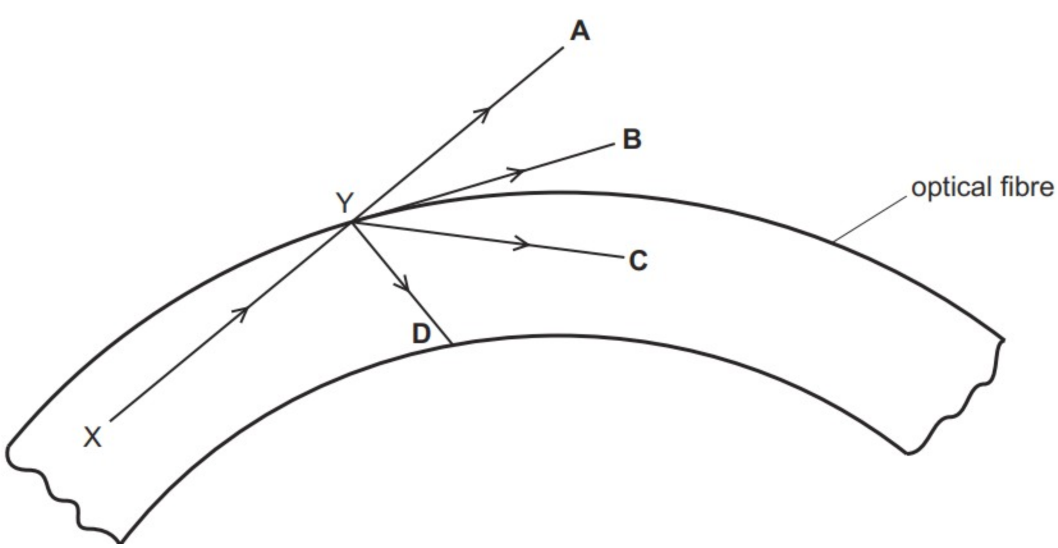
**C** virtual and erect

**D** virtual and inverted

**86. M/J/15/11/Q24**

A ray of light travels from X to Y along an optical fibre. The angle of incidence at Y is greater than the critical angle.

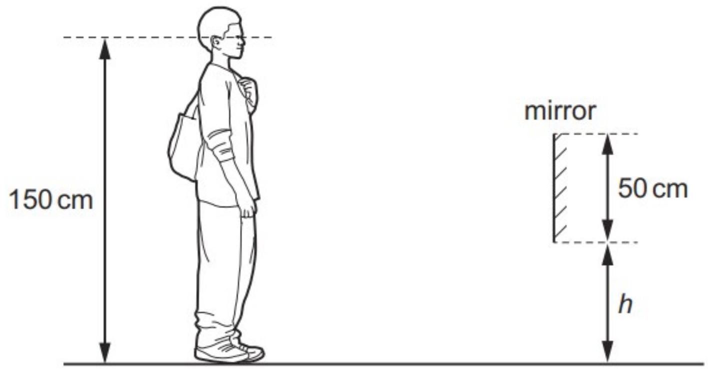
In which direction does the ray of light travel after reaching point Y?



## 87. O/N/14/11/Q20

A shoe shop puts a mirror on the wall so that customers can look at their shoes.

The length of the mirror is 50 cm. A customer has eyes 150 cm above ground level.



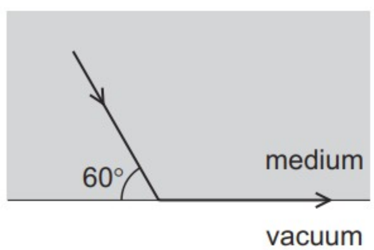

The bottom of the mirror is at height *h* above the ground.

What is the smallest value of *h* that allows the customer to see an image of his shoes in the mirror?

**A** 0 **B** 25 cm **C** 50 cm **D** 75 cm

## 88. O/N/14/11/Q21

The diagram shows light travelling through a medium. The light reaches the boundary with a vacuum as shown. The light emerges travelling along the surface.




What is the refractive index of the medium?

**A** $\dfrac{\sin 60°}{\sin 30°}$ **B** $\dfrac{\sin 60°}{\sin 90°}$ **C** $\dfrac{\sin 90°}{\sin 30°}$ **D** $\dfrac{\sin 90°}{\sin 60°}$

## 89. O/N/14/11/Q23

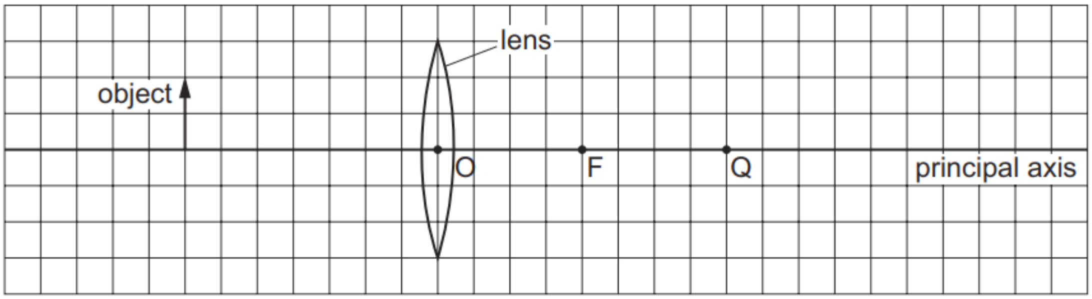
A digital camera uses a lens to produce a diminished (reduced in size) image on a light sensor.

Which row shows the correct type of lens and the nature of the image?

| | type of lens | nature of image |
|---|---|---|
| **A** | converging | inverted |
| **B** | converging | upright |
| **C** | diverging | inverted |
| **D** | diverging | upright |

**90. O/N/14/11/Q22**

The diagram shows an object on the principal axis of a converging (convex) lens. A principal focus of the lens is at F.




Where is the image formed by the lens?

**A** between O and F

**B** between F and Q

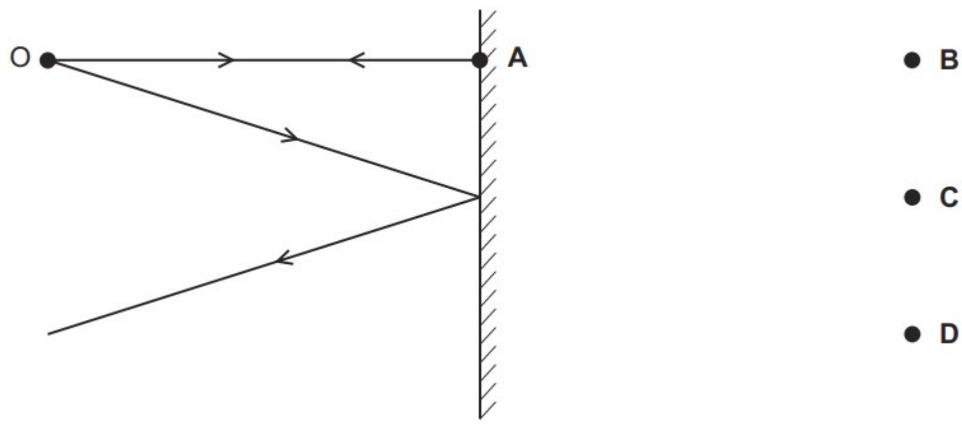
**C** at Q

**D** to the right of Q

**91. M/J/14/12/Q26**

The diagram shows two divergent rays of light from an object O being reflected from a plane mirror.

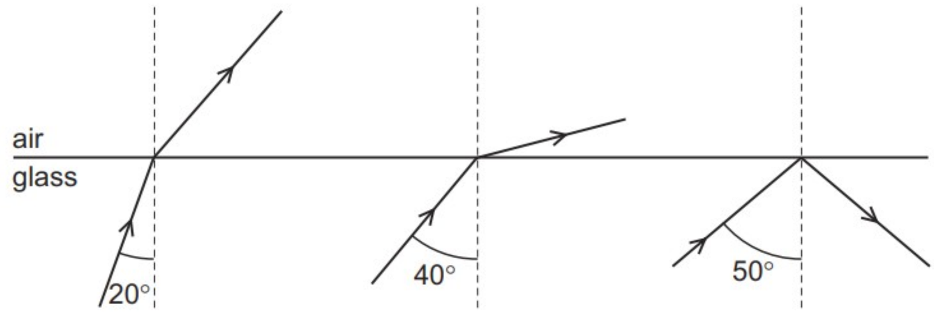
At which position is the image formed?



**92. M/J/14/11/Q25**

Three rays of light are incident on the boundary between a glass block and air.

The angles of incidence are different.



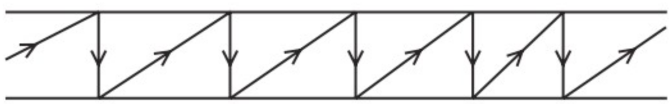

What is a possible critical angle for light in the glass?

**A** 15° **B** 30° **C** 45° **D** 60°

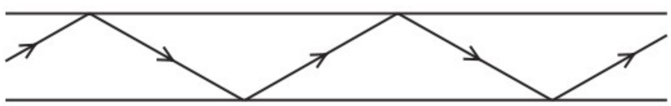
**93. O/N/13/11/Q20**

Which diagram represents the reflection of light along an optical fibre?

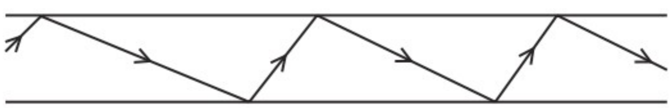
**A**



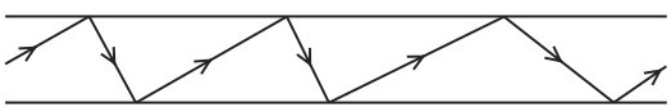
**B**



**C**

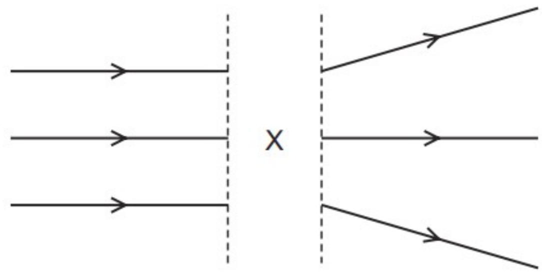


**D**



**94. O/N/13/11/Q21**

The diagram shows rays of light.




What is in the space labelled X?

**A** a converging lens

**B** a diverging lens

**C** a plane mirror

**D** a rectangular glass block

**95. O/N/13/12/Q24**

A ray of light strikes the surface of a glass block at an angle of incidence of 45°.

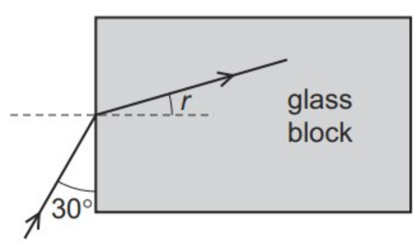
The refractive index of the glass is 1.8.

What is the angle of refraction inside the block?

**A** 23° **B** 25° **C** 45° **D** 81°

**96. M/J/13/11/Q20**

A ray of light meets the face of a glass block at an angle of 30° as shown.




The refractive index of the glass is 1.5.

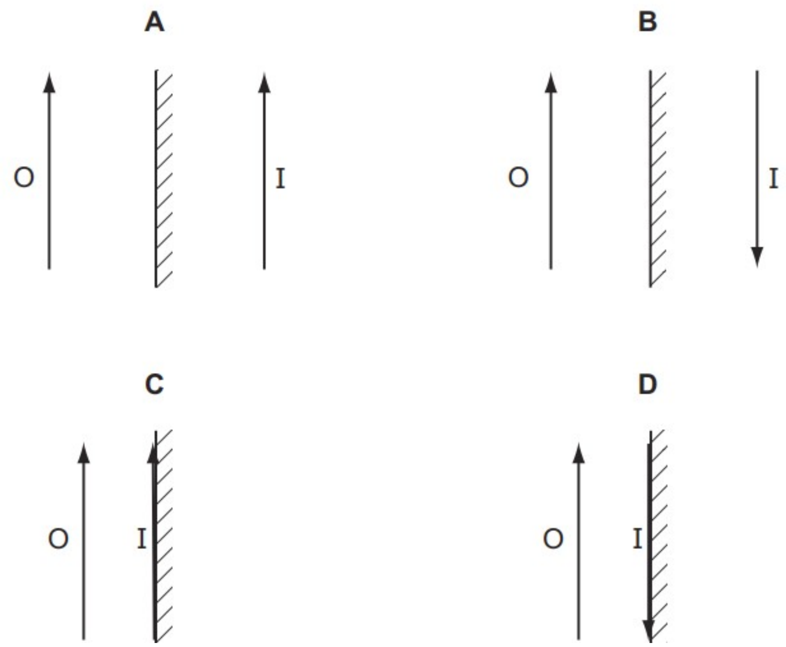
What is the angle of refraction *r* inside the glass block?

**A** 19° **B** 20° **C** 35° **D** 40°

**97. O/N/12/11/Q22**

An object O is placed in front of a plane mirror.

Which diagram correctly represents the image I formed by the mirror?




**98. O/N/12/11/Q25**

Red and violet are the colours at the ends of the visible spectrum.

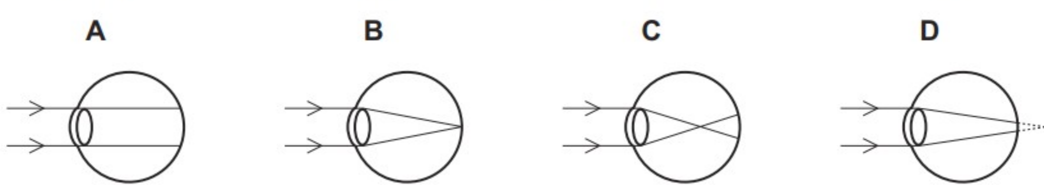
How do the frequencies and the wavelengths of these colours compare?

| | higher frequency | longer wavelength |
|---|---|---|
| **A** | red | red |
| **B** | red | violet |
| **C** | violet | red |
| **D** | violet | violet |

**99. M/J/12/12/Q24**

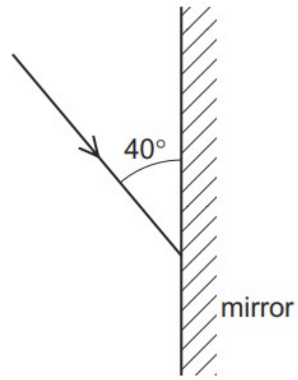
A man is short-sighted.

Which ray diagram shows what happens in his eye when he looks at a distant object?



## 100. M/J/12/12/Q22

The diagram shows a ray of light directed at a plane mirror.



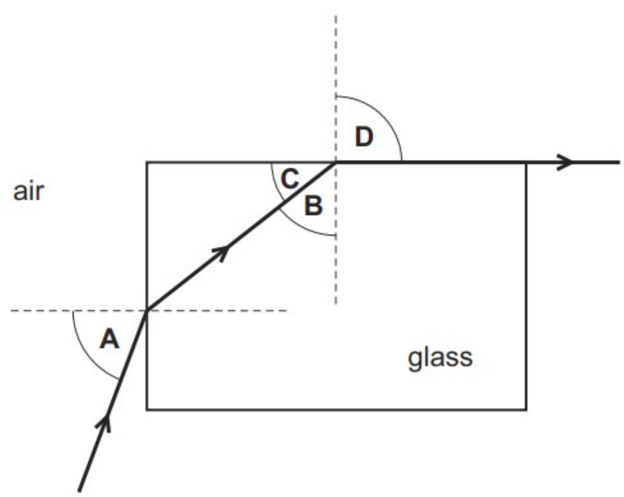

What are the angle of incidence and the angle of reflection?

| | angle of incidence | angle of reflection |
|---|---|---|
| **A** | 40° | 40° |
| **B** | 40° | 50° |
| **C** | 50° | 40° |
| **D** | 50° | 50° |

## 101. M/J/12/12/Q23

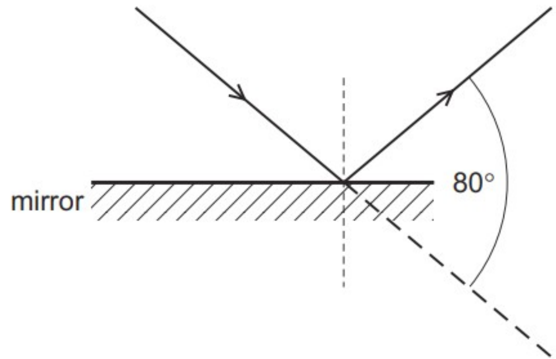
Light travels through a glass block as shown.

Which angle is the critical angle for light in the glass?



**102. M/J/12/11/Q22**

Light is incident on a mirror and is reflected as shown.




What is the angle of incidence and the angle of reflection?

| | angle of incidence / ° | angle of reflection / ° |
|---|---|---|
| **A** | 40 | 40 |
| **B** | 40 | 50 |
| **C** | 50 | 40 |
| **D** | 50 | 50 |

**103. M/J/12/11/Q23**

Light is incident on one face of a glass block at an angle of incidence of 40°. The glass block is in air.

The refractive index of the glass is 1.46.

What is the angle of refraction inside the glass block?

**A** 26° **B** 27° **C** 58° **D** 70°

**104. M/J/12/11/Q24**

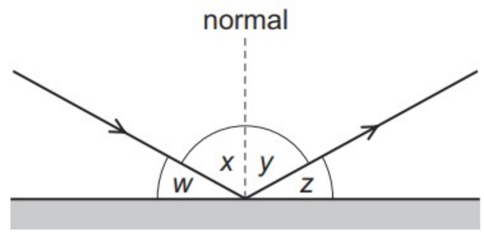
In a short-sighted eye, light from distant objects is not focused on the retina.

Where is this light focused and what type of lens is needed to correct the problem?

| | where focused | lens needed |
|---|---|---|
| **A** | behind the retina | converging lens |
| **B** | behind the retina | diverging lens |
| **C** | in front of the retina | converging lens |
| **D** | in front of the retina | diverging lens |

**105. M/J/11/12/Q23**

A ray of light strikes a plane mirror and is reflected.

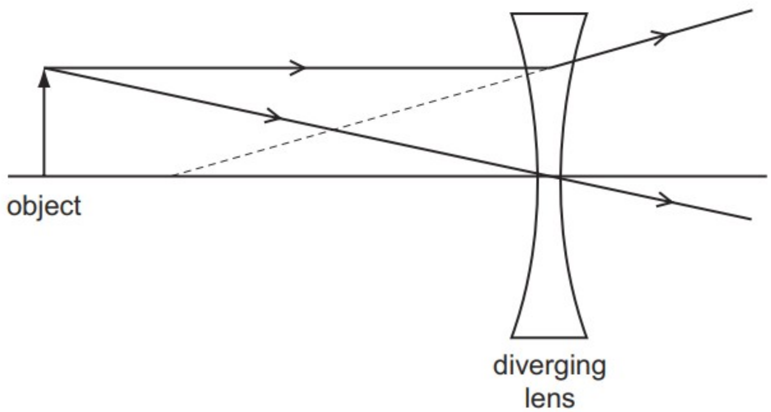



Which pair of angles **must** be equal in value?

**A** *w* and *x* **B** *w* and *y* **C** *x* and *y* **D** *x* and *z*

**106. M/J/11/12/Q24**

The ray diagram shows two rays from a point on an object placed in front of a diverging (concave) lens.




What are the properties of the image produced?

**A** real and larger than the object

**B** real and smaller than the object

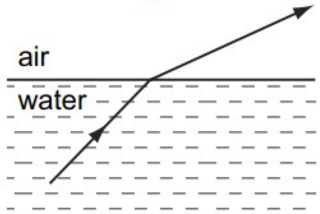
**C** virtual and larger than the object

**D** virtual and smaller than the object

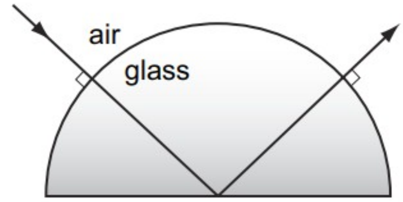
**107. M/J/11/11/Q22**

In which diagram is the path of the light ray **not** correct?

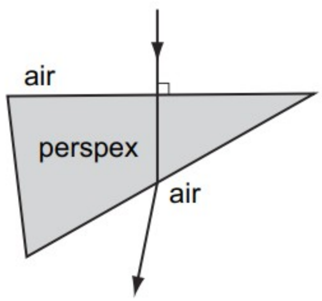
**A**



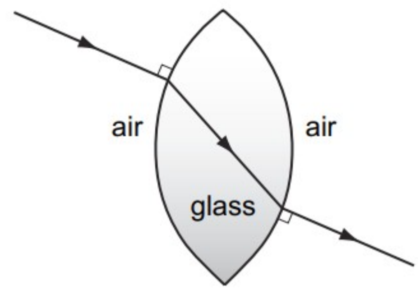

**B**




**C**




**D**




**108. O/N/10/11/Q22**

A ray of light strikes the surface of a glass block at an angle of incidence of 45°.

The refractive index of the glass is 1.5.

What is the angle of refraction inside the block?

**A** 28° **B** 30° **C** 45° **D** 67°

**109. O/N/10/11/Q23**

An object is viewed through a concave (diverging) lens.

What is the correct description of the image formed?

**A** real, inverted, magnified

**B** real, upright, diminished

**C** virtual, inverted, magnified

**D** virtual, upright, diminished

**110. M/J/10/12/Q22**

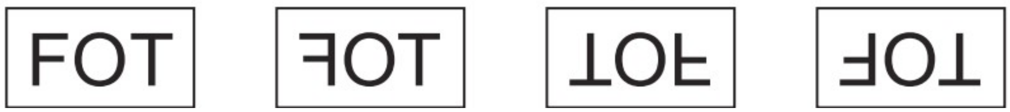
A student holds a sheet of paper with letters on it facing a plane mirror.

The letters on the paper are shown.

TOF

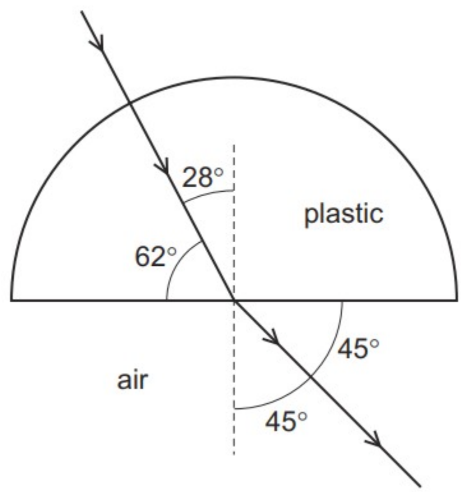
What does the student see in the mirror?

A B C D



**111. M/J/10/12/Q23**

A semi-circular block is made from a plastic. A ray of light passes through it at the angles shown.

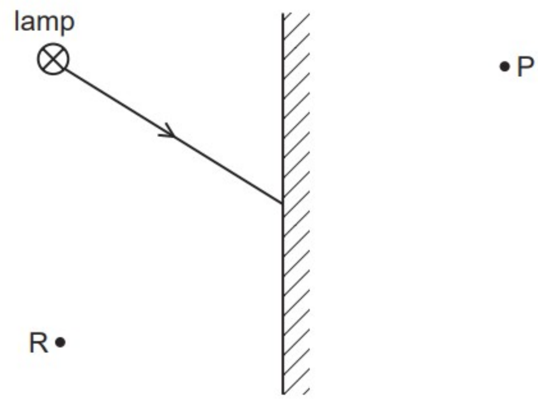



To two decimal places, what is the refractive index of the plastic?

**A** 1.25 **B** 1.41 **C** 1.51 **D** 1.61

**112. O/N/09/Q20**

The diagram shows a ray of light from one point on a lamp striking a plane mirror.




The image of the point on the lamp formed by the mirror is

**A** at P and is real.

**B** at P and is virtual.

**C** at R and is real.

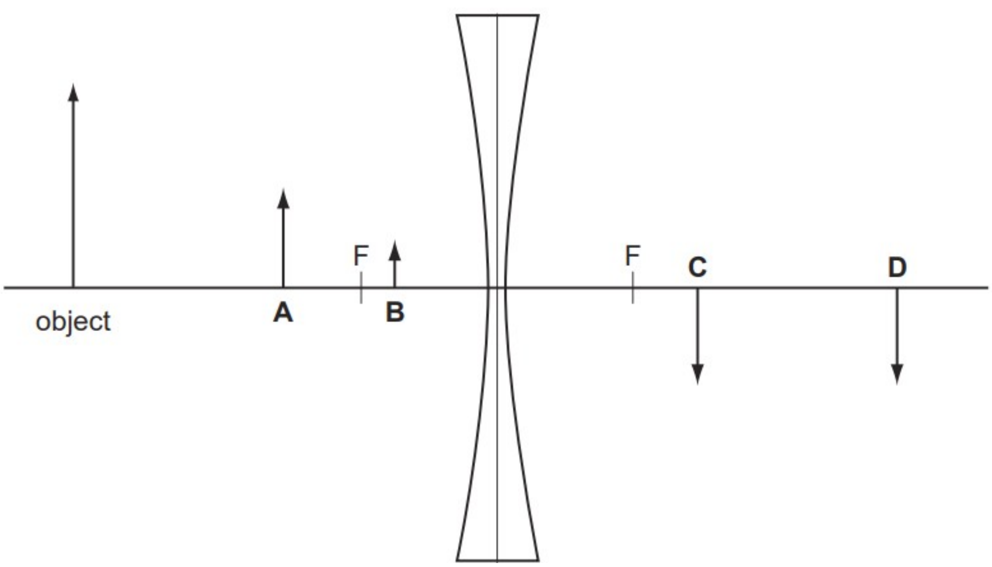
**D** at R and is virtual.

**113. O/N/09/Q22**

An object is placed in front of a diverging lens as shown on the scale diagram.

The principal focus F is marked on each side of the lens.

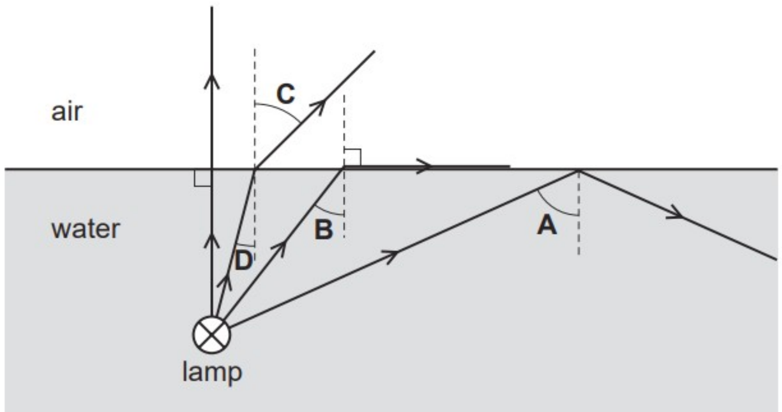
What is the position of the image formed by the lens?



**114. M/J/09/Q21**

The diagram shows four rays of light from a lamp below the surface of some water.

What is the critical angle for light in water?




**115. M/J/09/Q22**

When white light is dispersed by a prism, compared with blue light, the **red** light is

**A** slowed down less and refracted less.

**B** slowed down less and refracted more.

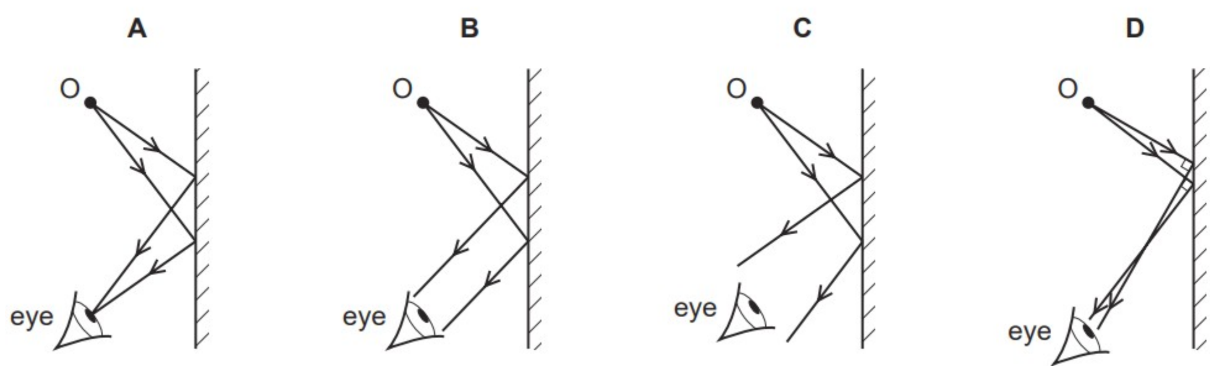
**C** slowed down more and refracted less.

**D** slowed down more and refracted more.

**116. O/N/08/Q23**

An eye views an object O by reflection in a plane mirror.

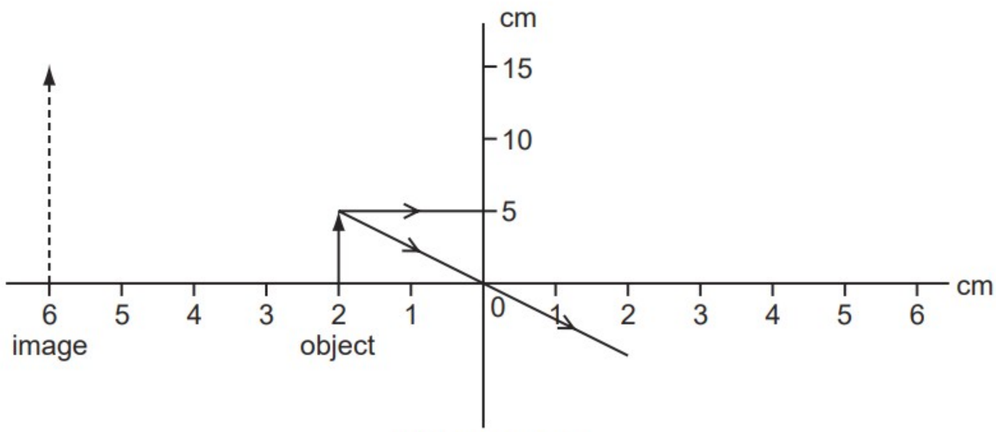
Which is the correct ray diagram?



**117. M/J/08/Q23**

An object 5.0 cm high is placed 2.0 cm from a converging (convex) lens which is being used as a magnifying glass.

The image produced is 6.0 cm from the lens and is 15 cm high.




What is the focal length of the lens?

**A** 2.0 cm **B** 3.0 cm **C** 4.0 cm **D** 6.0 cm

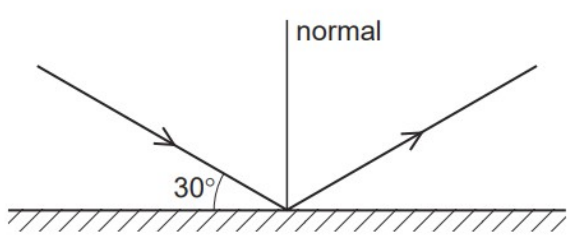
**118. M/J/08/Q25**

Which colour, red or blue, has the higher frequency and which has the longer wavelength?

| | higher frequency | longer wavelength |
|---|---|---|
| **A** | blue | blue |
| **B** | blue | red |
| **C** | red | blue |
| **D** | red | red |

**119. O/N/07/Q20**

The diagram shows a ray of light reflected from a plane mirror.

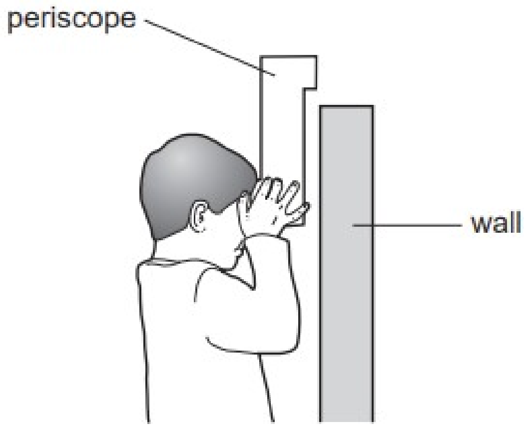



What is the angle of reflection?

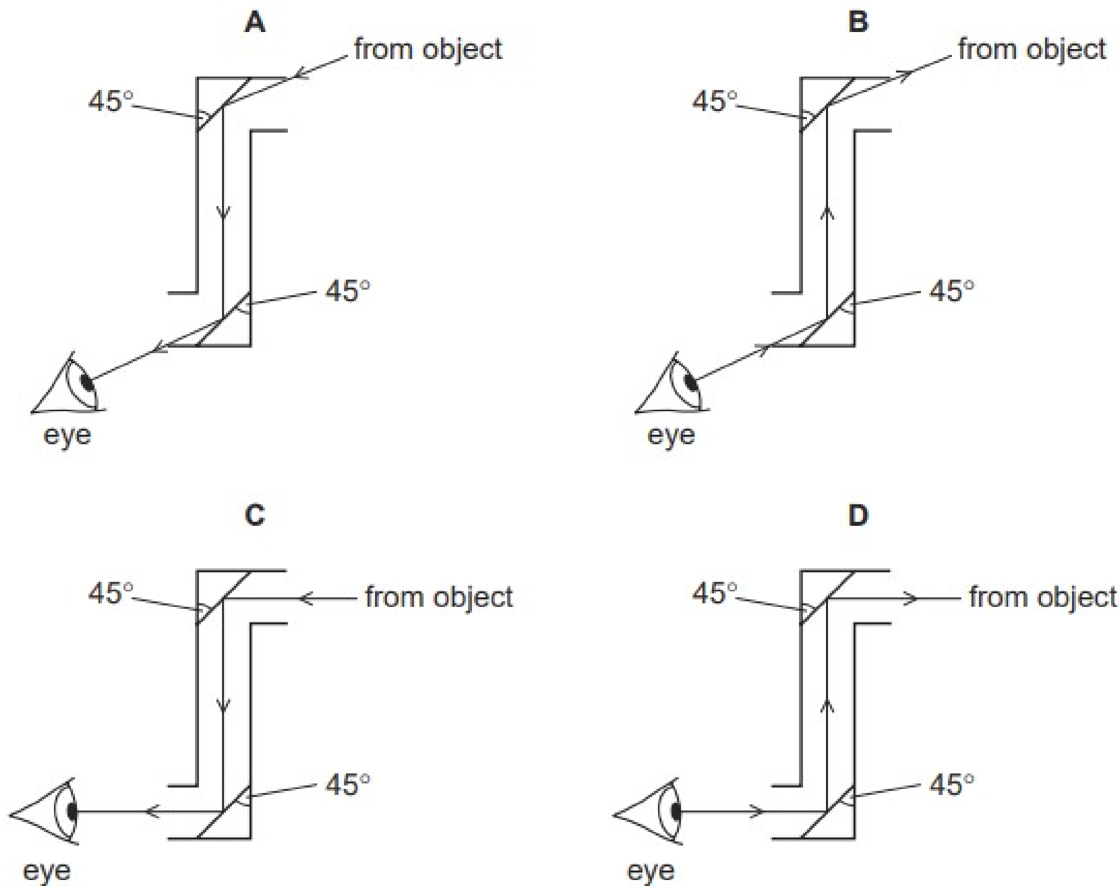
**A** 30° **B** 60° **C** 90° **D** 120°

**120. M/J/07/Q20**

The diagram shows a child using a periscope to look at an object on the other side of a wall.




Which diagram shows a correctly drawn ray of light from the object?



**121. M/J/07/Q21**

What happens to light as it passes from glass into air?

**A** Its frequency decreases because its speed decreases.

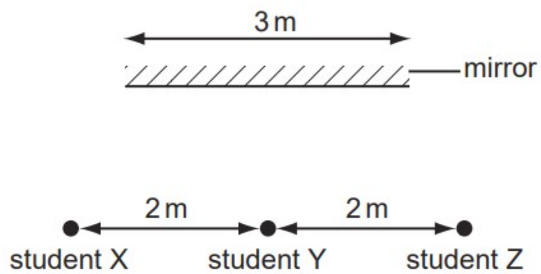
**B** Its frequency increases because its speed increases.

**C** Its wavelength decreases because its speed decreases.

**D** Its wavelength increases because its speed increases.

**122. O/N/06/Q21**

Three students stand 2 m apart in front of a plane mirror which is 3 m long.




Student Y is standing opposite the mid-point of the mirror.

How many students can see the images of the other two?

**A** 0 **B** 1 **C** 2 **D** 3

**123. O/N/06/Q22**

The human eye has a converging lens system that produces an image at the back of the eye.

An eye views a distant object. What type of image is produced?

**A** real, erect, same size

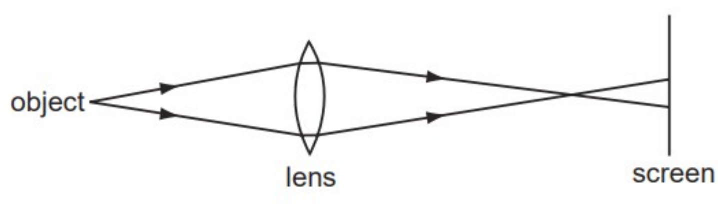
**B** real, inverted, diminished

**C** virtual, erect, diminished

**D** virtual, inverted, magnified

**124. M/J/06/Q23**

A lens forms a blurred image of an object on a screen.




How can the image be focussed on the screen?

**A** by moving the object away from the lens and screen

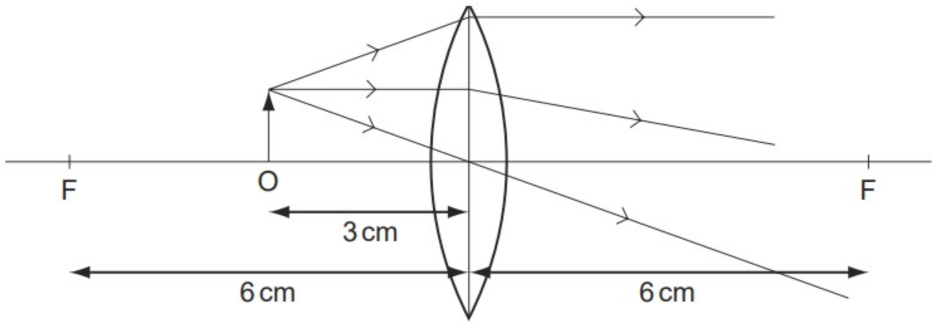
**B** by moving the screen away from the lens and object

**C** by using a brighter object at the same position

**D** by using a lens of longer focal length at the same position

## 125. O/N/05/Q23

The diagram shows an object O placed 3 cm away from a converging lens of focal length 6 cm.




What type of image is produced?

**A** real, erect and diminished

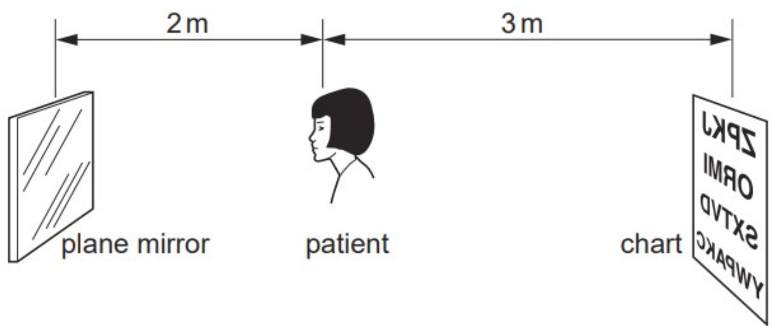
**B** real, inverted and magnified

**C** virtual, erect and magnified

**D** virtual, inverted and diminished

## 126. M/J/05/Q21

The diagram shows a patient having her eyes tested. A chart with letters on it is placed behind her and she sees the chart reflected in a plane mirror.



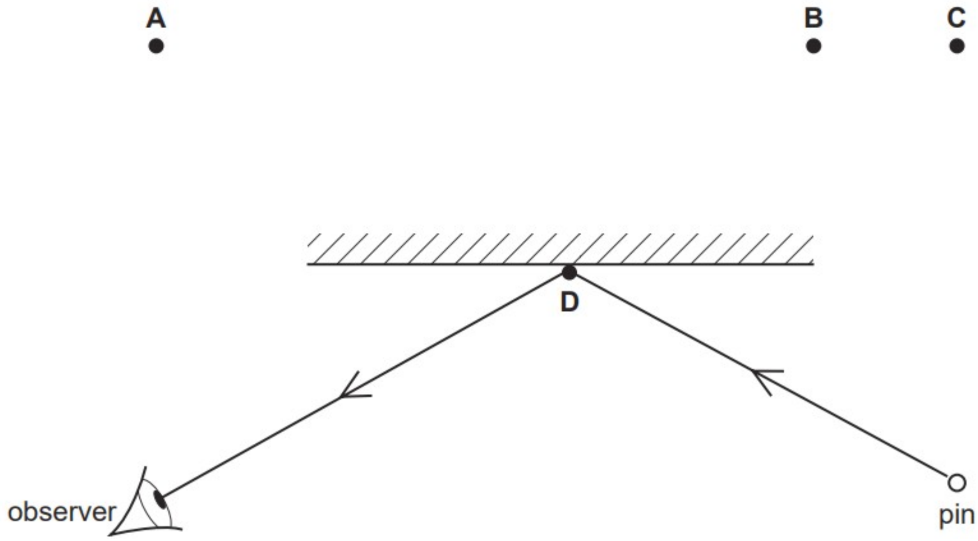

How far away from the patient is the image of the chart?

**A** 2 m  **B** 4 m  **C** 5 m  **D** 7 m

**127. O/N/04/Q24**

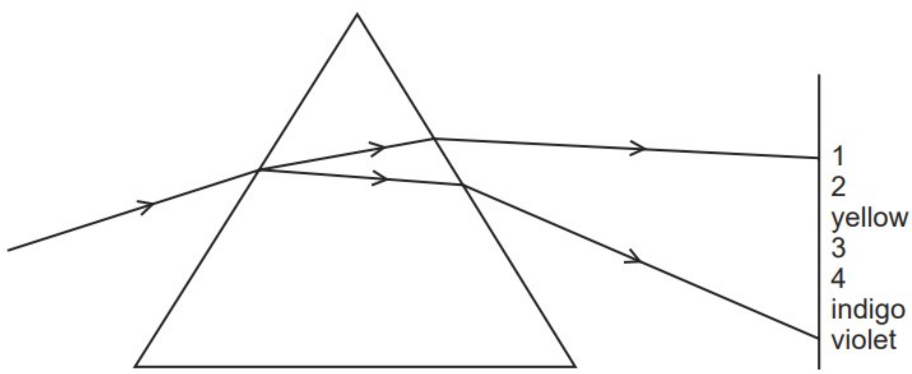
A pin is placed in front of, and to the right of, a plane mirror as shown.

Where is the image of the pin?




**128. O/N/04/Q26**

The diagram shows the spectrum produced when white light is dispersed by a glass prism.

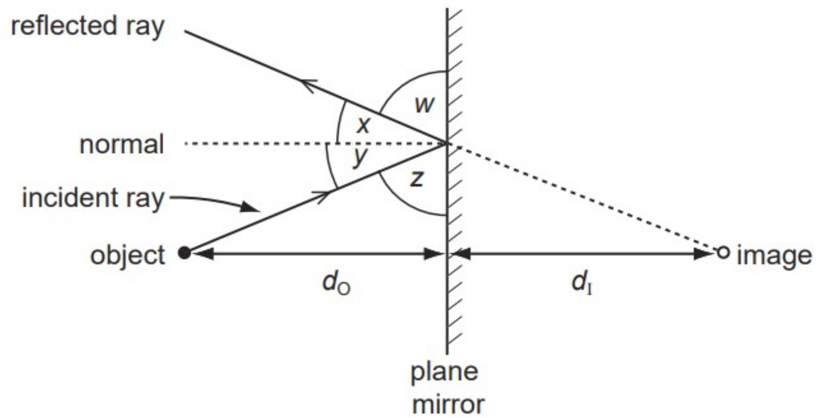



What are the numbered visible colours?

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **A** | infra-red | red | green | ultra-violet |
| **B** | red | green | orange | blue |
| **C** | red | orange | green | blue |
| **D** | red | orange | green | ultra-violet |

**129. M/J/04/Q20**

An image is formed in a plane mirror.

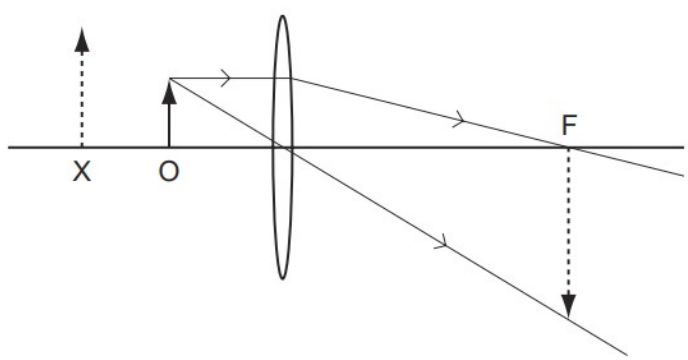



Which statement must be correct?

| | angles | distances |
|---|---|---|
| **A** | $w = y$ | $d_O = d_I$ |
| **B** | $w = z$ | $d_O$ is greater than $d_I$ |
| **C** | $x = y$ | $d_O = d_I$ |
| **D** | $x = z$ | $d_O$ is greater than $d_I$ |

**130. M/J/04/Q22**

A student starts to draw a ray diagram for an object at O, near a thin convex lens, but is not sure whether the image is formed at X or at F.




The correctly drawn image is

**A** real and formed at F.

**B** real and formed at X.

**C** virtual and formed at F.

**D** virtual and formed at X.

**ANSWERS**

1. D
2. D
3. C
4. C
5. A
6. D
7. D
8. B
9. C
10. C
11. B
12. C
13. C
14. C
15. C
16. A
17. B
18. D
19. D
20. B
21. B
22. C
23. D
24. D
25. A
26. D
27. C
28. D
29. D
30. A
31. B
32. B
33. D
34. C
35. B
36. C
37. B
38. C
39. D
40. C
41. A
42. D
43. C
44. D
45. B
46. D
47. A
48. D
49. A
50. B
51. B
52. D
53. C
54. D
55. A
56. A
57. D
58. D
59. B
60. C
61. D
62. C
63. C
64. A
65. A
66. A
67. B
68. A
69. B
70. C
71. C
72. C
73. B
74. B
75. A
76. B
77. B
78. A
79. C
80. A
81. B
82. D
83. D
84. C
85. C
86. C
87. B
88. C
89. A
90. D
91. B
92. C
93. B
94. B
95. A
96. C
97. A
98. C
99. C
100. D
101. B
102. D
103. A
104. D
105. C
106. D
107. D
108. A
109. D
110. B
111. C
112. B
113. B
114. B
115. A
116. C
117. B
118. B
119. B
120. C
121. D
122. D
123. B
124. D
125. C
126. D
127. C
128. C
129. C
130. D

**1. O/N/23/21/Q4**

Red light of frequency $4.7 \times 10^{14}$ Hz travels in air at a speed of $3.0 \times 10^{8}$ m/s.

**(a)** Calculate the wavelength in air of this red light.

wavelength = ..................................................... m [3]

**(b)** Fig. 4.1 shows a narrow beam of this red light striking the surface of a parallel-sided glass block.

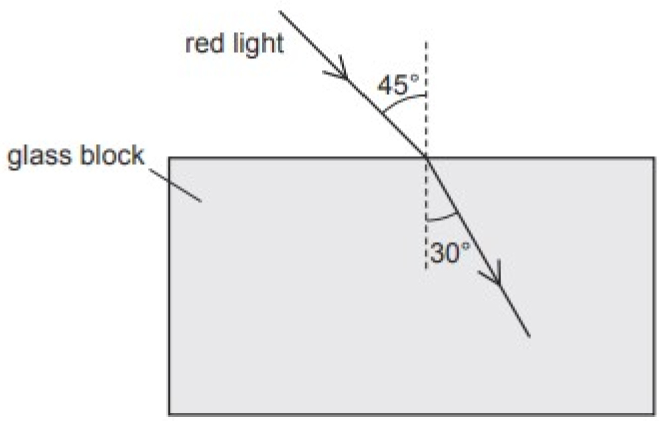



**Fig. 4.1**

**(i)** State what happens to the speed, to the frequency and to the wavelength of the light as it enters the block.

speed ..............................................................................................................

frequency ..............................................................................................................

wavelength ..............................................................................................................
[2]

**(ii)** Using the angles shown on Fig. 4.1, calculate the refractive index of the glass.

refractive index = ........................................................ [3]

**(iii)** The light continues along the path shown in Fig. 4.1 until it strikes the bottom surface of the block. Light then emerges into the air.

Draw on Fig. 4.1 to show the path taken by the light until it strikes the bottom surface and the path of the light that emerges into the air. [2]

[Total: 10]

**2. M/J/23/22/Q7**

**(a)** Ultraviolet radiation is one component of the electromagnetic spectrum.

**(i)** State the name of **two** components of the electromagnetic spectrum that have a smaller wavelength than ultraviolet radiation.

1 ..........................................................................................................................

2 ..........................................................................................................................
[1]

**(ii)** State **one** useful application of ultraviolet radiation.

.......................................................................................................................... [1]

**(iii)** Exposure to ultraviolet radiation from the Sun damages the eyes.

State **one** type of damage to the eye caused by ultraviolet radiation.

.......................................................................................................................... [1]

**(b)** Fig. 7.1 shows a ray of light. The ray passes into a semi-circular block of glass at A and leaves the glass at B, travelling along the surface to C.

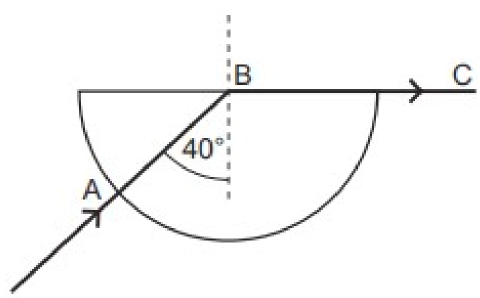



**Fig. 7.1**

**(i)** State the name given to the angle of incidence marked as 40°.

.......................................................................................................................... [1]

**(ii)** Calculate the refractive index of the glass.

refractive index = .......................................................... [2]

[Total: 6]

**3. M/J/23/21/Q6**

The virtual reality headset in Fig. 6.1 contains a display and a lens, as shown in Fig. 6.2.


**Fig. 6.1**

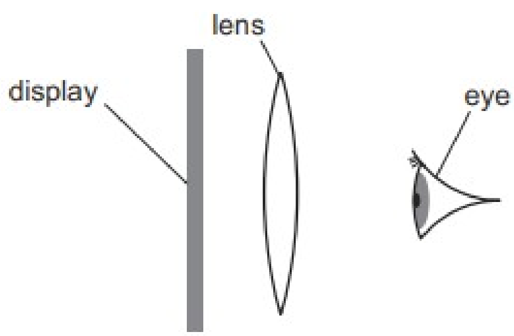



**Fig. 6.2** (not to scale)

The display is the object for the lens.

The lens acts as a magnifying glass and forms a virtual image of the display.

**(a)** **(i)** Describe where the display must be positioned relative to the focal length of the lens for the lens to act as a magnifying glass for the image on the display.

..........................................................................................................................................

.................................................................................................................................... [1]

**(ii)** Explain how a virtual image is formed.

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [1]

**(b)** An arrow on the display is 3.4 cm from the lens. The virtual image of the arrow is 22 cm from the lens.

Fig. 6.3 shows the arrow O, the lens L and the virtual image of the arrow I, drawn on a grid with a scale of 1:2.

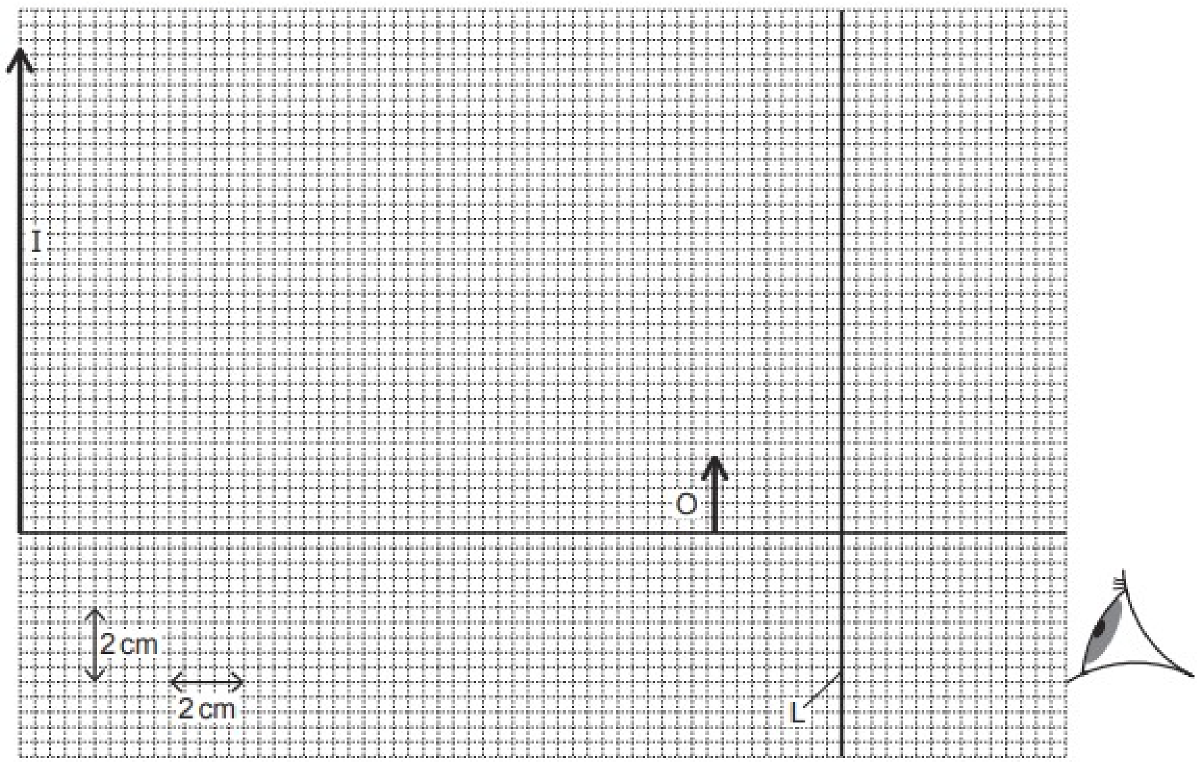



**Fig. 6.3** (scale: 1 cm represents 2 cm)

**(i)** On Fig. 6.3, draw a ray diagram to show the formation of the virtual image I. [3]

**(ii)** Determine the focal length of the lens.

focal length = ...................................................... cm [1]

[Total: 6]

**4. O/N/22/21/Q8**

A parallel beam of light travelling in air is incident on a glass lens.

The beam is perpendicular to the lens as shown in Fig. 8.1.

The dashed line P indicates the position of the lens. The centre of the lens is dot C.

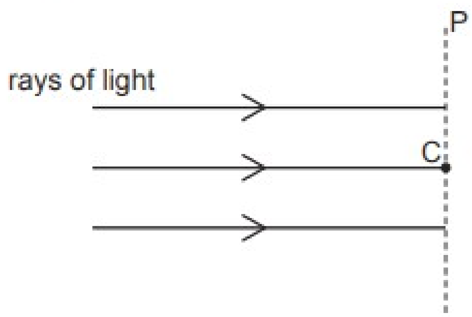



**Fig. 8.1**

**(a)** The lens is a diverging lens.

On Fig. 8.1:

**(i)** indicate the shape of the lens by drawing the outline of the lens around dashed line P [1]

**(ii)** draw the path taken by each ray of light after it passes through the lens. [2]

**(b)** Diverging lenses are used to correct short-sight.

**(i)** Fig. 8.2 is a simplified diagram of a short-sighted eye. Light from a distant object strikes the eye lens and enters the eye.

On Fig. 8.2, continue the three rays in the eye until they reach the back of the eye.

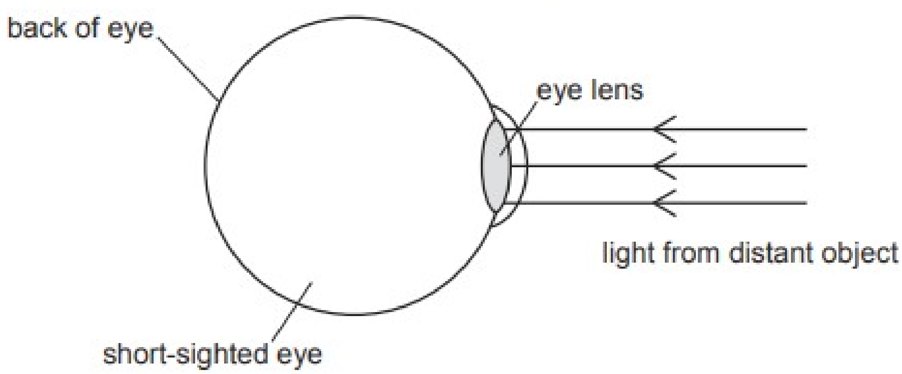



**Fig. 8.2**

[2]

**(ii)** State how the image of a distant object detected by a normal eye differs from the image detected by the short-sighted eye.

...................................................................................................................................

.......................................................................................................................... [1]

**(iii)** Explain how a diverging lens corrects the sight of a short-sighted eye viewing a distant object.

...................................................................................................................................

.......................................................................................................................... [1]

**(c)** The focal length of the diverging lens is 4.0 cm.

An object of height 3.5 cm is placed 6.0 cm from the centre of the lens.

**(i)** Fig. 8.3 is a full-scale diagram drawn on a grid, on which the dashed line L represents the lens and the arrow O the object.

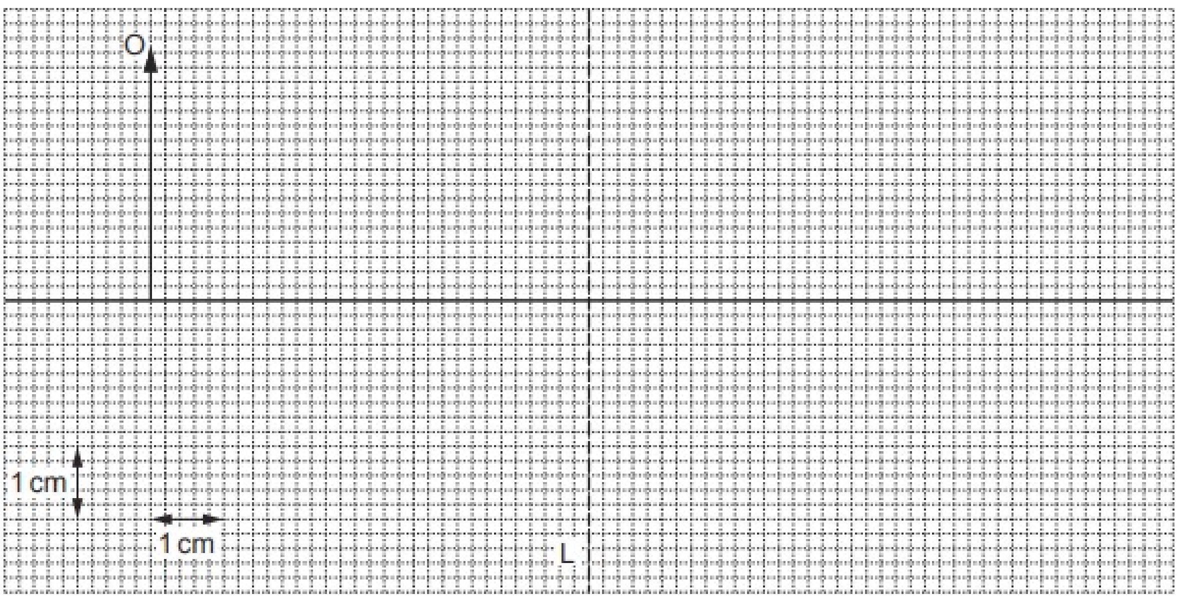



**Fig. 8.3**

By drawing on Fig. 8.3, find the position of the image I of object O. Draw image I and label it I. [4]

**(ii)** Explain whether the image produced is real or virtual.

...................................................................................................................................

.......................................................................................................................... [1]

**(iii)** On the grid in Fig. 8.3, write an E in a position from which an eye can see the image. [1]

**(iv)** Determine the linear magnification produced.

linear magnification = ............................................................ [2]

[Total: 15]

**5. M/J/22/22/Q4**

Fig. 4.1 shows a ray of white light incident on a glass prism.

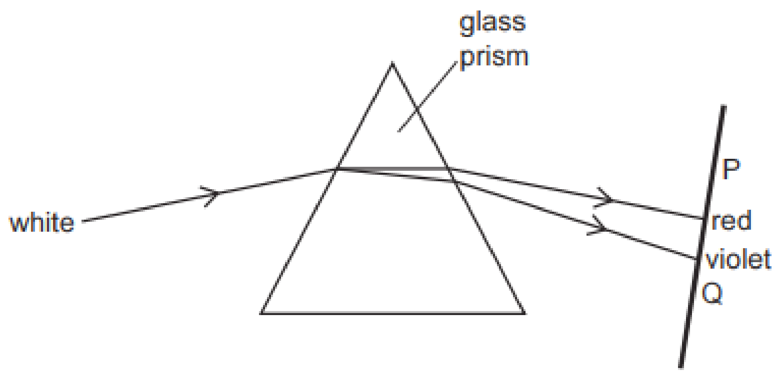



**Fig. 4.1** (not to scale)

Refraction causes the white light to separate into different colours.

**(a)** Define the term 'angle of refraction'.

..................................................................................................................................

.................................................................................................................................. [2]

**(b)** The angle of incidence of the white light as it enters the prism is 40° and the angle of refraction for the red light is 25°.

Calculate the refractive index of the glass for red light. Show your working.

refractive index = .................................................. [2]

**(c)** Using Fig. 4.1, state and explain how the refractive index for red light differs from the refractive index for violet light.

..................................................................................................................................

..................................................................................................................................

..................................................................................................................................

.................................................................................................................................. [2]

**(d)** The source of white light used in Fig. 4.1 produces other types of electromagnetic radiation as well as visible light.

State the name of the invisible radiation found at P and the invisible radiation found at Q.

at P .................................................. at Q .................................................. [1]

**6. M/J/22/21/Q4**

Two parallel rays of light, one red and one blue, enter a glass prism.

Fig. 4.1 shows both rays of light before they enter the prism. The blue ray is also shown incident on a different side of the prism after passing through the prism.

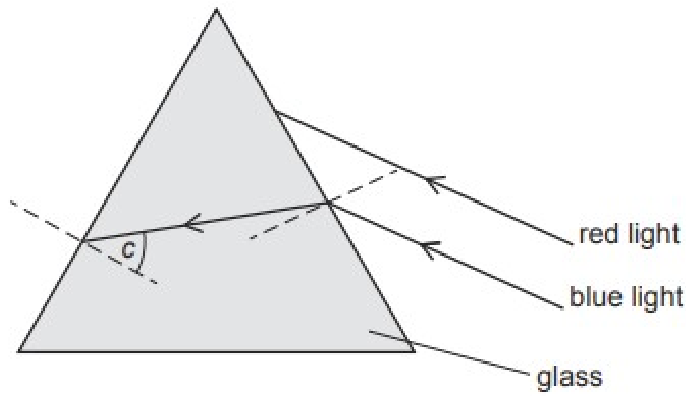



**Fig. 4.1** (not to scale)

The ray of blue light strikes the left side of the prism at an angle equal to its critical angle *c*.

**(a)** **(i)** On Fig. 4.1, mark and label the angle of incidence *i* and the angle of refraction *r* for the blue light as it enters the prism. [1]

**(ii)** On Fig. 4.1, continue the path of the blue light after it strikes the left side of the prism. [1]

**(iii)** The refractive index of glass for red light is smaller than the refractive index for blue light.

On Fig. 4.1, draw the path of the red light as it travels in the prism and after it strikes the left side of the prism. [2]

**(b)** **(i)** State what is meant by the critical angle.

..........................................................................................................................................

.................................................................................................................................... [2]

**(ii)** The refractive index of glass for blue light is 1.5.

Calculate the critical angle *c* for blue light in glass. Show your working.

*c* = ............................................................ [2]

[Total: 8]

**7. O/N/21/22/Q3**

Fig. 3.1 shows light entering a transparent block.

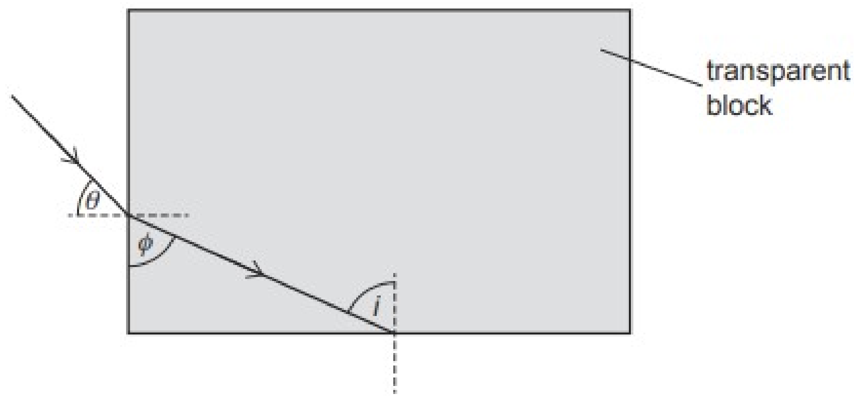



**Fig. 3.1** (not to scale)

The light enters the block at an angle $\theta$ to the normal and travels through the block until it meets the bottom surface.

The angle between the ray in the block and the vertical side of the block is $\phi$.

**(a)** Light travels more slowly in the block than in air.

**(i)** Explain how Fig. 3.1 shows this.

.......................................................................................................................................

.......................................................................................................................................

....................................................................................................................................... [2]

**(ii)** State what happens to the wavelength of the light and what happens to the frequency of the light as it enters the block.

wavelength .........................................................................................................................

frequency ........................................................................................................................... [2]

**(b)** The refractive index of the transparent material is 1.6. Angle $\theta$ is 45°.

**(i)** Determine angle $\phi$.

$\phi$ = ........................................................ [3]

**(ii)** The angle of incidence $i$ at the bottom surface is equal to $\phi$ and the critical angle for the material of the block in air is 39°.

Explain what happens to the light after it meets the bottom surface.

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [2]

**[Total: 9]**

**8. M/J/21/22/Q10(b)**

White light is made up of different colours.

**(i)** State the name of four of the colours in the visible spectrum and place them in order from the smallest wavelength to the largest wavelength.

smallest wavelength ............................................

............................................

............................................

largest wavelength ............................................

[2]

**(ii)** A narrow beam of white light can be split into different colours.

Fig. 10.4 shows rays of white light emitted from a lamp.

Complete Fig. 10.4 to show how a narrow beam is produced from these rays and how a spectrum is shown on the screen. Label your diagram.

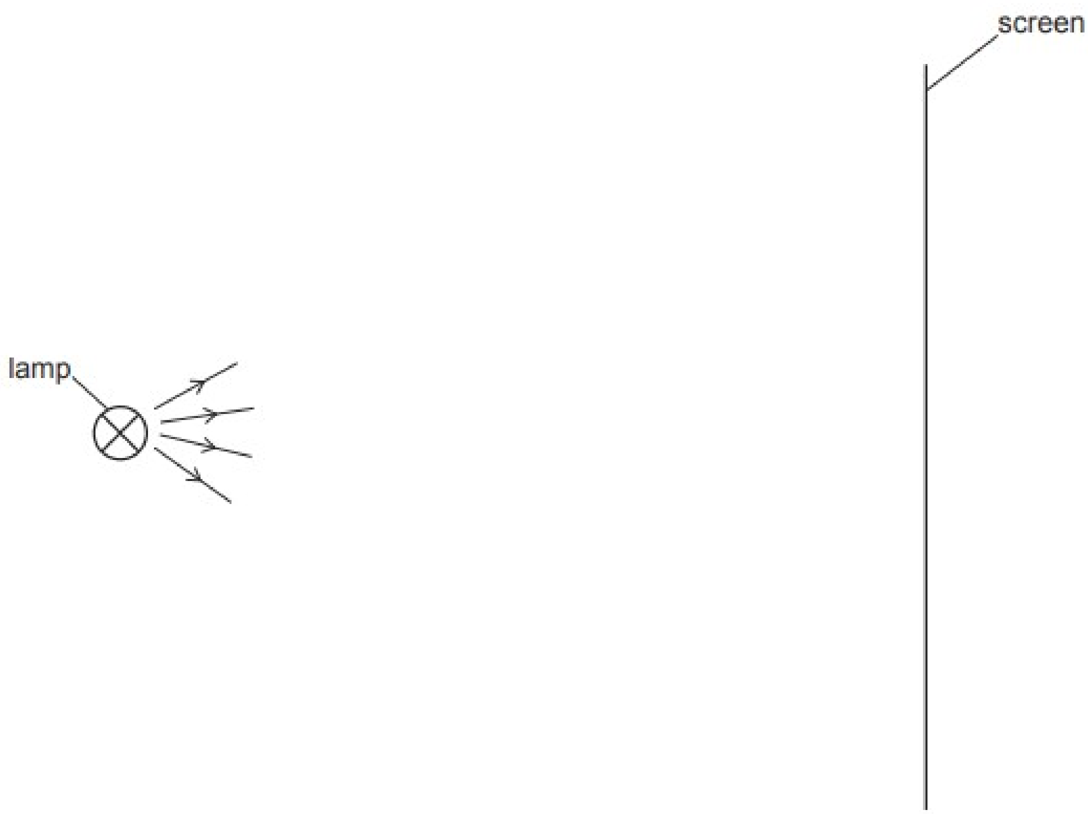



**Fig. 10.4**

[4]

**9. M/J/21/21/Q5**

Fig. 5.1 shows part of the ray diagram of a lens being used as a magnifying glass.

Three rays are shown coming from the top of an object O.

The points labelled F are one focal length from the lens.

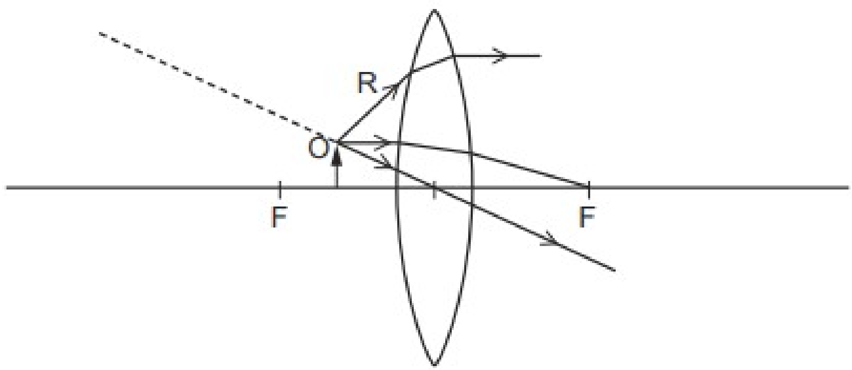



**Fig. 5.1**

**(a)** State the name of the type of lens shown in Fig. 5.1.

.............................................................................................................................. [1]

**(b)** Describe what happens to the ray of light R:

• as it enters the lens ..............................................................................................

..............................................................................................................................

• as it leaves the lens. ..............................................................................................

..............................................................................................................................

[2]

**(c)** Using all three rays from O, complete Fig. 5.1 to show the image formed. [2]

**(d)** Underline **all** of the words in the list that describe the image formed in **(c)**.

inverted upright real virtual

[1]

[Total: 6]

**10. O/N/20/22/Q8**

A thin converging lens is made of a transparent material of refractive index 1.4.

**(a)** A ray of light travelling in air strikes the surface of the lens at an angle of incidence of 55°.

**(i)** Calculate the angle of refraction.

angle of refraction = ........................................................... [2]

**(ii)** Place a tick (✓) in **one** of the boxes in the third column of Table 8.1 to indicate how the light ray deviates and what happens to the speed of the light in the ray as it enters the lens.

**Table 8.1**

| direction of deviation | speed of light | |
|---|---|---|
| away from the normal | decreases | |
| away from the normal | does not change | |
| away from the normal | increases | |
| towards the normal | decreases | |
| towards the normal | does not change | |
| towards the normal | increases | |

[1]

**(iii)** State what happens to the frequency of the light in the ray as it enters the glass.

........................................................................................................................ [1]

**(b)** The focal length of the lens is 2.5 cm.

**(i)** State what is meant by *focal length*.

..............................................................................................................................

........................................................................................................................ [1]

**(ii)** Fig. 8.1 is a full-scale diagram that shows an object O of height 3.0 cm and the lens.

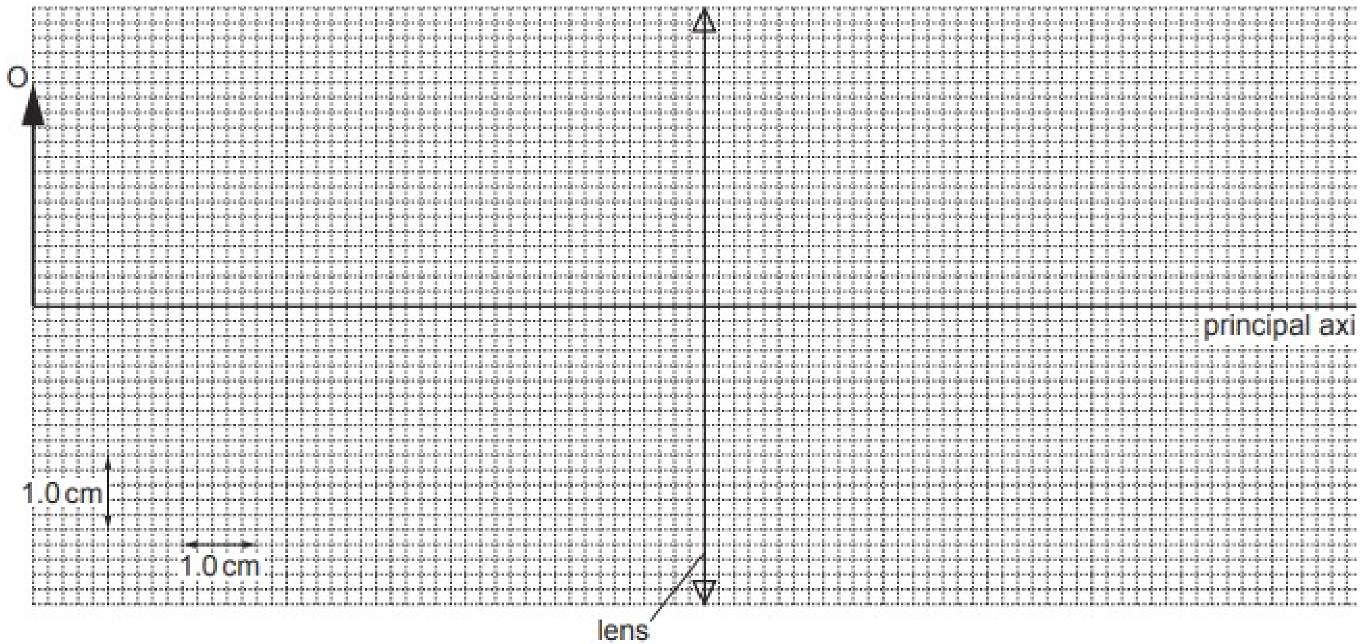



**Fig. 8.1** (full scale)

By drawing on Fig. 8.1, locate and mark the image I of O. [4]

**(iii)** Determine the distance of I from the lens and calculate the magnification of O produced by the lens.

distance = ...........................................................

magnification = ...........................................................
[3]

**(c)** Describe how a converging lens is used in a camera.

...................................................................................................................................

...................................................................................................................................

...................................................................................................................................

...................................................................................................................................

............................................................................................................................ [3]

[Total: 15]

## 11. O/N/20/21/Q4

The three angles of a glass prism are 45°, 45° and 90° as shown in Fig. 4.1.

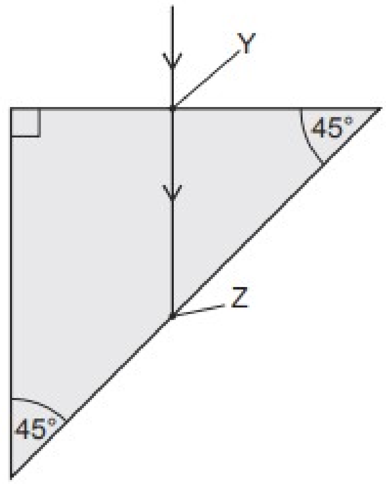



**Fig. 4.1**

At point Y, a ray of light of a single frequency travels in air and strikes the side of the prism at 90°. The ray passes into the glass prism.

**(a)** Light travels more slowly in glass than in air.

**(i)** State what happens to the wavelength of the light in the ray as it enters the glass.

............................................................................................................................ [1]

**(ii)** State what happens to the frequency of the light in the ray as it enters the glass.

............................................................................................................................ [1]

**(b)** The refractive index of glass is 1.6.

**(i)** Calculate the critical angle for light in glass.

critical angle = ........................................................... [2]

**(ii)** On Fig. 4.1, sketch the path of the light after it strikes the side of the prism at Z and after it returns to the air. [2]

[Total: 6]

**12. M/J/20/22/Q4**

**(a)** Two mirrors, A and B, are inclined at an angle of 60° to each other.

Light strikes mirror A at an angle of 30°, as shown in Fig. 4.1.

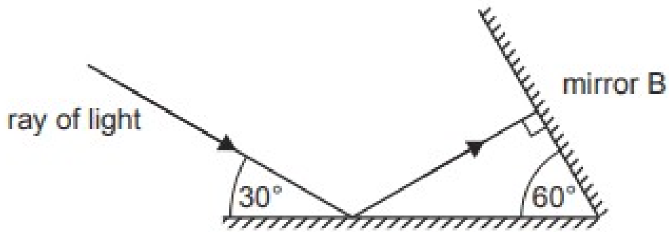



**Fig. 4.1**

**(i)** Determine the angle of incidence at mirror A.

angle of incidence at A = ............................................................ [1]

**(ii)** Determine the angle of incidence at mirror B.

angle of incidence at B = ............................................................ [1]

**(iii)** Describe the path of the reflected ray after it leaves mirror B.

..........................................................................................................................................

.......................................................................................................................................... [1]

**(b)** A plane mirror hanging on a wall is used to form the image of an object.

State **three** characteristics of the image formed.

1. ..........................................................................................................................................

2. ..........................................................................................................................................

3. ..........................................................................................................................................

[3]

[Total: 6]

**13. M/J/20/21/Q6**

A dentist uses a plane mirror to see the back of a tooth.

**(a)** A plane mirror produces an image of an object.

Describe the position of this image.

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [2]

**(b)** Fig. 6.1 shows the plane mirror used by the dentist to see the point labelled X on the tooth.

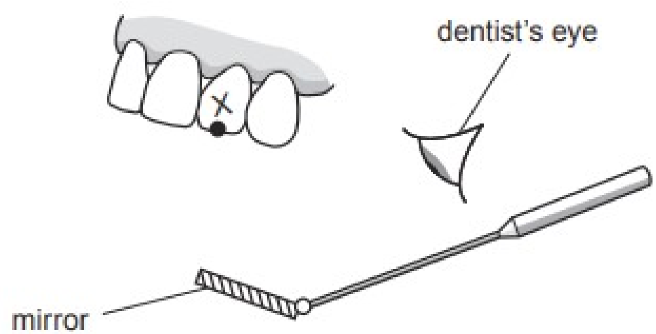



**Fig. 6.1**

On Fig. 6.1:

**(i)** mark the position of the image of X formed by the mirror [1]

**(ii)** draw a ray of light from X to show how the dentist can see the tooth. [2]

**(c)** State **one** characteristic of the image formed by the plane mirror other than its position.

..........................................................................................................................................

.................................................................................................................................... [1]

[Total: 6]

**14. O/N/19/21/Q10**

Visible light is one component of the electromagnetic spectrum.

**(a)** State the speed of light in a vacuum.

........................................................................................................................ [1]

**(b)** State the colours of the visible spectrum.

........................................................................................................................ [1]

**(c)** An object O of height 3.0 cm is placed 4.0 cm from the centre of a diverging lens.

Fig. 10.1 shows the object O, the diverging lens and the two focal points (principal focuses), $F_1$ and $F_2$, of the lens.

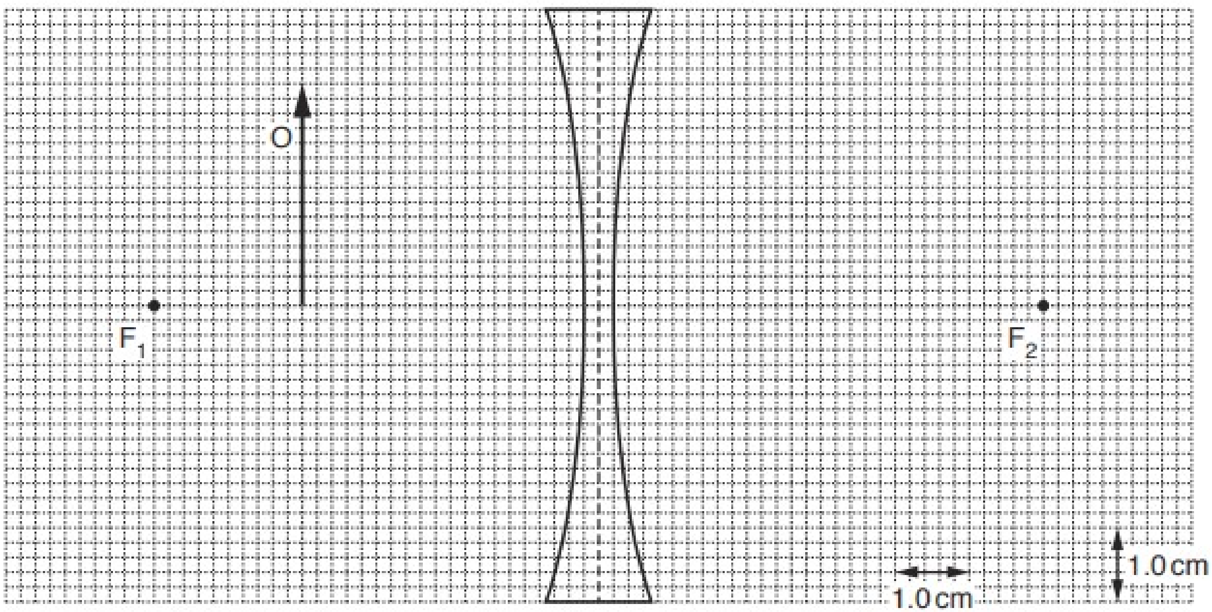



**Fig. 10.1**

**(i)** Determine the focal length of this lens.

focal length = ........................................................ [1]

**(ii)** The diverging lens produces an image of object O.

On Fig. 10.1, by drawing rays from the tip of O, locate this image and label it I. [4]

**(iii)** Determine the height of the image.

height = ........................................................ [1]

**(iv)** Determine the magnification of the image.

magnification = ................................................................ [1]

**(v)** State **two** ways in which a virtual image differs from a real image.

1. ..........................................................................................................................................

..........................................................................................................................................

2. ..........................................................................................................................................

..........................................................................................................................................

[2]

**(vi)** State one use for a diverging lens.

..........................................................................................................................................

.................................................................................................................................... [1]

**(d)** Light passing from air into glass refracts in a similar way to a water wave passing from deep water into shallow water.

Fig. 10.2 represents light passing from air into glass at an angle to the surface.

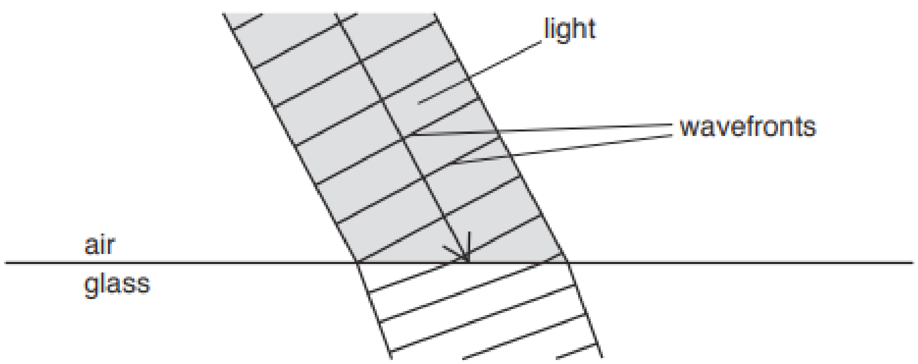



**Fig. 10.2**

One side of a wavefront strikes the glass before the other side.

Explain why the wavefronts change direction as the light enters the glass.

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [3]

**15. M/J/19/22/Q3**

Fig. 3.1 shows three rays of light travelling in water from a light source S.

One ray of light is totally internally reflected at the boundary between water and air.

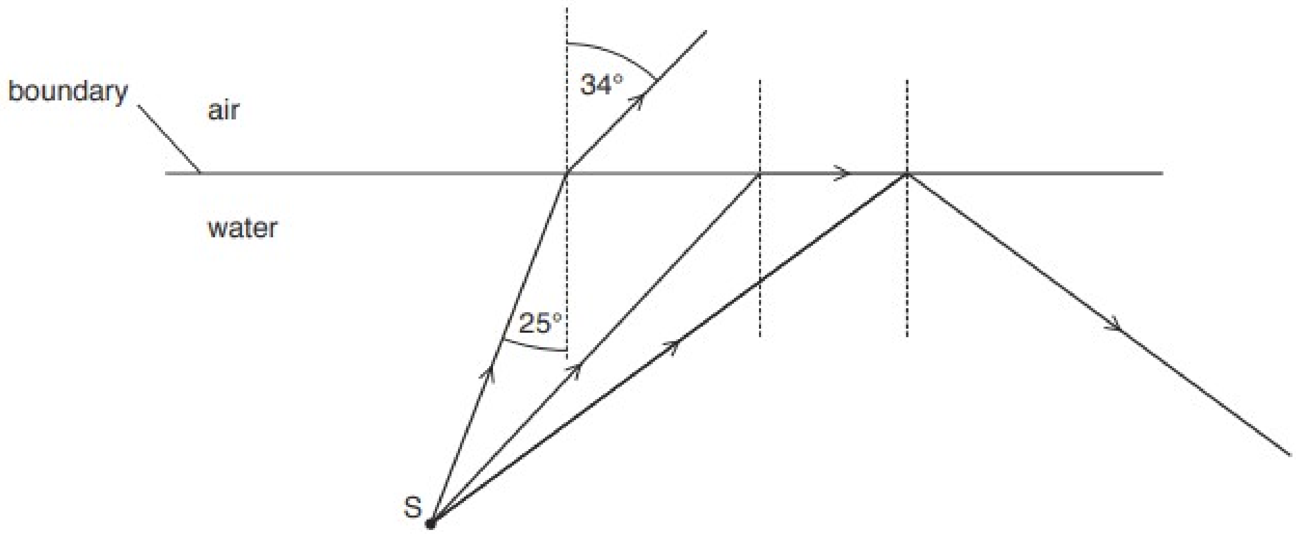



**Fig. 3.1** (not to scale)

**(a)** On Fig. 3.1, mark the critical angle and label it *c*. [1]

**(b)** State why only one of the three rays is totally internally reflected.

..........................................................................................................................................

.................................................................................................................................. [1]

**(c)** Calculate:

**(i)** the refractive index for water

refractive index = ........................................................ [2]

**(ii)** the critical angle for the boundary between water and air.

critical angle = ........................................................ [2]

[Total: 6]

**16. O/N/18/21/Q9(c)**

Fig. 9.2 shows light, in air, striking the vertical side of a rectangular glass block at an angle of incidence of 60°.

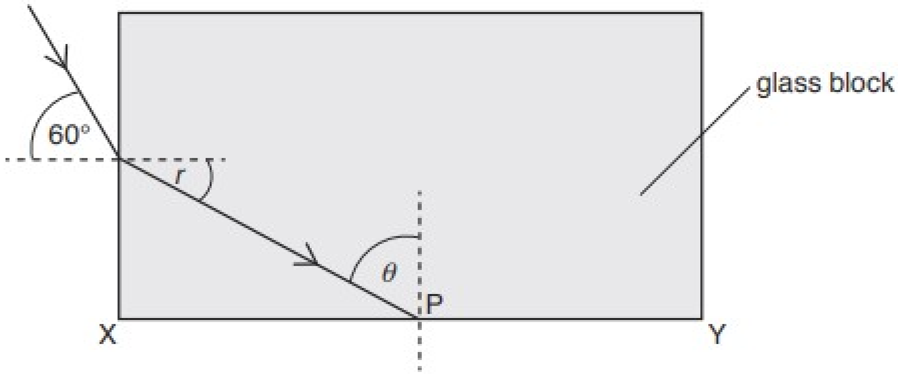



**Fig. 9.2**

The refractive index of the glass is 1.6. The light travels in the glass and strikes side XY at P.

**(i)** Underline all the terms that describe a light wave.

**electromagnetic** **longitudinal** **transverse**

[1]

**(ii)** At the point where the light enters the glass, the angle of refraction is *r*.

Calculate angle *r*.

*r* = ................................................................ [2]

**(iii) 1.** Calculate the critical angle *c* for light travelling in the block.

*c* = ................................................................ [2]

**2.** At P, the angle $\theta$ between the ray and the normal is given by $\theta = 90° - r$.

State and explain what happens to the light when it strikes side XY.

..........................................................................................................................

..........................................................................................................................

.......................................................................................................................... [2]

**3.** On Fig. 9.2, draw the path of the light after it strikes side XY at P and the path of the light when it is again travelling in the air. [2]

## 17. M/J/18/22/Q6

A converging lens has a focal length of 3.0 cm. An object of height 2.0 cm is placed 5.0 cm from the centre of the lens. Fig. 6.1 shows the arrangement of the object and the lens.

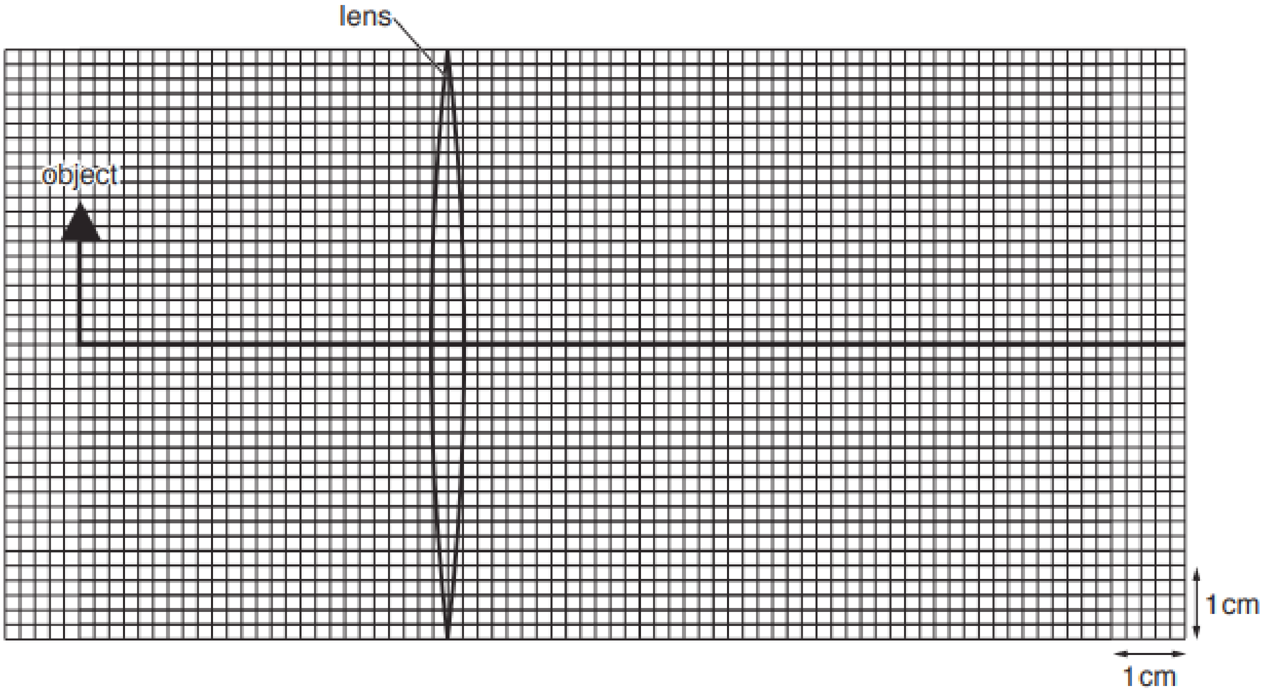



**Fig. 6.1** (to scale)

**(a)** On Fig. 6.1, draw rays from the top of the object to show how the lens forms an image of the object. Mark the image clearly. [2]

**(b)** The image is magnified. State **one** other feature of the image.

........................................................................................................................[1]

**(c)** Calculate the linear magnification produced by the lens in this case.

magnification = ........................................................ [2]

**(d)** State the name of **one** optical device that produces a magnified image as shown by Fig. 6.1.

........................................................................................................................[1]

**18. M/J/18/21/Q9**

Electromagnetic radiation is produced by the Sun and travels from the Sun to the Earth.

**(a)** Some of the statements below are true and some of them are false.

Put a tick (✓) in the box after each statement to show whether it is true or false.

| | true | false |
|---|---|---|
| Gamma rays are used to kill cancerous cells but can also cause cancer. | ☐ | ☐ |
| Infra-red is used in sun-beds. | ☐ | ☐ |
| Radio waves have the highest frequency in the electromagnetic spectrum. | ☐ | ☐ |
| The higher the frequency of the radiation the smaller is the wavelength in air. | ☐ | ☐ |

[2]

**(b)** The glass prism in Fig. 9.1 is used to split white light from the Sun into different colours.

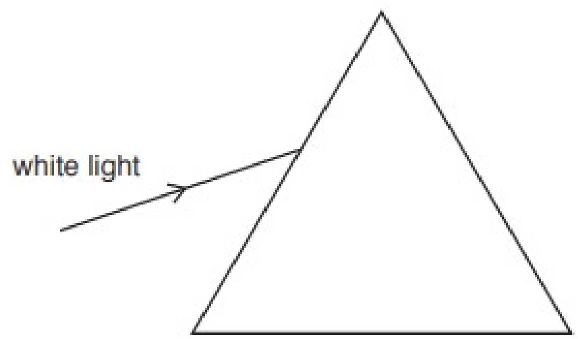



**Fig. 9.1**

**(i)** Draw, on Fig. 9.1, to show the action of the prism on the light as it passes through the prism. [3]

**(ii)** Explain why the prism splits white light into different colours.

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................

.......................................................................................................................... [2]

**(c)** The Earth is $1.5 \times 10^8$ km from the Sun.

**(i)** State the speed of light in a vacuum.

.......................................................................................................................................[1]

**(ii)** Calculate the time taken for light to travel from the Sun to the Earth.

time = ..............................................................[2]

**19. O/N/17/21/Q10(d)**

Fig. 10.2 shows a ray of green light in air striking the side of a glass prism.

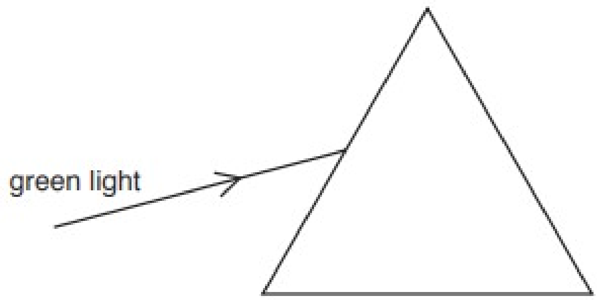



**Fig. 10.2** (not to scale)

The refractive index of the glass for green light is 1.5.

**(i)** On Fig. 10.2,

**1.** draw the normal for this ray,

**2.** mark the angle of incidence with a letter *i*.

[1]

**(ii)** This angle of incidence is 57°.

Calculate the angle of refraction in the glass.

angle of refraction = ........................................................... [2]

**(iii)** State what happens, as the light enters the glass, to the light's

**1.** frequency,

..............................................................................................................................

**2.** speed,

..............................................................................................................................

**3.** wavelength.

..............................................................................................................................

[2]

**(iv)** On Fig. 10.2, draw the path taken by the light as it passes through the glass and into the air. [2]

**20. M/J/17/22/Q9**

Fig. 9.1 shows a man in a room looking into a mirror, as viewed from above.

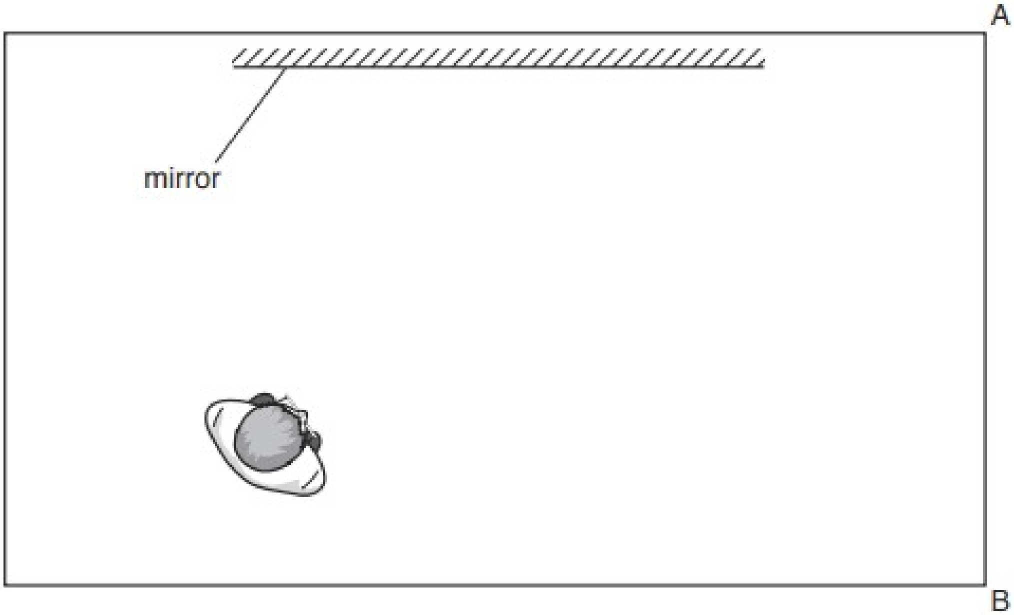



**Fig. 9.1**

**(a)** The man can see an image of part of the wall AB in the mirror.

Point C is the point on the wall closest to A that the man can see by reflection in the mirror. This point is not marked on Fig. 9.1.

**(i)** On Fig. 9.1, find point C and draw a ray of light from C which enables the man to see the image of C. [2]

**(ii)** On the ray that you have drawn on Fig. 9.1, label the angle of incidence *i* and the angle of reflection *r*. [2]

**(iii)** One characteristic of the image formed by the mirror is that it is virtual.

**1.** Explain what is meant by a *virtual* image.

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................[2]

**2.** State one other characteristic of an image formed by a plane mirror.

..........................................................................................................................

..........................................................................................................................[1]

**(b)** Light travels along optical fibres by total internal reflection.

**(i)** State two differences between total internal reflection and reflection in a mirror.

1. ..........................................................................................................................

..........................................................................................................................

2. ..........................................................................................................................

..........................................................................................................................
[2]

**(ii)** The critical angle for light travelling from the material of the fibre to air is 44°.

Calculate the refractive index of the material in the fibre.

refractive index = ............................................................[2]

**(iii)** Light enters the optical fibre from air with an angle of incidence of 50°.

Calculate the angle of refraction.

angle of refraction = ............................................................[2]

**(iv)** Optical fibres are used in telecommunications to send large amounts of data.

Pulses of light travel at almost the same speed along a fibre as electrical pulses travel along a metal wire.

State two advantages of using optical fibres rather than wires for transmitting data.

1. ..........................................................................................................................

..........................................................................................................................

2. ..........................................................................................................................

..........................................................................................................................
[2]

**21. M/J/17/21/Q4**

Fig. 4.1 shows a converging lens, an object O and the image I produced by the lens.

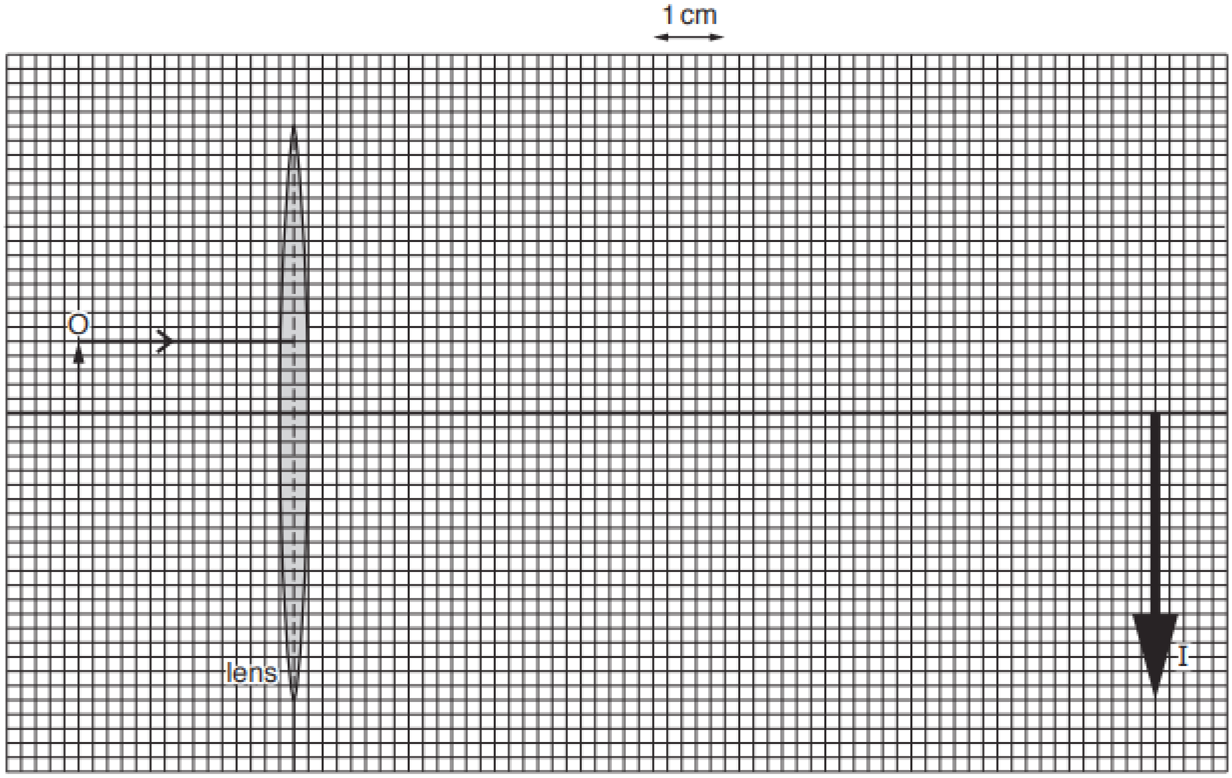



**Fig. 4.1**

**(a)** The image formed is real and has a linear magnification of 4.0.

**(i)** Define the term *linear magnification*.

..........................................................................................................................................

..........................................................................................................................................[1]

**(ii)** Explain what is meant by *real image*.

..........................................................................................................................................

..........................................................................................................................................[1]

**(b)** On Fig. 4.1 a horizontal ray has been drawn from the top of the object to the lens.

**(i)** Continue this ray until it meets the image.

**(ii)** Using Fig. 4.1, determine the focal length of the lens.

focal length = ..............................................................

**(iii)** Draw another two rays from the top of the object to show how the image is formed.

[3]

**22. M/J/17/21/Q10**

**(a)** Fig. 10.1 shows two rays from an object that is placed in front of a plane mirror.

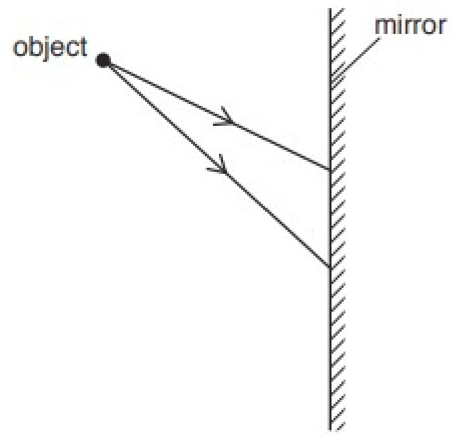



**Fig. 10.1**

**(i)** On Fig. 10.1, draw the two reflected rays and locate the position of the image. [2]

**(ii)** Apart from its position, state one characteristic of the image.

.............................................................................................................................[1]

**(b)** Fig. 10.2 shows wavefronts in a ripple tank. They move in the direction of the arrow.

The wave hits the boundary between two regions and the wave slows down as it enters the shaded region.

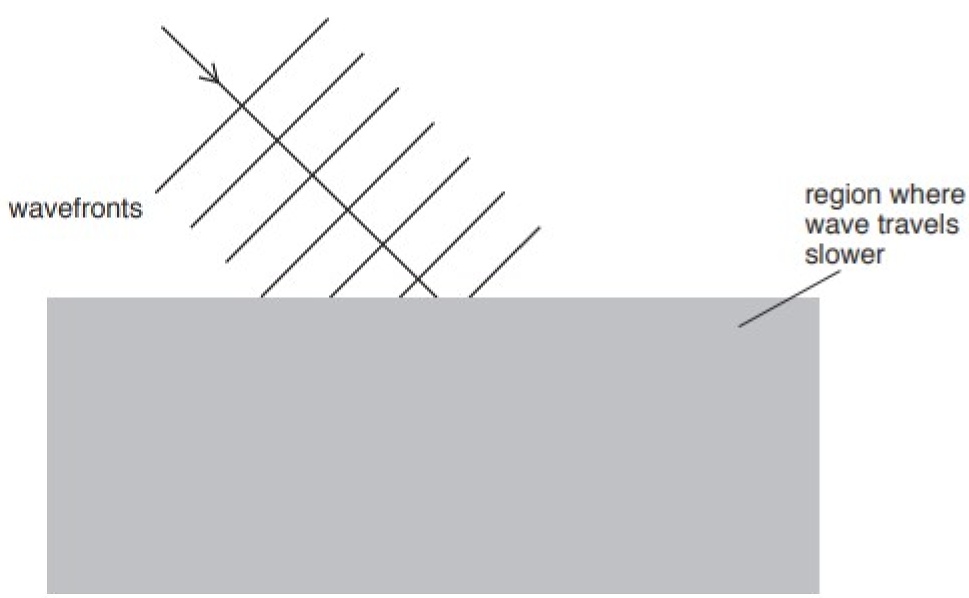



**Fig. 10.2**

**(i)** State what is meant by *wavefront*.

.............................................................................................................................

.............................................................................................................................[2]

(ii) Both reflection and refraction occur at the boundary.

1. On Fig. 10.2, draw the reflected wavefronts. [1]

2. On Fig. 10.2, draw the refracted wavefronts. [2]

(c) A loudspeaker produces a sound of frequency 2.0 kHz. The wavelength of this sound in air is 16 cm.

(i) Calculate the speed of sound in air.

speed = ........................................................[2]

(ii) 1. State the range of frequencies that can be heard by a healthy human ear.

..........................................................................................................[1]

2. Calculate the smallest wavelength of sound that can be heard by a healthy human ear.

wavelength = ........................................................[1]

(iii) Describe a simple experiment to show that sound waves obey the law of reflection. You may draw a diagram if you wish.

..........................................................................................................

..........................................................................................................

..........................................................................................................

..........................................................................................................[3]

**23. O/N/16/22/Q10**

A laser produces a beam of red light.

**(a)** The red light from the laser has a frequency of $4.3 \times 10^{14}$ Hz.

**(i)** State the speed of light in air.

.......................................................................................................................[1]

**(ii)** Calculate the wavelength of this red light in air.

wavelength = ...............................................................[2]

**(b)** Red light from the laser strikes one side of a glass prism at an angle of incidence *i*. The light refracts towards the normal as it enters the prism.

Fig. 10.1 shows the prism, the light and a screen.

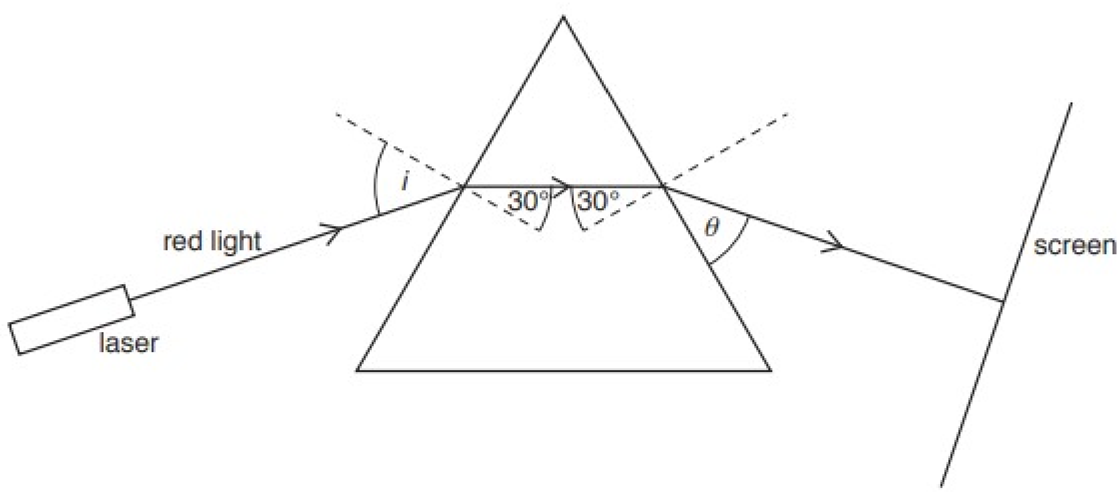



**Fig. 10.1** (not to scale)

**(i)** State what happens to the speed of light as it enters the glass.

.......................................................................................................................[1]

**(ii)** The refractive index of the glass is 1.5. The angle of refraction in the glass, where the light enters the prism, is 30°.

Calculate the angle of incidence *i*.

*i* = ...............................................................[2]

**(iii)** The light then passes back into the air and strikes the front of the white screen, as shown in Fig. 10.1.

Calculate $\theta$, the angle between the ray in air and the side of the prism.

$\theta$ = ........................................................... [1]

**(c)** The laser in **(b)** is replaced with a filament lamp and a slit, as shown in Fig. 10.2.

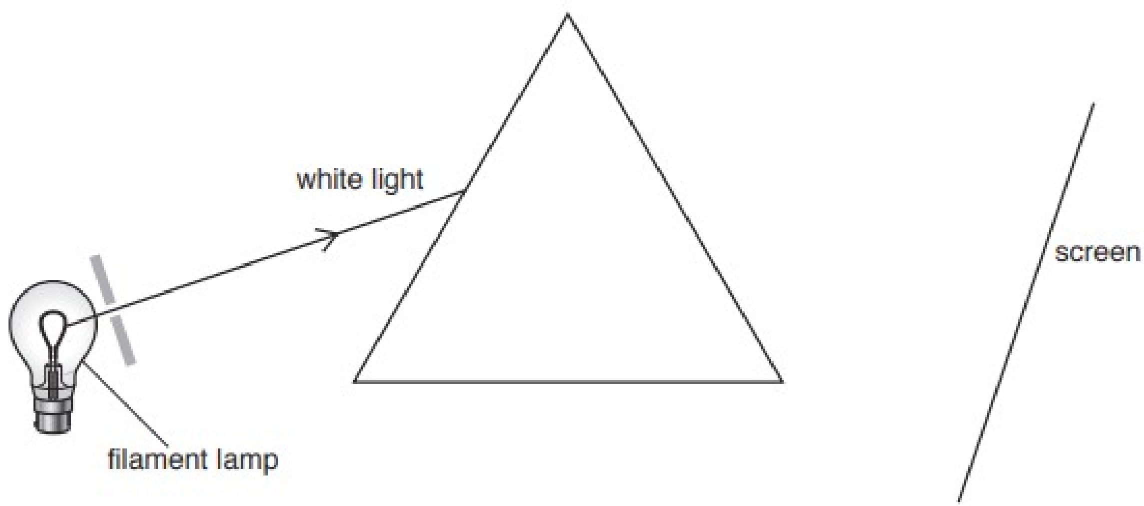



**Fig. 10.2** (not to scale)

A ray of white light now strikes the prism.

**(i)** On Fig. 10.2, draw what happens to the light as it passes through the prism and strikes the screen. [2]

**(ii)** Describe what is seen on the screen.

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................

.......................................................................................................................... [2]

**(iii)** In addition to visible light, the filament lamp also emits some infra-red radiation. This infra-red radiation is able to pass through glass.

**1.** On the screen in Fig. 10.2, mark an X to indicate a place where infra-red radiation strikes the screen. [1]

2. Infra-red radiation is often detected by using a sensitive thermometer with a bulb that has been painted black.

Explain why the blackened bulb makes the thermometer a good detector of infra-red radiation.

..........................................................................................................................................

..........................................................................................................................................[1]

**(d)** Explain the role played by infra-red radiation in intruder alarms.

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..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................[2]

**24. O/N/16/21/Q4**

A lamp is positioned at the bottom of a small pool of water. The critical angle for light passing from water into air is 49°.

**(a)** Explain what is meant by the term *critical angle*.

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

.....................................................................................................................................[2]

**(b)** The lamp sends light towards the surface of the pool.

Fig. 4.1 shows three rays of light that are at 30°, 60° and 90° to the horizontal.

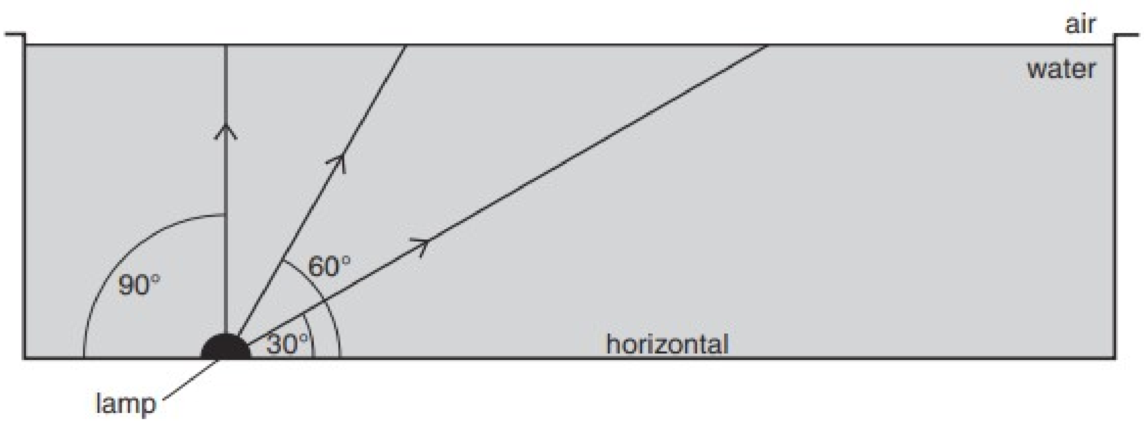


**Fig. 4.1**

On Fig. 4.1, draw the path taken by each of the three rays after they strike the surface of the water. [3]

**25. M/J/16/22/Q5**

A person can focus clearly on objects that are far away, but near objects appear blurred.

Fig. 5.1 shows three rays from a point on a near object as they pass through the person's eye. The refraction of the light as it enters the eye is ignored.

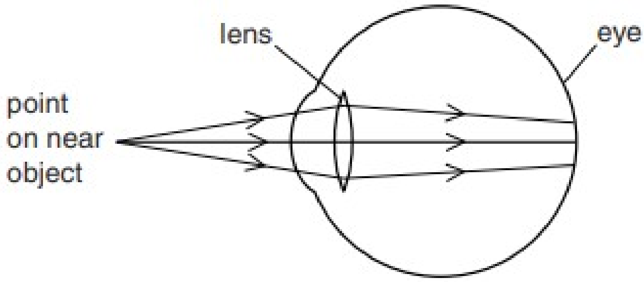


**Fig. 5.1**

**(a)** **(i)** State the name of the eye defect shown in Fig. 5.1.

..........................................................................................................................[1]

**(ii)** Explain what causes the image of the near object to appear blurred.

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................[2]

**(b)** A lens is used to correct the defect.

**(i)** On Fig. 5.2, draw a suitable lens placed in front of the eye and continue the path of the three rays to the back of the eye.

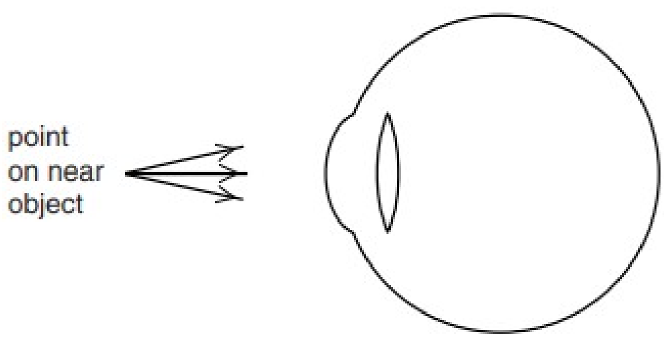


**Fig. 5.2** [2]

**(ii)** State what type of lens is used.

..........................................................................................................................[1]

**26. M/J/16/21/Q6**

**(a)** Explain what is meant by *critical angle*.

..........................................................................................................................................

..........................................................................................................................................

..........................................................................................................................................

.....................................................................................................................................[2]

**(b)** Diamonds are attractive because of their ability to reflect light. The critical angle for diamond in air is 24°.

**(i)** Calculate the refractive index of diamond.

refractive index = ..........................................................[2]

**(ii)** Fig. 6.1 shows two diamonds of different shapes. A ray of light enters each diamond, as shown.

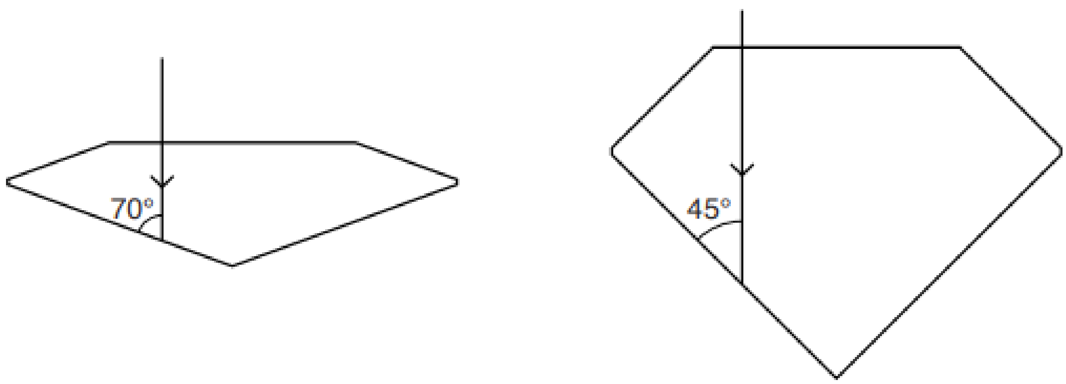


**Fig. 6.1**

On Fig. 6.1 continue the path of each ray of light until it emerges into the air. [3]

**27. O/N/15/22/Q7**

Light enters a parallel-sided glass block at A. The angle between the side of the block and the ray of light in air is 35°.

The light then strikes the edge of the block at B. Fig. 7.1 shows the ray of light and the glass block.

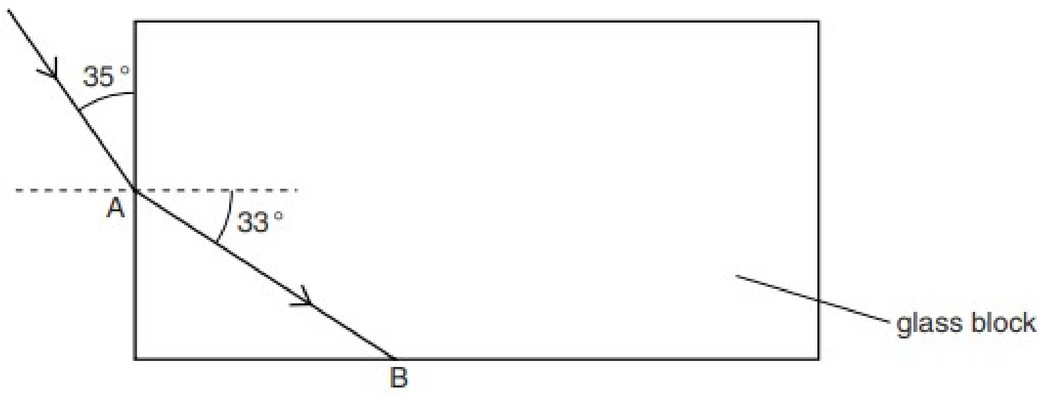


**Fig. 7.1**

The angle of refraction in the glass at A is 33°.

**(a)** Calculate the refractive index for light in the glass.

refractive index = .......................................................... [2]

**(b)** The light undergoes total internal reflection at B.

**(i)** State one condition necessary for total internal reflection to occur.

..........................................................................................................................................

..........................................................................................................................................

.......................................................................................................................... [1]

**(ii)** On Fig. 7.1, continue the ray to show the path of the light after total internal reflection at B and until it leaves the block. [1]

**28. O/N/15/21/Q6**

A burning candle is placed close to a thin converging lens. The candle acts as the object.

A white, vertical screen is moved to a position on the other side of the lens from the candle.

Fig. 6.1 is a full-scale diagram, on graph paper, of the lens and the screen.

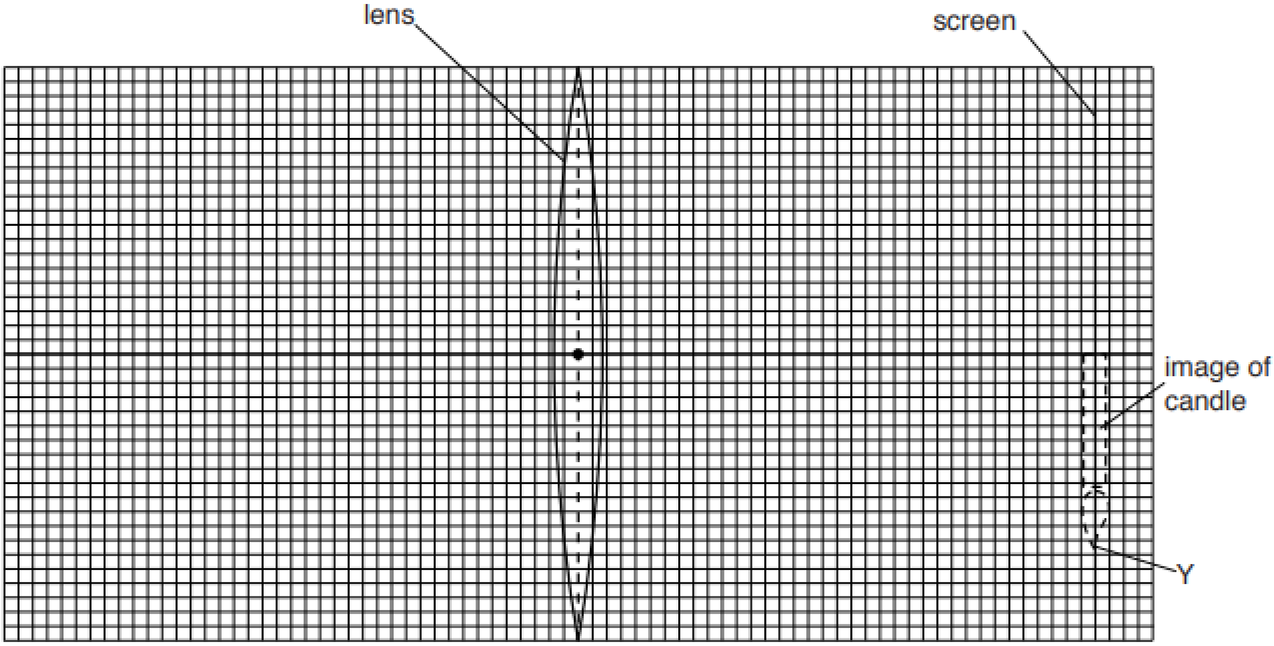


**Fig. 6.1**

The focal length of the lens is 2.4 cm. The screen is 7.2 cm from the centre of the lens. A sharply focused, inverted image of the candle is produced on the screen, as shown in Fig. 6.1.

**(a)** Define the term *focal length*.

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [1]

**(b)** **(i)** On Fig. 6.1, mark and label with an F, each of the two focal points (principal foci) of the lens. [1]

**(ii)** The point Y is the tip of the image.

On Fig. 6.1, draw a ray diagram to locate the position of the top of the object.
Label this point X. [3]

**(iii)** Using Fig. 6.1, determine the distance of the candle from the centre of the lens.

distance = ........................................................ [1]

## 29. M/J/15/21/Q5

Fig. 5.1 shows a ray of light entering and passing along an optical fibre.

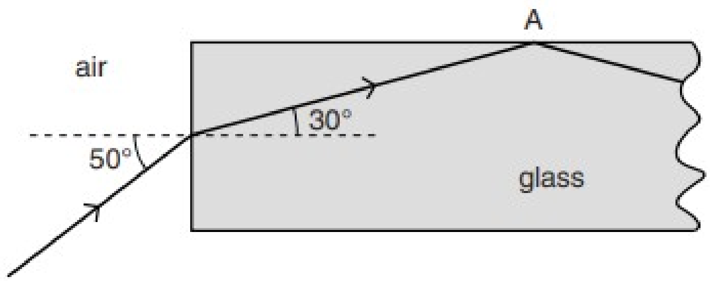


**Fig. 5.1**

**(a)** Calculate the refractive index of the glass in the optical fibre.

refractive index = ............................................................[2]

**(b)** Explain why the ray of light is totally internally reflected at A.

..........................................................................................................................................

..........................................................................................................................................

....................................................................................................................................[2]

**(c)** Both optical fibre and copper wire are used to transmit data.

Optical fibre is cheaper and can carry more data per second than copper wire.

State one other advantage of using optical fibre rather than copper wire to transmit data.

..........................................................................................................................................

....................................................................................................................................[1]

**30. O/N/14/22/Q10**

**(a)** State the speed of light in air.

.......................................................................................................................[1]

**(b)** Fig. 10.1 shows a ray of blue light passing from air into a glass block and refracting at the surface.

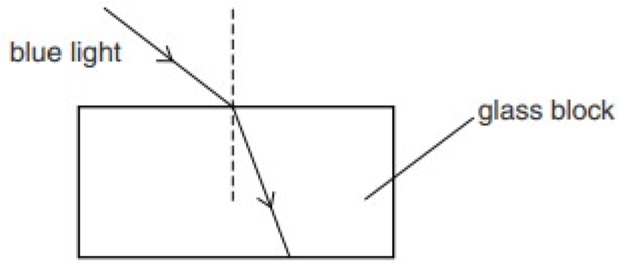


**Fig. 10.1** (not to scale)

**(i)** As the light enters the glass, state what happens to

**1.** the speed of the light,

.......................................................................................................................[1]

**2.** the frequency of the light,

.......................................................................................................................[1]

**3.** the wavelength of the light.

.......................................................................................................................[1]

**(ii)** On Fig. 10.1, mark and label the angle of incidence *i* and the angle of refraction *r*. [2]

**(c)** The refractive index of glass for blue light is 1.5.

**(i)** A ray of blue light strikes the surface of a glass block at an angle of incidence of 89°.

Calculate the angle of refraction of the light in the block.

angle of refraction = ...........................................................[3]

**(ii)** Explain why the angle of refraction of blue light in glass is always less than 45°.

.......................................................................................................................

.......................................................................................................................

.......................................................................................................................

**(d)** Blue light, travelling in air, strikes the side of a different glass block and continues in the same direction as it enters the glass block. Fig. 10.2 shows the ray of light and the shape of the glass block. The critical angle for this glass is 42°.

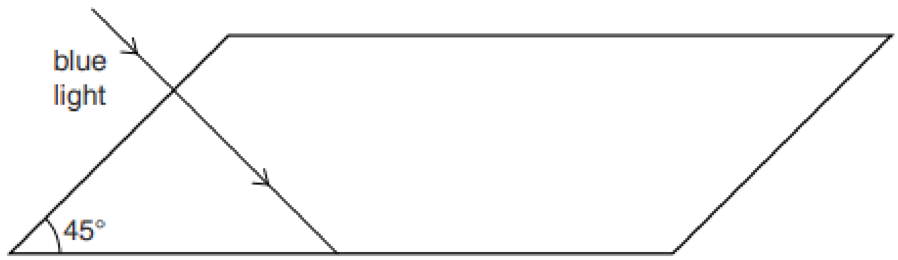


**Fig. 10.2**

**(i)** Explain why the light continues in the same direction as it enters the glass block.

..........................................................................................................................................

..........................................................................................................................................

.....................................................................................................................................[2]

**(ii)** On Fig. 10.2, complete the path of the light until it leaves the glass. [2]

**31. M/J/14/22/Q5**

Visible light, radio waves, X-rays, gamma rays and microwaves are some of the components of the electromagnetic spectrum.

**(a)** State two other components of the electromagnetic spectrum.

1. ..................................................................................................................................

2. ..................................................................................................................................
[1]

**(b)** White light is a mixture of different colours.

Fig. 5.1 shows a ray of white light entering a glass prism.

The white light separates into a number of colours. Only the blue light and the red light are shown.

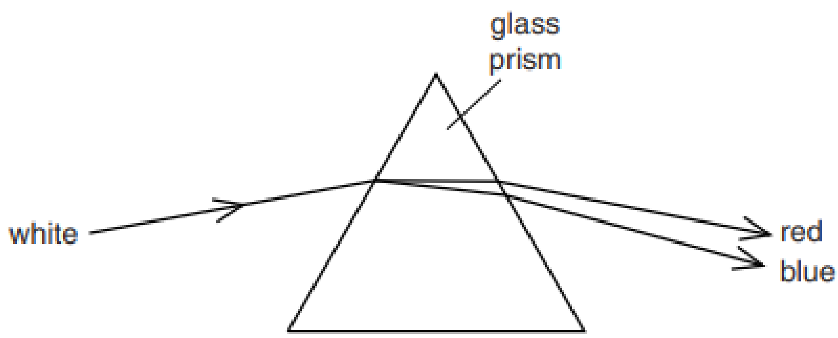


**Fig. 5.1**

Explain why the blue light and the red light separate as shown.

Use the term *refractive index* in your answer.

..................................................................................................................................

..................................................................................................................................

..................................................................................................................................

..................................................................................................................................

.......................................................................................................................... [3]

**32. M/J/14/22/Q6**

**(a)** A beam of parallel light strikes a converging lens of focal length 2.8 cm.

The width of the beam before it reaches the lens is 1.0 cm. The width changes on the other side of the lens.

State a distance from the lens where the width of the beam is

**(i)** less than 1.0 cm,

............................................................................................................................ [1]

**(ii)** more than 1.0 cm.

............................................................................................................................ [1]

**(b)** An object is placed 3.0 cm from a converging lens of focal length of 2.8 cm. Fig. 6.1 is an incomplete, full-scale ray diagram for this arrangement.

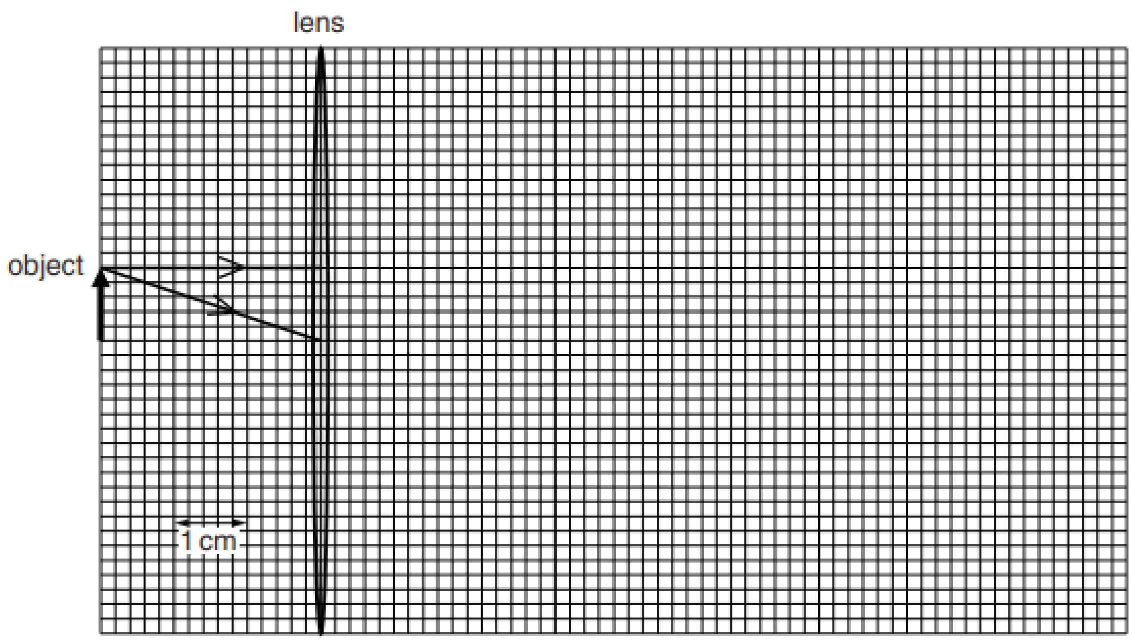


**Fig. 6.1** (full scale)

**(i)** On Fig. 6.1, draw the paths of the two rays after they pass through the lens. [2]

**(ii)** Explain how your ray diagram shows that the image is more than 11 cm from the lens.

..............................................................................................................................

............................................................................................................................ [1]

**(iii)** Underline **three** of the following words which describe the image.

diminished inverted magnified real upright virtual [1]

## 33. M/J/14/21/Q3

A collector views a postage stamp of height 1.5 cm through a lens. The lens is 2.0 cm from the stamp. The image has a linear magnification of 3.0.

The stamp, the image of the stamp and the position of the lens are shown full scale in Fig. 3.1.

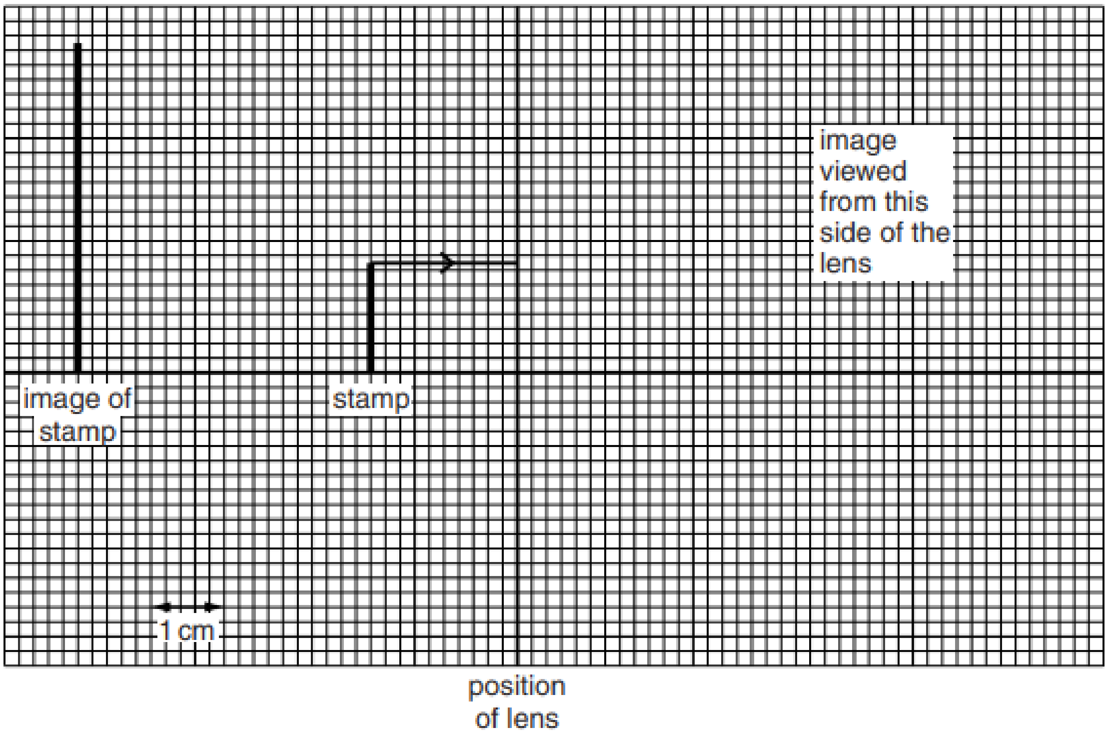


**Fig. 3.1** (full scale)

A ray of light from the top of the stamp to the lens is shown on Fig. 3.1.

**(a)** State the type of lens used.

.......................................................................................................................... [1]

**(b)** State what is meant by *linear magnification*.

..........................................................................................................................

.......................................................................................................................... [1]

**(c)** **(i)** On Fig. 3.1, complete the path of the ray from the top of the stamp after it passes through the lens. [1]

**(ii)** Use your drawing to determine the focal length of the lens.

.......................................................................................................................... [1]

**(iii)** On Fig. 3.1, draw two additional rays from the top of the stamp to show how the image is formed. [1]

**34. O/N/13/21/Q10**

A student goes for a walk in the mountains. During a storm, she sees lightning strike a hillside in the distance. Several seconds later, she hears the thunder caused by the lightning.

**(a)** **(i)** Explain why she hears the thunder several seconds after she sees the lightning.

..........................................................................................................................

.......................................................................................................................... [1]

**(ii)** The student knows the distance to the hillside. She waits for lightning to strike the hillside again. Describe how she can determine a value for the speed of sound in air.

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................

.......................................................................................................................... [2]

**(b)** After the storm, the student sees a rainbow.

**(i)** State the speed of light in air.

speed = ..................................................... [1]

**(ii)** Calculate the wavelength in air of light of frequency $7.5 \times 10^{14}$ Hz.

wavelength = ..................................................... [2]

**(iii)** State the colours of the spectrum in order of increasing wavelength.

..........................................................................................................................

.......................................................................................................................... [2]

**(c)** In the laboratory, the student sends blue light into a glass prism placed on a sheet of paper. The arrangement is shown in Fig. 10.1.

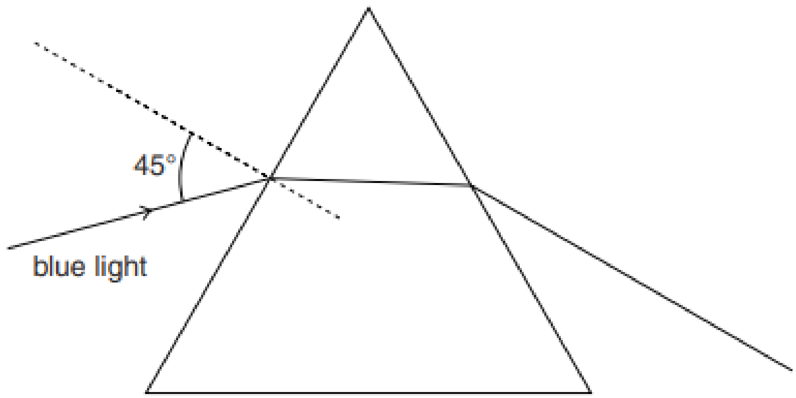


**Fig. 10.1**

The blue light enters the prism with an angle of incidence of 45°.

**(i)** On Fig. 10.1, mark the angle of refraction in the glass and label it *r*. [1]

**(ii)** The student wishes to determine the angle of refraction. On the sheet of paper, she draws the path the ray takes in the glass. Describe how she could do this.

..........................................................................................................................

..........................................................................................................................

.................................................................................................................... [2]

**(iii)** **1.** State the formula that relates the angle of incidence *i*, the angle of refraction *r* and the refractive index *n*.

..........................................................................................................................

.................................................................................................................... [1]

**2.** The angle of refraction in the glass is 28°.

Calculate the refractive index of the glass.

refractive index = .................................................. [1]

**(iv)** The student sends a ray of red light to the glass prism along the same path as the blue light.

On Fig. 10.1, mark the path taken by the red light after it enters the prism. [2]

**35. M/J/13/22/Q10**

A student traces the path of a ray of blue light as it enters and as it leaves a glass prism. Fig. 10.1 shows the trace obtained by the student.

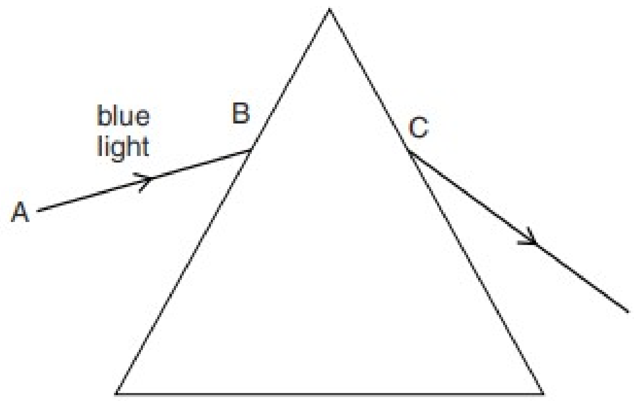


**Fig. 10.1**

**(a)** On Fig. 10.1, draw and label, at the point B, the normal, the angle of incidence *i* and the angle of refraction *r*. [3]

**(b)** State, in terms of the properties of light waves, why the light refracts at B.

..........................................................................................................................................

.................................................................................................................................... [1]

**(c)** The angle of incidence for the ray of blue light at B is 45°. The refractive index of the glass is 1.5. Calculate the angle of refraction at B.

angle of refraction = ................................................... [3]

**(d)** The student performs another experiment with a ray of red light along the line AB.

On Fig. 10.1, show the path taken by this ray of light as it passes through and leaves the prism. [2]

**(e)** The student performs another experiment with a semicircular glass block and a ray of white light. Fig. 10.2 shows the path taken by this ray of light as it enters the glass at P until it hits the straight edge at Q.

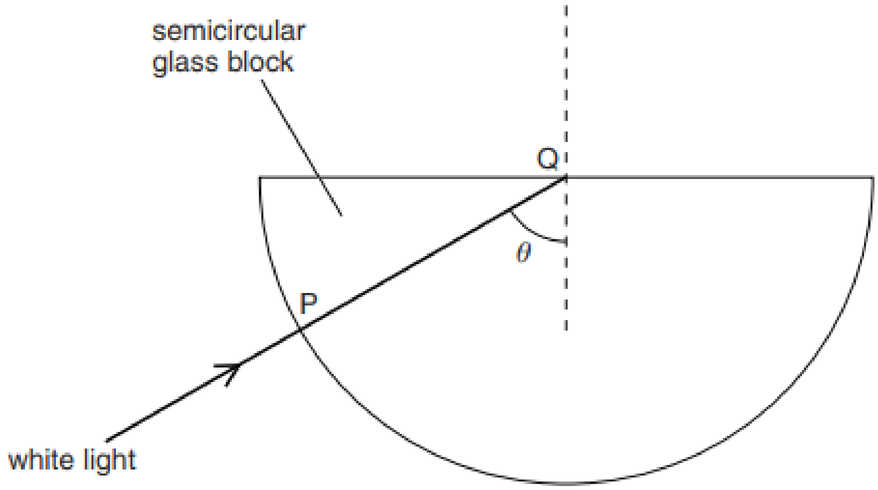


**Fig. 10.2**

The student finds that there is no change in direction as the ray enters the glass at P and that no light passes out of the glass at Q.

**(i)** Explain why the ray does not change direction at P.

..........................................................................................................................

.................................................................................................................. [1]

**(ii)** Explain why no light passes out of the glass at Q.

..........................................................................................................................

..........................................................................................................................

.................................................................................................................. [2]

**(iii)** On Fig. 10.2, draw the complete path followed by this ray. [1]

**(iv)** The student directs the ray of white light into the glass along different paths, so that the angle $\theta$ is slowly reduced.

Describe what happens to the ray at Q.

..........................................................................................................................

..........................................................................................................................

.................................................................................................................. [2]

**36. M/J/13/21/Q5**

Fig. 5.1 shows a ray of light that enters a semicircular glass block at A.
At B, some of the light is reflected and some light leaves the glass and travels along the surface.

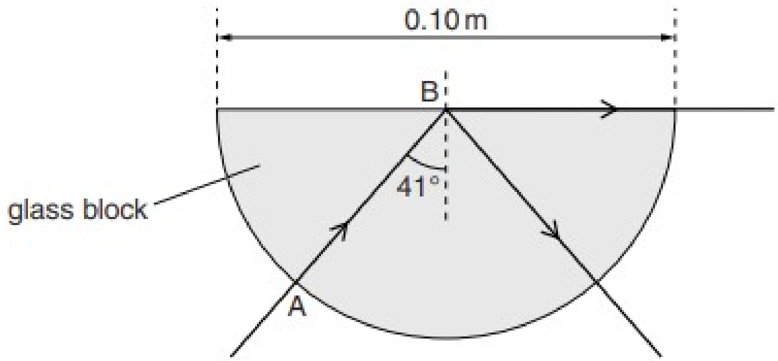


**Fig. 5.1**

**(a)** State the name of the angle of incidence marked 41°.

.............................................................................................................................[1]

**(b)** Rays of light are incident at B with different angles of incidence.

**(i)** On Fig. 5.2a, the angle $i_1$ is less than 41°.
Draw the path taken by the ray of light after B.

**(ii)** On Fig. 5.2b, the angle $i_2$ is greater than 41°.
Draw the path taken by the ray of light after B.

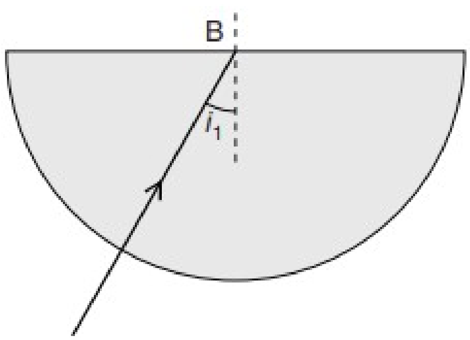


**Fig. 5.2a**

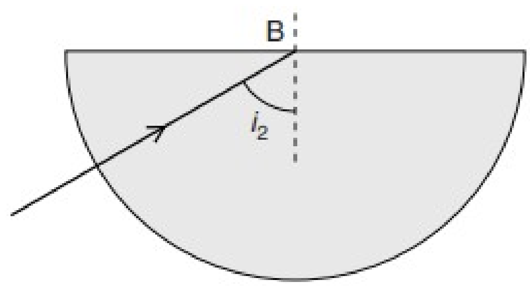


**Fig. 5.2b**

[2]

**(c)** The speed of light in the glass block is $2.0 \times 10^8$ m/s.
The diameter of the glass block is 0.10 m.

Calculate the time taken for the light to travel from A to B.

time = ..................................................... [2]

**37. O/N/12/22/Q10**

A laser produces red light of frequency $4.7 \times 10^{14}$ Hz. The speed of light in glass is $2.0 \times 10^{8}$ m/s.

**(a)** Calculate the wavelength in glass of light from this laser.

wavelength = ...................................................... [2]

**(b)** Describe an experiment to verify *the law of reflection* for light. You may include a diagram in your answer.

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..........................................................................................................................................

.......................................................................................................................................... [5]

**(c)** Fig. 10.1 shows a ray of light travelling in an optical fibre. The ray strikes the side of the fibre at P.

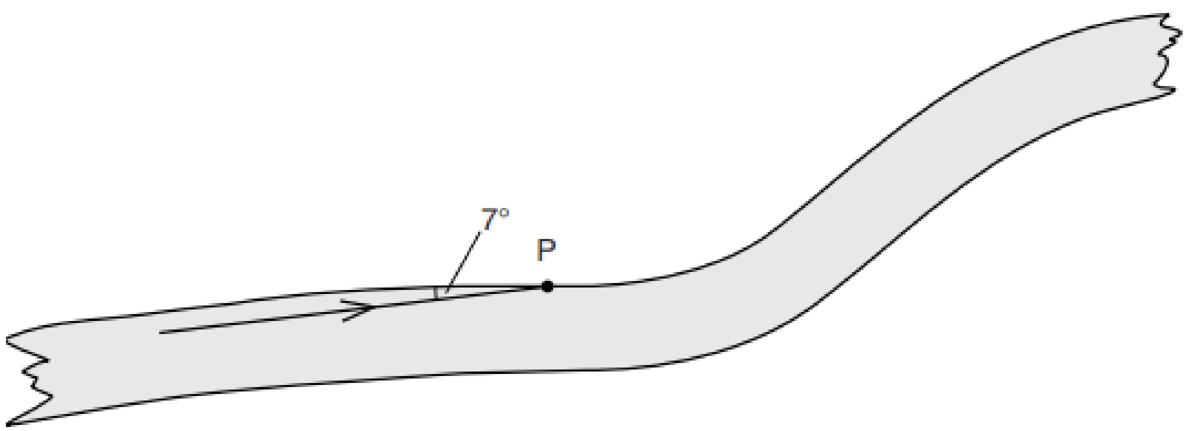


**Fig. 10.1**

The angle between the ray and the side of the fibre is 7°.

(i) Determine the angle of incidence of the ray at P.

angle = ........................................................[1]

(ii) State and explain what happens to the ray at P.

..........................................................................................................................

..........................................................................................................................

......................................................................................................................[2]

(d) A room is illuminated by wall lamps. Fig. 10.2 shows a mirror on the wall behind one of the lamps.

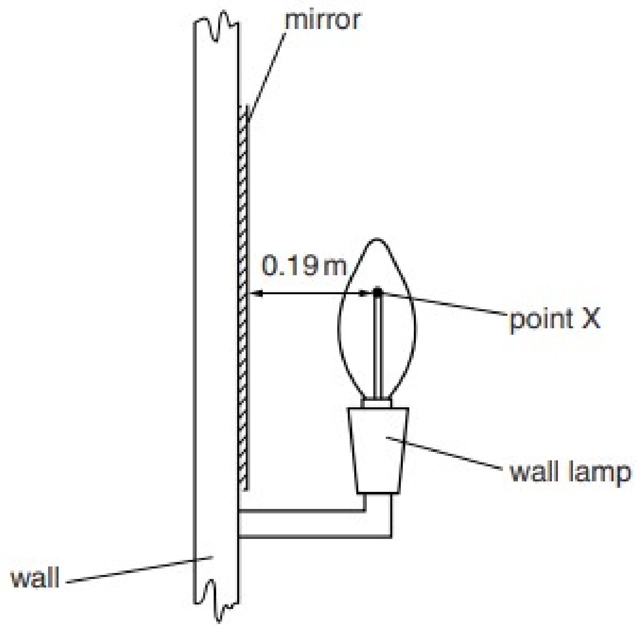


**Fig. 10.2** (not to scale)

X is a point on the filament of the lamp. It is 0.19 m in front of the mirror.

(i) On Fig. 10.2, draw rays from X and locate the image of X. Label the image I. [3]

(ii) State the distance between I and the mirror.

distance = ........................................................[1]

(iii) Suggest one advantage of placing a mirror behind the lamp in the room.

..........................................................................................................................

......................................................................................................................[1]

**38. M/J/12/22/Q5**

Optical fibres are used to transmit telephone signals. Fig. 5.1 shows a ray of light that strikes the inside surface of an optical fibre at P.

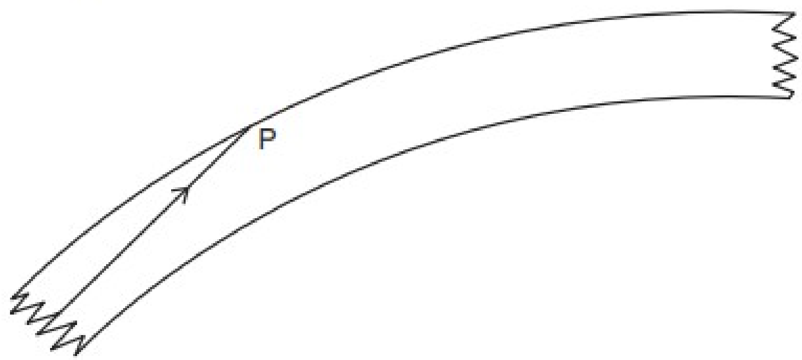


**Fig. 5.1**

**(a)** State one advantage of using optical fibres to transmit telephone signals.

..........................................................................................................................................

.....................................................................................................................................[1]

**(b) (i)** On Fig. 5.1, draw a normal at P and mark the angle of incidence with the letter *i*. [1]

**(ii)** State and explain what happens to the ray at P. Use the term *critical angle* in your answer.

..........................................................................................................................................

..........................................................................................................................................

.....................................................................................................................................[2]

**(c)** The optical fibre is made of glass of refractive index 1.5.
At the start of the optical fibre, the ray enters the glass from air.
The angle of incidence in the air is 60°.

Calculate the angle of refraction in the glass.

angle = ...................................................................[2]

**39. M/J/12/21/Q10**

**(a)** Fig. 10.1 shows the path of a ray of blue light as it passes through a glass prism.

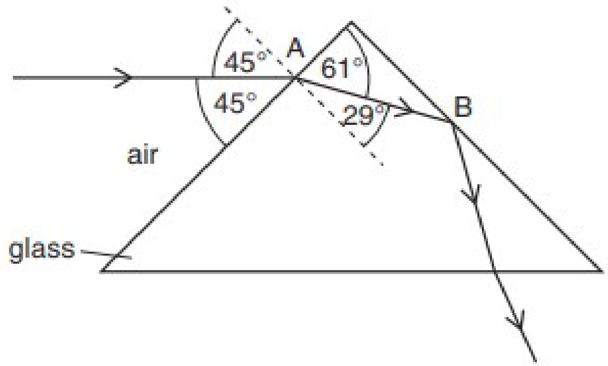


**Fig. 10.1**

**(i)** State the wave term used to describe what happens to the ray of light at A.

..............................................................................................................................[1]

**(ii)** Using angles from Fig. 10.1, calculate the refractive index of the glass.

refractive index = ...........................................................[3]

**(iii)** Explain why the ray does not emerge from the prism at B.

..............................................................................................................................

..............................................................................................................................[2]

**(iv)** Fig. 10.2 shows a second, horizontal, ray of blue light striking the prism at point C.

On Fig. 10.2, continue the path of the second ray through and out of the glass prism. [2]

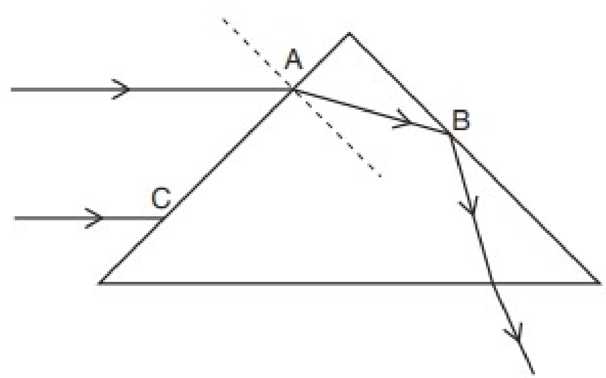


**Fig. 10.2**

**(b)** The camera lens shown in Fig. 10.3 is used to photograph the object O.

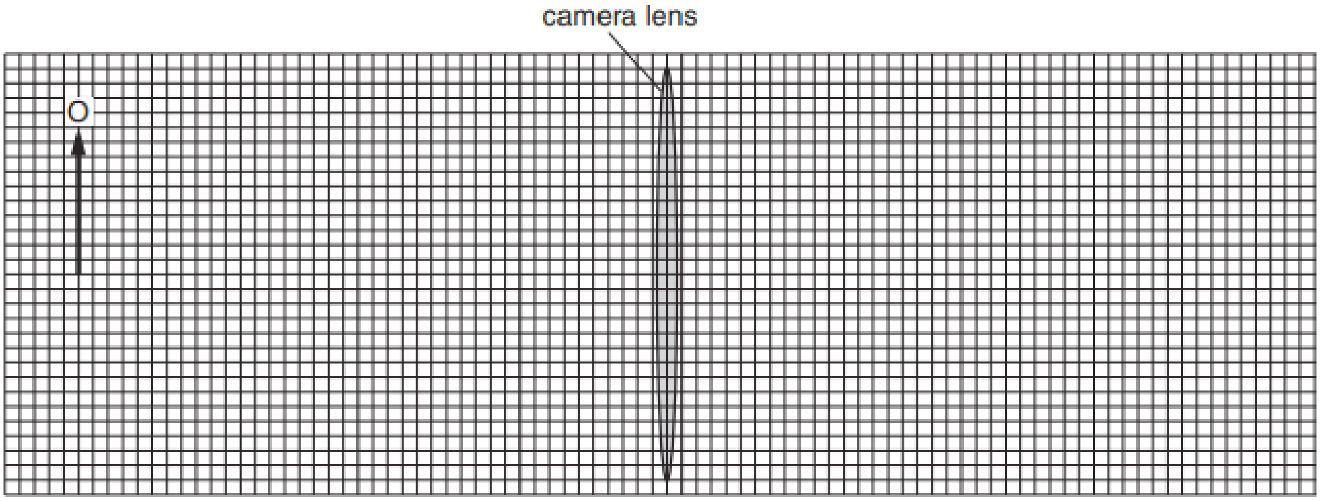


**Fig. 10.3** (full scale)

The object O is 2.0 cm high and is placed 8.0 cm from the centre of the lens. The lens has a focal length of 3.0 cm.

**(i)** Draw rays on Fig. 10.3 to find the position and height of the image formed by the lens. Label the image I. [3]

**(ii)** Determine the height of the image.

.......................................................................................................................[1]

**(iii)** The image formed by the lens is a real image.

**1.** Explain the difference between a real image and a virtual image.

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.......................................................................................................................[1]

**2.** Explain how a converging lens is used to produce and view a virtual image.

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.......................................................................................................................

.......................................................................................................................[2]

**40. O/N/11/22/Q8**

Fig. 8.1 shows a short-sighted eye.

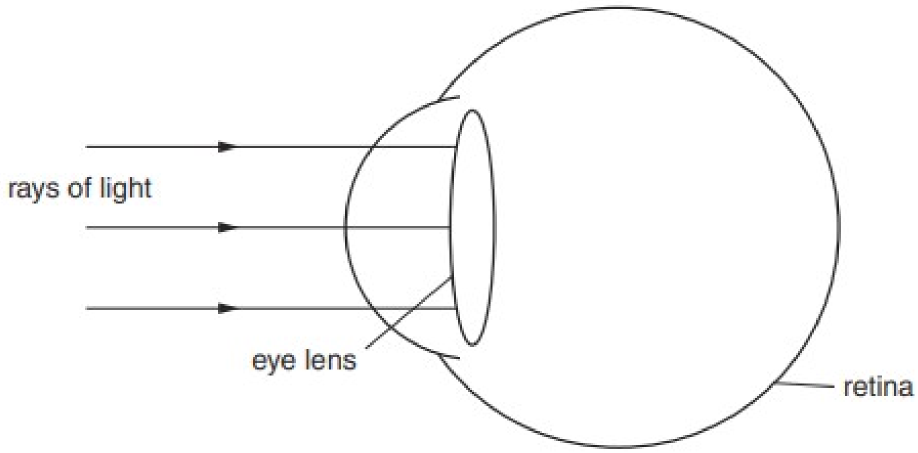


**Fig. 8.1**

Rays of light from a distant star are parallel as they reach the lens of the eye. Refraction of light as it enters the eye has been ignored in Fig. 8.1.

**(a) (i)** On Fig. 8.1, continue the rays to show their paths inside the short-sighted eye until they strike the retina. [2]

**(ii)** Explain how your diagram shows that the image of the star seen by the observer is blurred.

..................................................................................................................................

..................................................................................................................................[1]

**(b)** Fig. 8.2 shows three parallel rays of light.

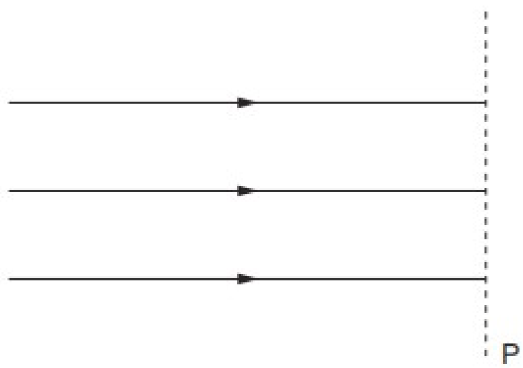


**Fig. 8.2**

**(i)** On the line P in Fig. 8.2, draw the shape of a lens that is used to correct short sight. [1]

**(ii)** On Fig. 8.2, continue the three rays through the lens that you have drawn. [1]

**41. O/N/11/21/Q4**

Fig. 4.1 shows a glass lens in air and its two focal points $F_1$ and $F_2$.

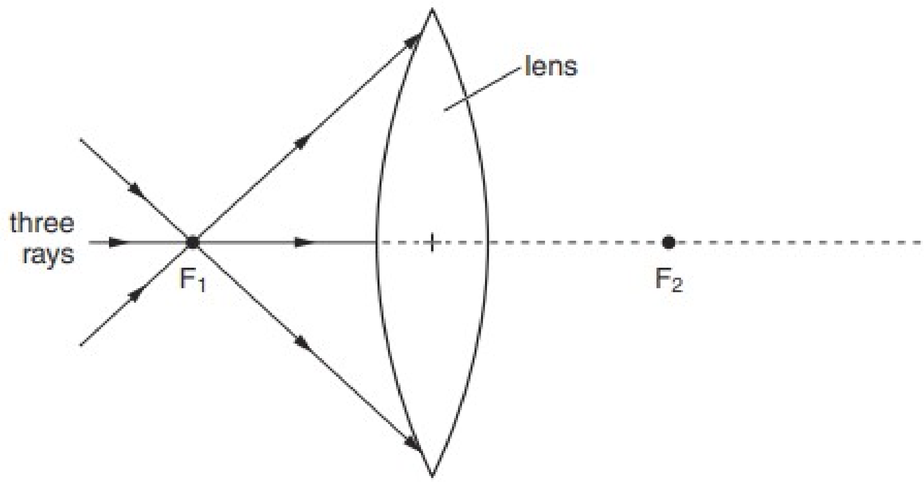


**Fig. 4.1**

Three rays of light pass through $F_1$ to the lens.

**(a)** On Fig. 4.1, continue the three rays through the lens and into the air. [2]

**(b)** State what happens to the speed of light on

**(i)** entering the glass lens from air,

.......................................................................................................................[1]

**(ii)** leaving the lens and returning to the air.

.......................................................................................................................[1]

**(c)** Light of wavelength $6.0 \times 10^{-7}$ m travels in air at a speed of $3.0 \times 10^{8}$ m/s.

**(i)** Calculate the frequency of this light.

frequency = ......................................................[2]

**(ii)** State the effect, if any, on the frequency as the light enters the glass from air.

.......................................................................................................................[1]

**42. M/J/11/22/Q10**

**(a)** Describe an experiment to measure the critical angle for light in glass or perspex.

Your answer should include a labelled diagram.

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......................................................................................................................................[5]

**(b)** Fig. 10.1 represents a simple camera.

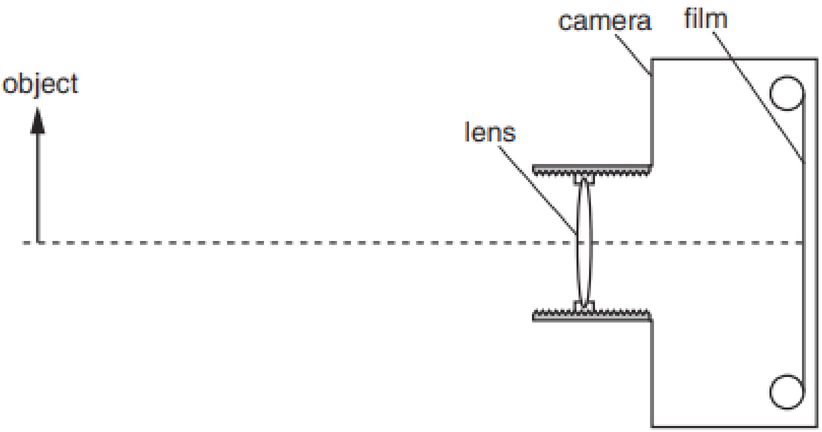


**Fig. 10.1** (to scale)

**(i)** State the type of lens used in this simple camera.

..........................................................................................................................[1]

**(ii)** Draw two rays from the top of the object to show how the image is formed on the film. Mark and label the image on the film. [3]

**(iii)** Define the term *linear magnification*.

..........................................................................................................................

..........................................................................................................................[1]

**(iv)** Fig. 10.1 is drawn to scale. Determine the linear magnification of the object shown in Fig. 10.1.

magnification = ....................................................[1]

**(v)** Apart from its size, state one other property of the image formed by the lens.

..........................................................................................................................[1]

**(vi)** Explain why, when taking photographs of other objects, it may be necessary to move the lens towards the film.

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..........................................................................................................................[3]

**43. O/N/10/22/Q9**

Fig. 9.1 shows a very large plane mirror, inclined at 45° to the horizontal, beneath a pattern on the high ceiling of a hall.

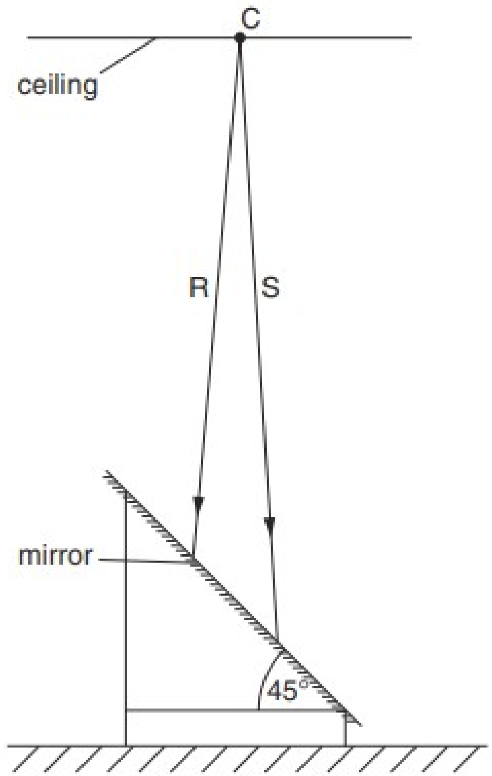


**Fig. 9.1**

The mirror is set on a stand at head-height immediately below the centre C of the pattern. R and S are two rays of light from C that strike the mirror.

**(a)** **(i)** On Fig. 9.1, continue the rays R and S after they strike the mirror. [1]

**(ii)** On Fig. 9.1, show how these rays are used to locate the image of C and mark and label the position of this image with the letter I. [2]

**(iii)** State two characteristics of this image.

1. ..............................................................................................................................

2. ..............................................................................................................................
[2]

**(iv)** Suggest how the mirror helps visitors to the hall to see the pattern on the ceiling.

..............................................................................................................................

..............................................................................................................................[1]

**(b)** Violet light from C has a wavelength of $4.0 \times 10^{-7}$ m.

**(i)** Calculate the frequency of this light, clearly stating the value of any constant used in the calculation.

frequency = .................................................. [3]

## 44. O/N/10/21/Q9

**(a)** Fig. 9.1 shows a young boy lying on his back on the bottom of a swimming pool. He is holding his breath and his eyes are open. A red light is positioned on the ground at Q.

At first the boy's head is touching the pool wall. He notices that, as he slides away from the pool wall, his eye reaches a point P where he first sees the light at Q. Fig. 9.1 shows the boy in this position.

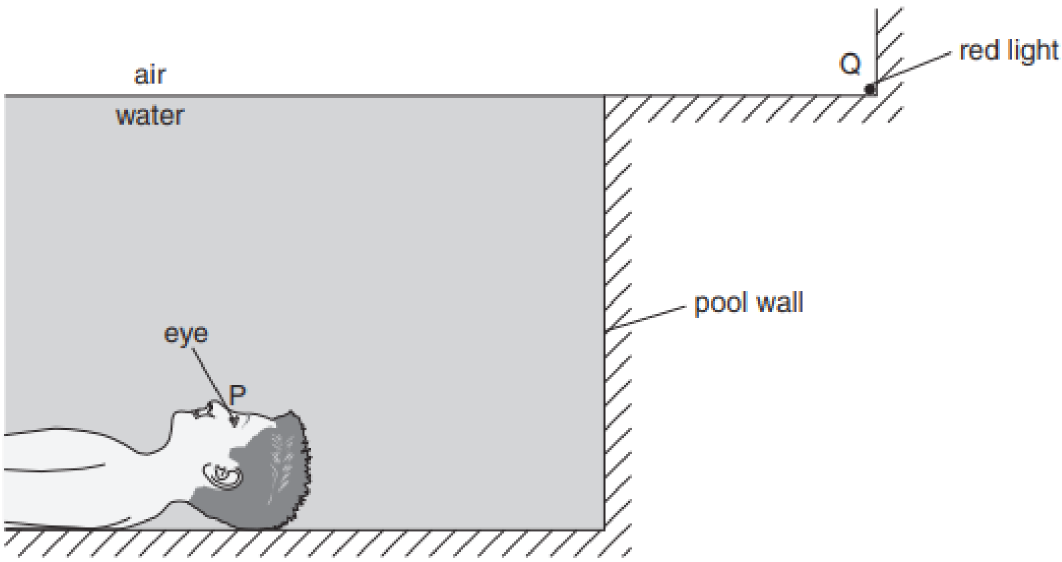


**Fig. 9.1**

**(i)** On Fig. 9.1, draw the ray of light travelling from Q to P. Mark the critical angle for light in water and label it C. [2]

**(ii)** Explain why the boy is unable to see the red light at Q when his eye is closer to the pool wall than P.

..........................................................................................................................

..........................................................................................................................

.................................................................................................................... [2]

**(iii)** The critical angle is 49°. Calculate the refractive index of water.

refractive index = ........................................... [2]

**(iv)** The red light is now replaced with a blue light. State the effect of this on the wavelength of the light in the air.

.................................................................................................................... [1]

**(b)** A small, very brightly illuminated display is located at the back of a projector. The projector lens produces an inverted and magnified image of the display on a white classroom wall.

Fig. 9.2 is a scale diagram showing the position and size of both the display and the image on the wall. R is a point on the display.

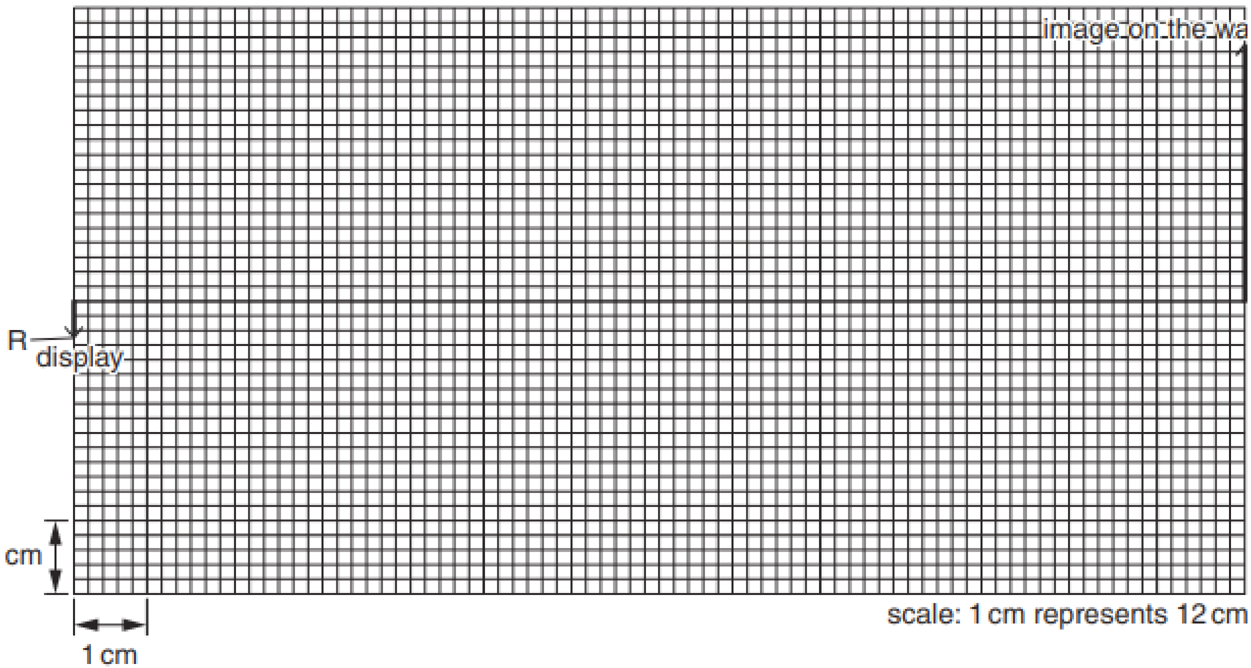


**Fig. 9.2**

**(i)** The image is inverted and magnified. State two other properties of the image.

1. ..............................................................................................................

2. ..............................................................................................................
[2]

**(ii)** On Fig. 9.2, draw the straight-line ray from R to the image. [1]

**(iii)** On Fig. 9.2, draw a vertical line representing the lens and label it L. [1]

**(iv)** A second ray from R to the image passes through a focal point (principal focus) of the lens. On Fig. 9.2, draw this ray and use it to mark this focal point. Label this focal point F. [2]

**(v)** Determine the focal length of the lens.

focal length = ........................................... [2]

## 45. M/J/10/22/Q5

Fig. 5.1 shows a ray of light entering a semi-circular glass block and striking the glass surface at M, the mid-point of the straight face.

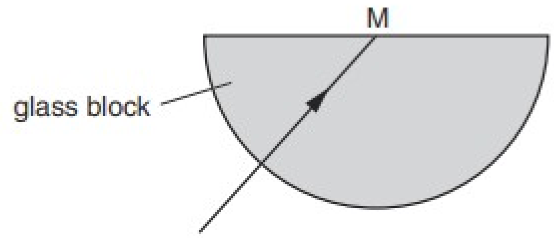


**Fig. 5.1**

**(a)** The ray of light strikes the glass surface at M with an angle of incidence $C$ equal to the critical angle of light in glass.

**(i)** State what is meant by *critical angle*.

..........................................................................................................................................

.......................................................................................................................................[1]

**(ii)** On Fig. 5.1, mark and label the angle $C$. [1]

**(iii)** On Fig. 5.1, continue the ray of light after it strikes the glass surface at M. [1]

**(b)** Fig. 5.2 shows a second ray of light striking M.

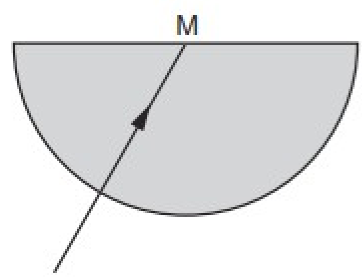


**Fig. 5.2**

This ray has an angle of incidence at M smaller than the critical angle.
On Fig. 5.2, continue this ray of light after it strikes the glass surface at M. [1]

**(c)** The refractive index of this glass is 1.5. A third ray of light enters the block from air with an angle of incidence of 50°. Calculate the angle of refraction.

angle= ..................................[2]

**46. M/J/10/21/Q5**

Fig. 5.1 shows a ray of white light incident on a glass prism.

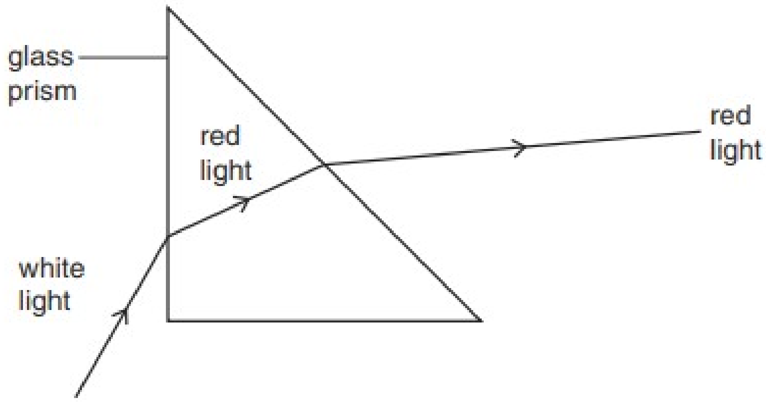


**Fig. 5.1**

Fig. 5.1 is not complete; it shows only the ray of red light produced from the white light.

**(a)** **(i)** On Fig. 5.1, draw the ray of blue light produced in, and beyond, the prism. [2]

**(ii)** State **two** colours of the spectrum found between the red and blue rays.

..................................................................................................................[1]

**(b)** Fig. 5.2 shows the same prism with the ray of white light incident at a different angle.

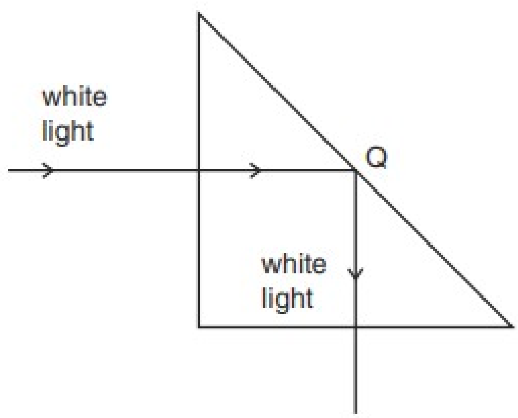


**Fig. 5.2**

Fig. 5.2 is complete; no spectrum is produced.

**(i)** Explain why no light emerges from the prism at Q.

..................................................................................................................

..................................................................................................................[1]

**(ii)** Explain why no spectrum is produced.

..................................................................................................................

**47. O/N/09/Q4**

Fig. 4.1 shows an old coin displayed in a museum.

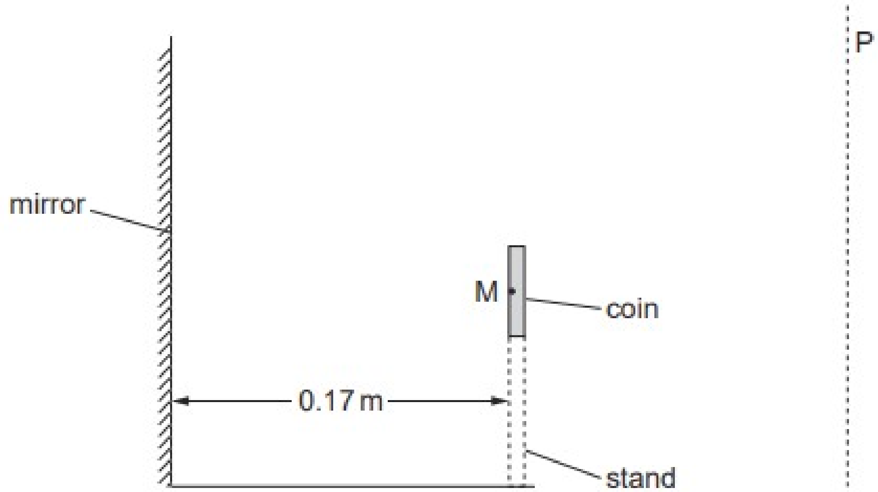


**Fig. 4.1**

The coin is vertical and is supported by a transparent stand. A vertical mirror 0.17 m behind the coin ensures that the back of the coin can be seen by a visitor looking from the line P.

M is a point on the back of the coin.

**(a)** On Fig. 4.1,

**(i)** draw **two** rays of light from M to show how its image is produced, [2]

**(ii)** label the image I. [1]

**(b)** State the distance from point M on the coin to its image.

distance = ...................................................... [1]

**48. M/J/09/Q5**

Fig. 5.1 shows a man looking at his reflection in a rectangular plane mirror.

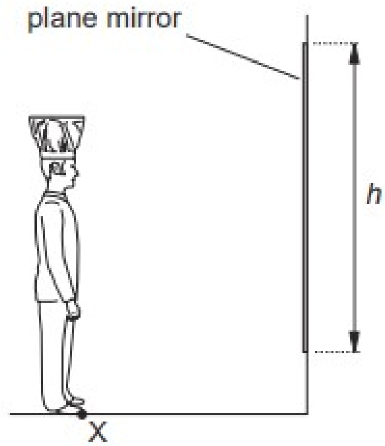


**Fig. 5.1**

The vertical side of the mirror has length *h*.

**(a)** **(i)** On Fig. 5.1, draw a ray of light from point X that is reflected by the mirror to the man's eye. [1]

**(ii)** On Fig. 5.1, mark the angle of incidence of your ray at the mirror.

Label this angle *i*. [1]

**(iii)** Define the *angle of incidence*.

..........................................................................................................................

..........................................................................................................................

.......................................................................................................................... [1]

**(b)** On Fig. 5.1, draw a ray of light from the top of the man's hat that is reflected by the mirror to his eye.

Use your rays to determine the smallest value of *h* that allows the man to see all of the image in the mirror, from the top of his hat to his toes.

On the diagram, 1 cm represents 0.5 m.

*h* = ........................................ [2]

**49. O/N/08/Q6**

Fig. 6.1 shows a ray of white light from a ray-box passing into a glass prism. A spectrum is formed between P and Q on the screen.

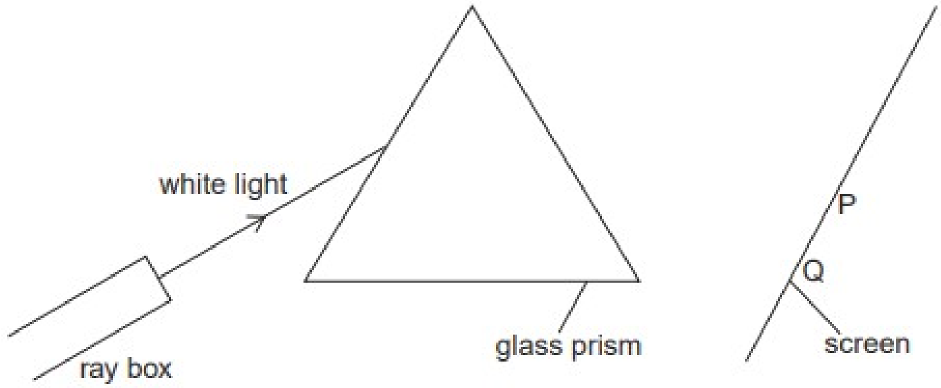


**Fig. 6.1**

**(a)** State the colour of the light at end P of the spectrum.

............................................................................................................................ [1]

**(b)** State whether the value of each of these properties for blue light is greater than, equal to or less than the value for red light.

**(i)** speed in a vacuum ............................................................................................ [1]

**(ii)** wavelength .................................................................................................... [1]

**(c)** Fig. 6.2 shows the ray passing through a red filter before it reaches the prism.

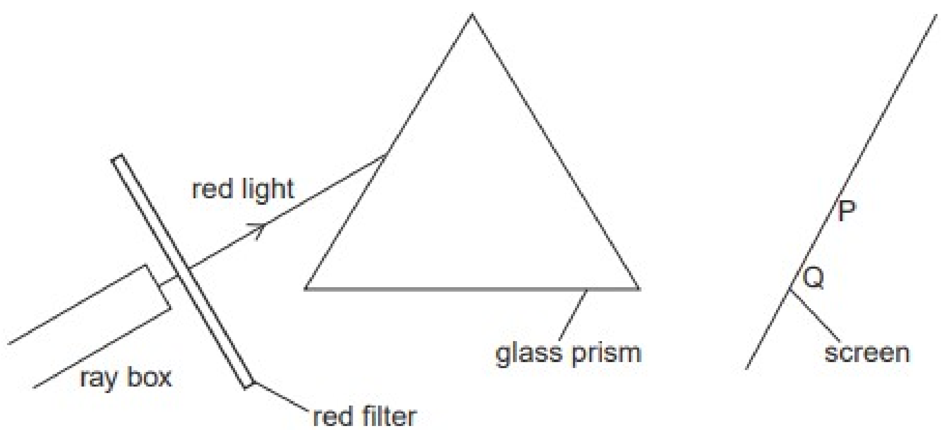


**Fig. 6.2**

Complete Fig. 6.2 to show the ray of red light passing through and emerging from the prism. [2]

**50. M/J/07/Q4**

Fig. 4.1 shows part of an optical fibre.

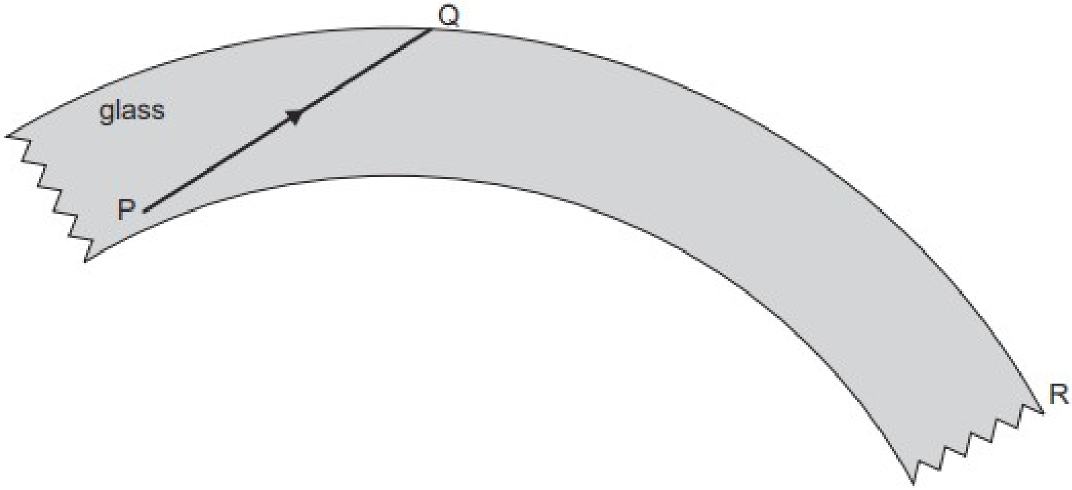


**Fig. 4.1**

The ray PQ undergoes total internal reflection in the optical fibre.

**(a)** On Fig. 4.1, continue the path of ray PQ until it reaches end R. [1]

**(b)** Explain what is meant by *total internal reflection*.

..........................................................................................................................................

..........................................................................................................................................

.................................................................................................................................... [1]

**(c)** Optical fibres are cheaper and lighter than copper wires. State one other advantage of using optical fibres rather than copper wires for telephone communications.

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..........................................................................................................................................

.................................................................................................................................... [1]

**(d)** The light in the optical fibre is travelling at a speed of $2.1 \times 10^8$ m/s and has a wavelength of $6.4 \times 10^{-7}$ m.

Calculate the frequency of the light.

frequency = ................................................. [2]

**51. M/J/07/Q5**

Fig. 5.1 is drawn full scale. The focal length of the lens is 3.0 cm.

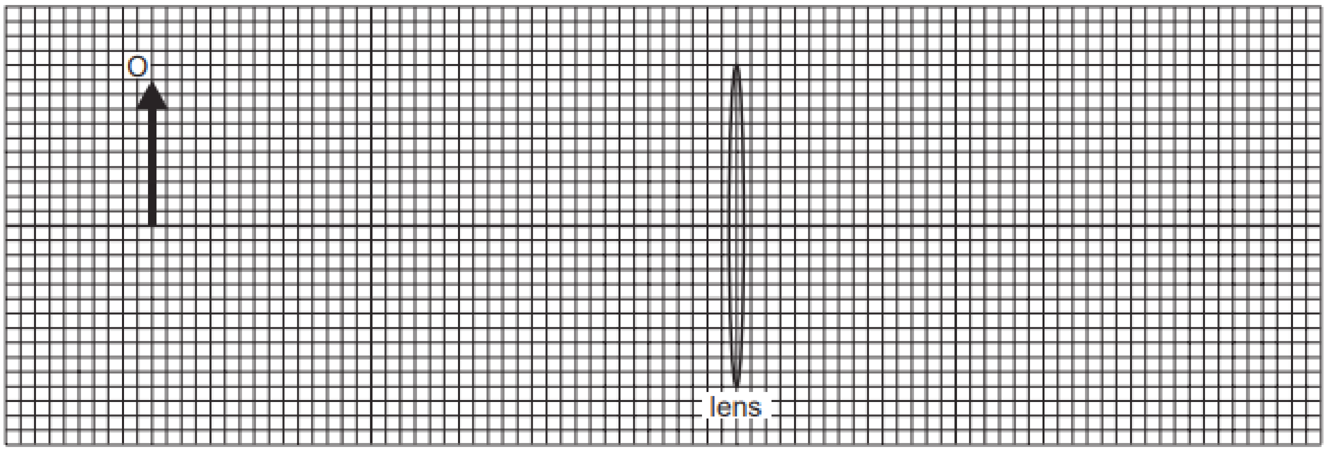


**Fig. 5.1**

**(a)** On Fig. 5.1, draw two rays from the top of the object O that meet at the image. [2]

**(b) (i)** Define the term *linear magnification*.

..........................................................................................................................

..........................................................................................................................

.................................................................................................................... [1]

**(ii)** Determine the magnification produced by the lens in Fig. 5.1.

magnification = ................................................ [1]

**(c)** Fig. 5.2 shows a normal eye viewing an object close to it. Fig. 5.3 is a long-sighted eye viewing an object at the same distance.

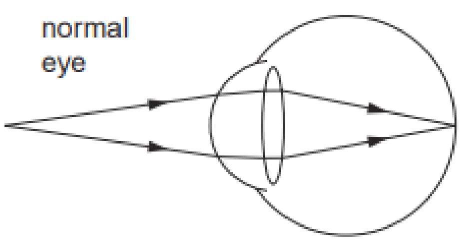


**Fig. 5.2**

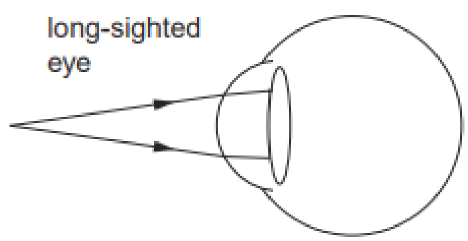


**Fig. 5.3**

Complete Fig. 5.3 to show the rays travelling through the eye. [1]

**52. O/N/06/Q4**

**(a)** Fig. 4.1 shows a ray of light incident on a mirror at X. The incident ray makes an angle of 50° with the surface of the mirror.

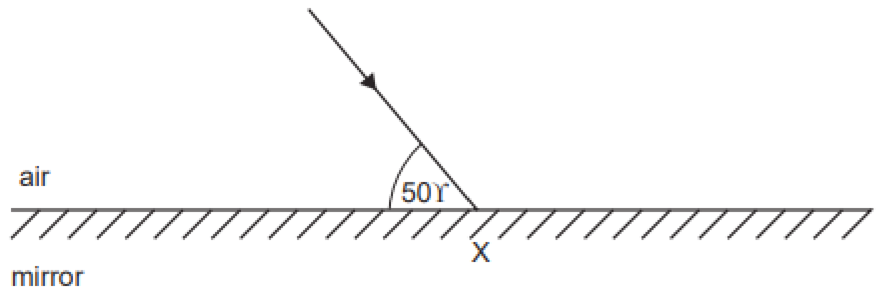


**Fig. 4.1**

**(i)** Complete Fig. 4.1 to show the normal and the reflected ray at X. [1]

**(ii)** State the values of

**1.** the angle of incidence, ............................................... [1]

**2.** the angle of reflection. ............................................... [1]

**(b)** Describe with the help of a diagram how you would find the position of the image produced by a plane mirror.

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.................................................................................................................................... [3]

**53. M/J/06/Q4**

Fig. 4.1 and Fig. 4.2 show rays of light passing through the same semi-circular block of plastic.

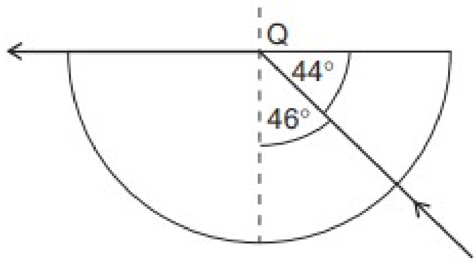


**Fig. 4.1**

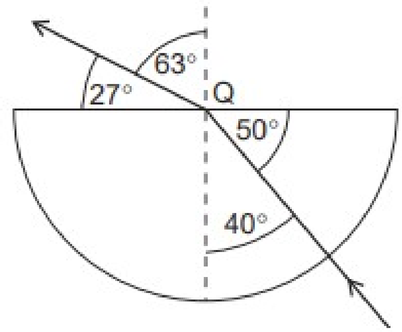


**Fig. 4.2**

Q is the centre of the straight side of the block.

**(a)** State the value of the critical angle in the plastic.

critical angle = .............................................. [1]

**(b)** Explain what is meant by the *critical angle*.

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................

.......................................................................................................................... [2]

**(c)** Calculate the refractive index of the plastic. State the formula that you use.

refractive index = ...................................... [3]

**(d)** Some light reflects back into the plastic at Q.

On Fig. 4.1, draw the reflected ray at Q. [1]

**54. O/N/04/Q4**

Fig. 4.1 shows an air bubble in water. The rays of light are incident on the air bubble.

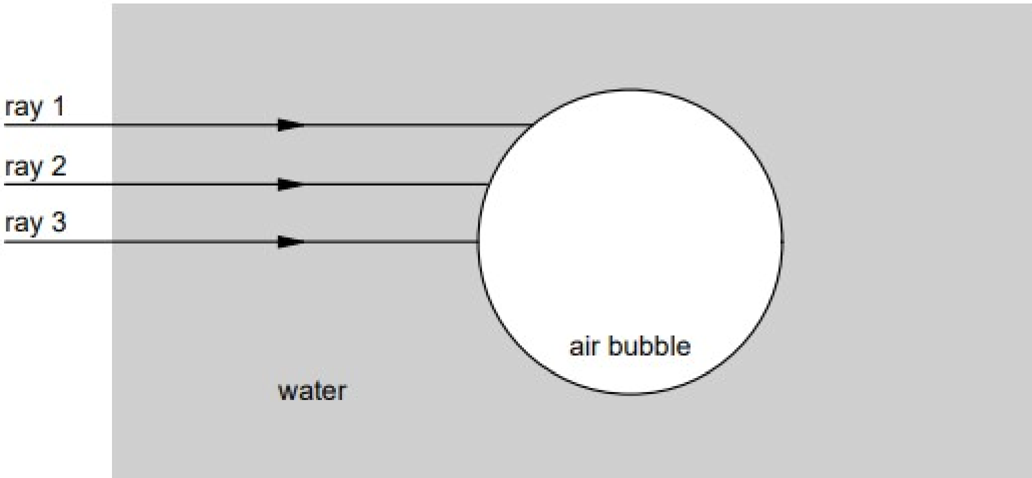


**Fig. 4.1**

The angle of incidence of ray 1 on the air bubble is greater than the critical angle.
The angle of incidence of ray 2 on the air bubble is less than the critical angle.
Ray 3 is perpendicular to the surface of the bubble.

The angle of incidence of ray 2 on the air bubble is 27° and the angle of refraction of ray 2 inside the air bubble is 37°.

**(a)** On Fig. 4.1, at the point where ray 1 meets the air bubble, mark

**(i)** the normal to the surface,

**(ii)** the angle of incidence.

[2]

**(b)** Complete Fig. 4.1 to show how all three rays continue after they meet the air bubble. [3]

**(c)** **(i)** Define what is meant by the *refractive index* of water.

..........................................................................................................................

..........................................................................................................................

..........................................................................................................................

**(ii)** Calculate the refractive index of water.

refractive index = ..................

# ANSWERS

## 1.

| Question | Answer | Marks |
|---|---|---|
| 4(a) | ($\lambda$ =) $v$ / $f$ **or** $3.0 \times 10^8$ / $4.7 \times 10^{14}$ | **C1** |
| | $6.4 \times 10^N$ (m) or 3 / $4.7 \times 10^6$ | **C1** |
| | $6.4 \times 10^{-7}$ (m) | **A1** |
| 4(b)(i) | speed: decreases<br>frequency: stays the same<br>wavelength: decreases | **B2** |
| 4(b)(ii) | $i$ = 45° **and** $r$ = 30° | **C1** |
| | ($n$ =) sin($i$) / sin($r$) **or** sin(45°) / sin(30°) | **C1** |
| | 1.4 | **A1** |
| 4(b)(iii) | straight continuation of ray in block **and** refraction away from normal in air | **B1** |
| | emergent ray parallel to initial incident ray | **B1** |

## 2.

| Question | Answer | Marks |
|---|---|---|
| 7(a)(i) | X-rays **and** gamma (radiation / rays) | **B1** |
| 7(a)(ii) | security marking / detecting counterfeit bank notes / sterilising water / fluorescent effects (e.g. fingerprints, blood, biological fluids) / (sun) tanning / sun beds / cancer treatment / curing inks and resins (e.g. in dental fillings) / production of vitamin (D) / generating electricity / solar panels / disinfection / sterilisation | **B1** |
| 7(a)(iii) | (skin) cancer / cataracts / burns or destroys the retina or cornea | **B1** |
| 7(b)(i) | critical angle | **B1** |
| 7(b)(ii) | ($n$ =) sin$i$ / sin$r$ **or** 1 / sin$c$ | **C1** |
| | 1.6 | **A1** |

## 3.

| Question | Answer | Marks |
|---|---|---|
| 6(a)(i) | object within focal length / same distance from lens as the focal length | **B1** |
| 6(a)(ii) | from divergent / parallel rays<br>**or** rays followed back (behind lens meet at point on image<br>**or** rays do not meet on a screen / do not converge<br>**or** rays appear to come from image | **B1** |
| 6(b)(i) | ray through optical centre from top of object | **B1** |
| | ray parallel to axis from top of object to lens **and** line from junction with lens to top of image | **B1** |
| | both rays extended back to image **and** ray parallel to axis correctly refracted through lens to appear to come from top of image | **B1** |
| 6(b)(ii) | 4(.0 cm) | **B1** |

## 4.

| Question | Answer | Marks |
|---|---|---|
| 8(a)(i) | vertical biconcave lens shape along P | **B1** |
| 8(a)(ii) | on right of P, middle ray continues in straight line | **B1** |
| | other two rays diverge from middle ray | **B1** |
| 8(b)(i) | continuations of the three rays meet at a point **and** on continuation of middle ray | **B1** |
| | meeting point within eye **and** rays strike back of eye at different places | **B1** |
| 8(b)(ii) | image is sharp / focused / not blurred | **B1** |
| 8(b)(iii) | light diverges before entering eye **or** meeting point now on the back of the eye | **B1** |
| 8(c)(i) | **one** point on principal axis 4.0 cm from the centre of the lens indicated | **B1** |
| | any **two** rays from:<br>paraxial ray from tip of object to lens **and** seems to come from left-hand principal focus after refraction<br>ray from tip of object towards right-hand principal focus (until lens) **and** paraxial after refraction<br>ray from tip of object <u>reaching</u> the centre of the lens | **B2** |
| | image marked at correct intersection | **B1** |
| 8(c)(ii) | virtual **and** light only seems to come from the image **or** light does not pass through image | **B1** |
| 8(c)(iii) | E marked on the right of the dashed line | **B1** |
| 8(c)(iv) | height of candidate's I / 3.5 | **C1** |
| | $0.30 \leq$ magnification $\leq 0.50$ | **A1** |

## 5.

| Question | Answer | Marks |
|---|---|---|
| 4(a) | clear angle is to / from the normal<br>**or** 90°– angle between ray and surface | **C1** |
| | angle between/with the refracted <u>ray</u> / <u>beam</u> and the normal or vv<br>**or** angle from the normal to the refracted <u>ray</u> / <u>beam</u> or vv | **A1** |
| 4(b) | (refractive index = ) sin*i* / sin*r* algebraic or numerical in any form | **M1** |
| | 1.5 | **A1** |
| 4(c) | refractive index smaller (for red ) | **B1** |
| | larger angle of refraction (for red)<br>**or** (red) bends / refracts less (towards normal)<br><br>assume comparison is for light **entering** glass at equal incident angles unless clear otherwise | **B1** |
| 4(d) | P infrared **and** Q ultraviolet | **B1** |

## 6.

| Question | Answer | Marks |
|---|---|---|
| 4(a)(i) | correct *i* and *r* marked | **B1** |
| 4(a)(ii) | reflection shown at approximately the correct angle<br>**or** ray along surface | **B1** |
| 4(a)(iii) | correct refraction as red light enters prism showing less refraction than blue light | **B1** |
| | correct refraction as red light emerges from left side of prism | **B1** |

| | | |
|---|---|---|
| 4(b)(i) | angle of incidence | B1 |
| | smallest angle when ray totally internally reflects<br>**or** (when ray is) just internally reflected<br>**or** when ray (refracted/emerges) along surface / at 90° to normal / angle of refraction is 90°<br><br>**OR** angle of refraction when the angle of incidence is 90° B2 | B1 |
| 4(b)(ii) | sin *c* = 1 / *n* in any form numerical or algebraic | B1 |
| | 42° | B1 |

## 7.

| Question | Answer | Marks |
|---|---|---|
| 3(a)(i) | it / light refracts / bends / moves towards normal **or** angle of incidence greater than angle of refraction | B1 |
| | change in direction of the ray / refraction shows a change in speed (at the boundary) **or** sin(*i*) / sin(*r*) is ratio of speeds **or** top of ray slows down first **and** enters the block first | B1 |
| 3(a)(ii) | (wavelength) decreases | B1 |
| | (frequency) does not change. | B1 |
| 3(b)(i) | (sin(*r*) =) sin(*i*) / *n* **or** sin(45°) / 1.6 **or** 0.71 / 1.6 **or** 0.44 **or** (*r* =) 21(°) | C1 |
| | 26(°) | C1 |
| | 64° | A1 |
| 3(b)(ii) | 64° / angle of incidence / *i* is greater than the critical angle / 39° **or** angle of incidence = 64° | B1 |
| | it undergoes total internal reflection | B1 |

## 8.

| | | |
|---|---|---|
| 10(b)(i) | four colours from<br>violet, indigo, blue, green, yellow, orange, red | B1 |
| | correct order of four of the above colours | B1 |
| 10(b)(ii) | prism | B1 |
| | slit (to produce narrow beam) | B1 |
| | dispersion shown at first or second face, i.e. a ray becoming at least two rays at least one face | B1 |
| | correct refraction on first and second faces for all rays shown | B1 |

## 9.

| Question | Answer | Marks |
|---|---|---|
| 5(a) | convex **or** converging | B1 |
| 5(b) | as it enters – refracts (towards normal) **or** changes speed | B1 |
| | as it leaves, one of:<br>• bends / refracts away from normal<br>• speed increases<br>• wavelength increases<br>• emerges parallel to the (principal) axis | B1 |
| 5(c) | either emergent ray R or the ray through F extended to the left of the lens | B1 |
| | both ray through F and R shown and image marked at intersection of extended rays | B1 |
| 5(d) | upright **and** virtual | B1 |

## 10.

| Question | Answer | Marks |
|---|---|---|
| 8(a)(i) | $n = \sin i / \sin r$ **or** $1.4 = \sin 55° / \sin r$ **or** ($r$ =) $\sin^{-1}(\sin i / n)$ **or** ($r$ =) $\sin^{-1}(\sin 55° / 1.4)$ | **C1** |
| | 36° | **A1** |
| 8(a)(ii) | towards (the normal) **and** speed decreases (i.e. fourth box **only**) | **B1** |
| 8(a)(ii) | stays constant | **B1** |
| 8(b)(i) | distance between (one) focal point / principal focus and (the centre of) lens | **B1** |
| 8(b)(ii) | at least one correct focal point labelled / clear / used | **B1** |
| | any **two** from drawn:<br>paraxial ray from tip of O to lens and refracted through correct far F<br>ray from tip of O through optical centre and undeviated<br>ray from tip of O through correct near F to lens and refracted paraxially | **B2** |
| | I marked at intersection of correct rays | **B1** |
| 8(b)(ii) | 3.4 cm ⩽ distance ⩽ 3.5 cm | **B1** |
| | (magnification =) candidate's distance / 9.0 **or** candidate's height / 3.0 | **C1** |
| | 0.37 ⩽ magnification ⩽ 0.40 **c.a.o.** | **A1** |
| 8(c) | on next page | |
| 8(c) | any **three** from:<br>light (from subject / object) focussed / refracted by lens<br>by moving / sliding / adjusting lens<br>real image (of subject / object) formed<br>on digital light detector / film | **B3** |

## 11.

| Question | Answer | Marks |
|---|---|---|
| 4(a)(i) | it decreases | **B1** |
| 4(a)(ii) | it does not change | **B1** |
| 4(b)(i) | (c =) $\sin^{-1}(1 / n)$ **or** $\sin^{-1}(1 / 1.6)$ **or** $\sin^{-1}(0.625)$ | **C1** |
| | 39° | **A1** |
| 4(b)(ii) | reflected ray at Z **and** $i = r$ **and** no refracted ray | **B1** |
| | ray strikes vertical face perpendicularly **and** undeviated in air | **B1** |

## 12.

| Question | Answer | Marks |
|---|---|---|
| 4(a)(i) | 60° | **B1** |
| 4(a)(ii) | 0(°) | **B1** |
| 4(a)(iii) | back along the initial path | **B1** |
| 4(b) | Any three of<br>• same size (as object)<br>• same distance (from mirror as object)<br>• upright<br>• virtual<br>• laterally inverted | **B3** |

## 13.

| Question | Answer | Marks |
|---|---|---|
| 6(a) | behind the mirror **or** on the opposite side (of mirror) to object | B1 |
| | same distance as object (from mirror) | B1 |
| 6(b)(i) | image marked at correct position by eye | B1 |
| 6(b)(ii) | ray from tooth to mirror **and** ray from mirror to eye | M1 |
| | angles of incidence and reflection equal (by eye) | A1 |
| 6(c) | virtual **or** same size (as object) **or** laterally inverted | B1 |

## 14.

| Question | Answer | Marks |
|---|---|---|
| 10(a) | $3.0 \times 10^8$ m / s | B1 |
| 10(b) | red orange yellow green blue indigo violet (any order) | B1 |
| 10(c)(i) | (–)6.(0) cm | B1 |
| 10(c)(ii) | any **two** rays drawn from:<br>paraxial ray that refracts and seems to come from $F_1$<br>ray through the optical centre of lens<br>ray that aims for $F_2$ but refracts and emerges paraxially | B2 |
| | rays traced back to point | B1 |
| | (point) labelled I **and** rest of image drawn down to the principal axis | B1 |
| 10(c)(iii) | 1.7–1.9 cm | B1 |
| 10(c)(iv) | candidate's **10(c)(iii)** / 3.0 evaluated | B1 |
| 10(c)(v) | any **two** from:<br>a real image can be projected on to a screen<br>light actually passes through a real image<br>on same side (of lens) as object **or** on opposite side of <u>mirror</u> to object | B2 |
| 10(c)(vi) | correction of short-sight / myopia | B1 |
| 10(d) | light travels more slowly in glass **or** light changes speed | B1 |
| | one side / left-hand side of wavefront slows down first | B1 |
| | wavelength decreases **or** wavefront travels a shorter distance in the same time | B1 |

## 15.

| Question | Answer | Marks |
|---|---|---|
| 3(a) | middle ray angle of incidence labelled c | B1 |
| 3(b) | angle of incidence (of right-hand ray only) greater than the critical angle | B1 |
| 3(c)(i) | (n=) sin i / sinr algebraic or numerical | C1 |
| | 1.3 | A1 |
| 3(c)(ii) | sin c = 1 / n algebraic or numerical | C1 |
| | answer in range 49° to 50.3° | A1 |

## 16.

| | | |
|---|---|---|
| 9(c)(i) | electromagnetic **and** transverse underlined | **B1** |
| 9(c)(ii) | $n = \sin(i) / \sin(r)$ **or** $(r =) \sin^{-1}(\sin(i) / n)$ **or** $(r =) \sin^{-1}(\sin(60°) / 1.6)$ | **C1** |
| | 33(°) | **A1** |
| 9(c)(iii) | 1 $n = 1 / \sin(c)$ **or** $(c =) \sin^{-1}(1 / n)$ **or** $\sin^{-1}(1 / 1.6)$ | **C1** |
| | 39° | **A1** |
| | 2 total internal reflection | **B1** |
| | $\theta > c$ **or** 57(°) > 39(°) **or** angle of incidence greater than the critical angle | **B1** |
| | 3 reflected ray at P (correct angle by eye) | **B1** |
| | light in air at 30° (by eye) to vertical surface | **B1** |

## 17.

| Question | Answer | Marks |
|---|---|---|
| 6(a) | **any** two of<br>• ray <u>through</u> middle of lens undeviated<br>• ray parallel to axis passes <u>through</u> focus, 3 cm from lens<br>• ray <u>through</u> focus on left of lens parallel to axis after lens | **B2** |
| 6(b) | inverted **or** real | **B1** |
| 6(c) | image size / object size **or** image distance / object distance<br>numerical or algebraic | **C1** |
| | rays from lens intersect on diagram (for image) **and** 1.3–1.7 | **A1** |
| 6(d) | projector **or** photographic enlarger | **B1** |

## 18.

| Question | Answer | Marks |
|---|---|---|
| 9(a) | **true false false true**<br>all 4 correct 2 marks | **B1** |
| | 3 correct 1 mark | **B1** |
| 9(b)(i) | correct refraction towards normal on entering prism | **B1** |
| | splitting into colours occurs at first surface | **B1** |
| | red at top and blue / violet at bottom <u>inside</u> and outside prism if shown | **B1** |
| 9(b)(ii) | (different colours have) different speeds in the glass<br>**or** different refractive indices | **B1** |
| | different (angles of ) <u>refraction</u><br>**or** <u>refract</u> at different angles | **B1** |
| 9(c)(i) | $3(.0) \times 10^8$ m / s | **B1** |
| 9(c)(ii) | (t=) d / s in any form algebraic or numerical, e.g. | **C1** |
| | 500 s | **A1** |
| 9(d)(i) | (I=) Q / t in any form algebraic or numerical, e.g. 180 / 2 | **C1** |
| | 1.5 A | **A1** |
| 9(d)(ii) | 1st column helium nucleus **or** 2 protons and 2 neutrons | **B1** |
| | 2nd column positive **and** neutral / no / charge / none | **B1** |
| | 3rd column paper / <u>thin</u> metal / skin / few cm air etc.<br>**and** (thick) lead / thick metal / concrete / stone / earth OR never completely stopped / nothing | **B1** |

## 19.

| | | |
|---|---|---|
| 10(d)(i) | normal indicated **and** angle of incidence indicated | **B1** |
| 10(d)(ii) | $n = \sin i / \sin r$ **or** $1.5 = \sin 57(°) / \sin r$ **or** $(r = )\sin^{-1}(\sin 57(°) / n)$ **or** $\sin^{-1}(\sin 57(°) / 1.5)$ | **C1** |
| | 34° | **A1** |
| 10(d)(iii) | **1** no change | **B1** |
| | **2** **3** decreases **and** decreases | **B1** |
| 10(d)(iv) | ray in glass between normal and continuation of the incident ray | **B1** |
| | ray in air between continuation of the refracted ray and side of prism | **B1** |

## 20.

| Question | Answer | Marks |
|---|---|---|
| 9(a)(i) | ray from right-hand corner of mirror to eye | **B1** |
| | any incident and corresponding reflected ray correct by eye | **B1** |
| 9(a)(ii) | normal drawn at any intersection of incident and reflected ray | **C1** |
| | both *r* and *i* labelled correctly with normal | **A1** |
| 9(a)(iii)1 | cannot be formed on a screen<br>**or** nothing at the image (position) | **B1** |
| | rays do not come (all the way) from the image<br>**or** rays only appear to come from image | **B1** |
| 9(a)(iii)2 | (same distance) behind the mirror<br>**or** same size (as object)<br>**or** upright / erect<br>**or laterally** inverted | **B1** |
| 9(b)(i) | reflection in mirror occurs at any angle<br>**or** total internal reflection (TIR) only occurs for *i* > critical angle<br>**or** there is no critical angle for the mirror B1 | **any 2**<br><br>**B2** |
| | TIR occurs from dense to less dense medium<br>**or** in the dense(r) medium<br>**or** from glass to air **or** inside / does not escape glass<br>**or** from slow to fast (media)<br>**or** mirror reflection from air to glass B1 | |
| | (mirror) reflection is not total, e.g., not all reflected<br>**or** better quality of image<br>**or** multiple images from a mirror B1 | |
| 9(b)(ii) | (n=) 1 / sinC **or** 1 / sin44 | **C1** |
| | 1.4 | **A1** |
| 9(b)(iii) | n = sini / sinr in any form, e.g. sinr = sin50 / n | **C1** |
| | 32° – 33° | **A1** |
| 9(b)(iv) | ANY 2 lines from<br>• more data per second **or** per unit time<br>• less decrease in strength / amplitude / attenuation<br>• less heat / power produced / wasted<br>• less need for repeating or amplification stations<br>• less interference / noise<br>• more secure / less chance of cross-talk<br>• lighter / less heavy | **B1**<br>**B1** |

## 21.

| Question | Answer | Marks |
|---|---|---|
| 4(a)(i) | image height / object height **or** image distance / object distance | **B1** |
| 4(a)(ii) | rays go to / meet at image **or** can be formed on a screen | **B1** |
| 4(b)(i) | horizontal ray continued to bottom of I | **B1** |
| 4(b)(ii) | 2.4 cm | **B1** |
| 4(b)(iii) | any two other rays drawn, e.g. through centre of lens | **B1** |

## 22.

| Question | Answer | Marks |
|---|---|---|
| 10(a)(i) | both reflected rays correct by eye | **B1** |
| | image in correct position shown by continuation of rays behind mirror | **B1** |
| 10(a)(ii) | virtual **or** upright **or** laterally inverted **or** same size (as object) | **B1** |
| 10(b)(i) | line ( joining points ) | **C1** |
| | line joining points / particles on wave with same phase **or** line joining points along a crest etc. | **A1** |
| 10(b)(ii)1 | correct angle to surface ± 7° with correct orientation **and** similar wavelength (by eye) | **B1** |
| 10(b)(ii)2 | at least two lines at smaller angle to surface with correct orientation in slower medium | **B1** |
| | showing smaller and constant wavelength with wavefronts deviated in correct direction | **B1** |
| 10(c)(i) | ($v$ = ) $f\lambda$ **or** 2(000) × 16 | **C1** |
| | 32 000 cm / s **or** 320 m / s | **A1** |
| 10(c)(ii)1 | a range with 15–25 Hz as the lowest frequency **and** 15–30 kHz as the highest | **B1** |
| 10(c)(ii)2 | 1.6 cm (if highest frequency is 20 000 Hz) | **B1** |
| 10(c)(iii) | tube or other method to produce (narrow) beam of sound on source and/or on detector | **B1** |
| | stated detector **or** stated reflector | **B1** |
| | detector moved to find maximum loudness **or** angles *i* and *r* measured with suitable experiment | **B1** |

## 23.

**(a) (i)** $3.0 \times 10^8$ m / s — B1

**(ii)** ($\lambda$ = ) $c/f$ **or** $3.0 \times 10^8 / 4.3 \times 10^{14}$ — C1
$7.0 \times 10^{-7}$ m — A1

**(b) (i)** decreases — B1

**(ii)** sin(i) = n × sin(r) **or** 1.5 × sin(30°) **or** 0.75 — C1
49° — A1

**(iii)** 41° — B1

**(c) (i)** dispersion at **both** surfaces **and** refractions in correct direction — B1
violet / blue light below the red light shown — B1

| | | |
|---|---|---|
| **(ii)** | spectrum **or** band of (continuous) colours **or** colours of rainbow | B1 |
| | red, orange, yellow, green, blue, (indigo, violet) | B1 |
| **(iii)** | **1** X marked above red | B1 |
| | **2** it is / black surfaces are good absorbers (of IR radiation) | B1 |

| | | | | | |
|---|---|---|---|---|---|
| **(d)** | intruder / human being emits IR | IR beam broken | | IR reflected | B1 |
| | **or** | | **or** | | |
| | intruder warm **or** IR detected | does not reach detector | | change detected | B1 |

**24.**

| | | |
|---|---|---|
| **(a)** | smallest angle for total internal reflection **or** angle for refraction along surface | B1 |
| | angle of <u>incidence</u> in (optically) <u>denser</u> medium | B1 |
| **(b)** | vertical ray continues undeviated | B1 |
| | second ray (60° to horizontal) refracts away from normal into the air | B1 |
| | third ray reflects internally **and** $i = r$ by eye **not** if any refracted ray | B1 |

**25.**

| | | | |
|---|---|---|---|
| **(a)** | **(i)** | long-sight **or** far-sight **or** hypermetropia | B1 |
| | **(ii)** | rays do not come together (on back of eye)<br>**or** rays do not converge (on retina)<br>**or** it / the image is not formed on retina / back of eye<br>**or** it / the image is formed behind retina / back of eye | B1 |
| **(b)** | **(i)** | lens between rays and eyeball **and** a converging lens shown | B1 |
| | **(ii)** | converging or convex | B1 |

**26.**

**(a)** angle of incidence — B1

smallest angle for light to be totally internally reflected — B1
**or** largest angle (of incidence) for ray to be refracted / emerge
**or** when light emerges along surface
**or** when angle of refraction is 90°

**(b) (i)** n = 1 / sinC algebraic or numerical — C1
2.5 **or** 2.46 **or** 2.458(59) — A1

**(ii)** *left hand diagram* ray refracts away from normal and emerges into air at bottom left surface — B1

*right hand diagram* reflected horizontal ray (by eye) — B1

*right hand diagram* rest of ray completely correct to emerge into air at top face without refraction (by eye) — B1

**27.**

**(a)** ($n$ = )sin $i$ / sin $r$ **or** sin 55° / sin 33° — M1
1.5(040274) — A1

**(b) (i)** angle of incidence greater than critical angle **or** denser to rarer medium — B1

**(ii)** reflected ray in correct direction (by eye) to edge of block **and** no second TIR (**ign** marked values) — B1

**28.**

**(a)** **distance** from (optical) centre to focal point (principal focus) — B1

**(b) (i)** **both** Fs correctly positioned at ±1 mm — B1

**(ii)** **two** of: paraxial ray to lens through focal point to image
ray through optical centre
ray through focal point and then paraxial to image (**ign.** arrows) — M2
X at crossing point of rays — A1

**(iii)** 3.4–3.8 cm — B1

**29.**

**(a)** sin *i* / sin *r* or sin 50 / sin 30 — C1
1.5(321) — A1

**(b)** moving from more dense to less dense medium — B1
**or** moving to lower refractive index (air)
**angle of incidence** is greater than critical angle — B1

**(c)** less heat loss / more efficient — B1
less chance of hacking / more secure / less interference
less reduction in signal / less need for boosting / larger distances possible / thinner or less bulky

**30.**

**(a)** $3.0 \times 10^8$ m / s — B1

**(b) (i)** **1.** decreases **cao** — B1
**2.** no change **cao** — B1
**3.** decreases **cao** — B1

**(ii)** **1.** *i* correctly marked (to normal) — B1
**2.** *r* correctly marked (to normal) — B1

**(c) (i)** sin *i* / sin *r* = *n* **or** sin *i* / sin *r* = 1.5 — C1
sin 89 / sin *r* = 1.5 **or** sin 89 / 1.5 **or** 0.67(0.666565) — C1
42° **or** 41.8025° — A1

**(ii)**

| | | | |
|---|---|---|---|
| *i* equal to / close to 90° **and** *r* less than 45° | sin *i* / sin 45 = 1.5 | $\sin^{-1}(1/n) / \sin^{-1}(1/1.5)$ **and** 41.8° | B1 |
| | **or** | **or** | |
| *i* never bigger than 89° / 90° | sin *i* >1 | *r* not be more than *c* | B1 |

**31.**

**(a)** ultra violet **and** infra-red — [B1]

**(b)** blue refracts / bends / deviates more — [B1]
blue slows more (than red when entering glass) or blue and red have different speeds (from each other in glass) — [B1]
blue and red have different refractive indices — [B1]

**32.**

**(a)** **(i)** any single value between 0 and 5.6 cm or a range all of whose values are correct [B1]

**(ii)** any value beyond 5.6 cm [B1]

**(b)** **(i)** ray through optical centre undeviated [B1]
other ray correct through or to axis 2.8 cm (± ½ small square) from lens [B1]

**(ii)** lines drawn meet after 11 cm or rays do not meet (on page) or rays almost parallel [B1]

**(iii)** inverted, magnified, real all 3 needed and none wrong [B1]

**33.**

**(a)** converging **or** convex [B1]

**(b)** image height ÷ object height [B1]

**(c)** **(i)** line when extended back joins top of image with intersection of ray and lens [B1]

**(ii)** 3.0 ± 0.1 cm ecf from diagram [B1]

**(iii)** any two further lines from top of stamp that appear to come from the top of the image [B1]

**34.**

**(a)** **(i)** speed of sound is (much) less than the speed of light (accept quoted values) B1

**(ii)** **measure** the time delay (between the lightning and thunder) B1
divide distance by time/delay B1

**(b)** **(i)** $3.0 \times 10^8$ m/s B1

**(ii)** ($\lambda$ = ) $c/f$ **or** $3.0 \times 10^8 / 7.5 \times 10^{14}$ C1
$4.0 \times 10^{-7}$ m A1

**(iii)** (in any order) blue, green, orange, red, yellow, (indigo), (violet) **or** VIBGYOR C1
violet, indigo, blue, green, yellow, orange, red A1

(c) (i) correct angle clear/labelled *r* — B1

(ii) mark/determine entrance and exit points (e.g. trace rays back to glass) — B1
join/draw line between entrance and exit points — B1

(iii) **1.** $n = \sin i / \sin r$ — B1

**2.** 1.5/1.51/1.506176 with no unit
(**not** just 1.5 without working out) — B1

(iv) correct direction of refraction at **both** faces — M1
completely correct (above blue) — A1

**35.**

**(a)** correct normal by eye — B1
correct angle of incidence between candidate's normal and incident ray — B1
correct angle of refraction marked between candidate's normal and BC — B1

**(b)** decrease / change in speed / wavelength — B1

**(c)** $n = \sin i/\sin r$ seen in any form — C1

($\sin r =$) $\sin 45°/1.5$ — C1
**or** 0.47(14) seen

28(.1)° — C1

**(d)** refracts less at first face and on correct side of normal — B1

refraction at second face away from normal so that red ray and blue ray are diverging — B1

**(e) (i)** angle of incidence is 0 — B1
**or** ray along normal/perpendicular to glass

**(ii)** angle of incidence/$\theta$ is larger than critical angle — B1
total internal reflection occurs — B1

**(iii)** reflected ray drawn correctly **and** emerging without refraction from block — B1

**(iv)** (eventually) light emerges (into air at Q) — B1
**or** light refracts (out at Q)
**or** (weak) refracted ray appears

light emerging at Q coloured in some way — B1
**or** correct description of movement of reflected ray (as $\theta$ decreases)

**36.**

| | | |
|---|---|---|
| **(a)** | critical angle | B1 |
| **(b) (i)** | light refracted out into air **and** bent away from normal (ignore reflected ray) | B1 |
| **(ii)** | correct internal reflection (by eye) **and** no refracted ray (**not** at 90°) | B1 |
| **(c)** | ($t$ =) distance/speed in any form numerical or algebraic (e.g. d/s, s/v $10/2 \times 10^8$) | C1 |
| | $2.5 \times 10^{-10}$ s | A1 |

**37.**

| | | |
|---|---|---|
| **(a)** | ($\lambda$ = )$v/f$ **or** $2 \times 10^8/4.7 \times 10^{14}$ | C1 |
| | $4.3 \times 10^{-7}$ m | A1 |

**(b)**

| | | | |
|---|---|---|---|
| raybox/light source/ laser **and** mirror | | pin(s) **and** mirror | B1 |
| shine ray at mirror | | place two pins | B1 |
| mark rays | **or** | two more pins in line with image | B1 |
| measure $i$ and $r$ **and** equal | | measure $i$ and $r$ **and** equal | B1 |
| repeat | | repeat | B1 |

| | | |
|---|---|---|
| **(c) (i)** | 83° | B1 |
| **(ii)** | total internal reflection **or** TIR **cao** | B1 |
| | **angle of incidence** exceeds critical angle | B1 |
| **(d) (i)** | (at least) one ray from X to mirror | M1 |
| | (at least) **two** rays from X to mirror **and** correct reflections | A1 |
| | rays traced back to marked I **or** I marked in correct place (by eye) | B1 |
| **(ii)** | 0.19 m | B1 |
| **(iii)** | less/no light wasted **or** hall brighter | B1 |

**38.**

**(a)** more telephone signals (at one time)
OR great(er) bandwidth; more data (per sec); more signals
OR faster data/information transfer
OR less attenuation; less energy/power/signal loss;
OR long(er) distance (before regeneration)
OR (more) secure
OR less noise/interference OR high(er) quality/clear(er) B1

**(b)** **(i)** correct normal and angle marked B1

**(ii)** total internal reflection B1
**angle of incidence** is larger than critical angle B1

**(c)** ($n$ =) sin $i$/sin $r$ in any form numerical or algebraic C1
35(.2644)° **unit ° needed** A1

**39.**

**(a)** **(i)** refraction B1

**(ii)** ($n$ =) sin $i$/sin $r$ C1
sin 45°/sin 29° C1
1.4585 to more than 1 sig. fig. A1

**(iii)** the angle of incidence/incident angle is greater than the critical angle B1
total internal reflection occurs B1

**(iv)** correct refraction at C with ray parallel to AB B1
correct reflection (and correct refraction on other face i.e. downwards) B1

**(b)** **(i)** Any TWO of:
undeviated ray through centre of lens
ray parallel to axis through point 3 cm from lens on right after lens
ray through point 3 cm to left of lens parallel to axis after lens M2
rays converge and vertical image drawn and labelled I A1

**(ii)** 1.2 ± 0.2 cm B1

**(iii)** **1.** real image (can be) formed on screen; virtual image not found on screen; rays converge on real image; rays do not converge on virtual image; rays only appear/seem to come from a point on virtual image B1

**2.** place object within focal length; between lens and focal point/principal focus B1
view from other side of lens; look through lens; image same side as/behind object B1

**40.**

**(a)** **(i)** central ray undeviated emerging from lens — M1

two outer rays meet the central ray at a point inside the eye **and** carry on to strike the retina — A1

**(ii)** light (from a single point) is spread over an area (on the retina)
**or** rays do not meet at a point on the retina
**or** image formed/rays meet/principal focus off retina — B1

**(b)** **(i)** **any** diverging lens: biconcave, planoconcave, convexoconcave – i.e. lens **clearly** thinner at the centre — B1

**(ii)** **all** rays diverge — B1

**41.**

**(a)** one outer ray parallel to principal axis — C1

three rays parallel to the principal axis — A1

**(b)** **(i)** (speed) reduced **or** slows down — B1

**(ii)** (speed) returns to original value/$3.0 \times 10^8$ m/s — B1

**(c)** **(i)** ($f$ =) $c/\lambda$ **or** $3.0 \times 10^8 / 6.0 \times 10^{-7}$ — C1

$5(.0) \times 10^{14}$ Hz — A1

**(ii)** no effect/unchanged/($f$ =) $5(.0) \times 10^{14}$ Hz — B1

**42.**

**(a)** suitable block (semicircular/rectangular/prism) B1
suitable source of rays (e.g. ray box; pins on incident ray; laser **not** torch) **must** be labelled on diagram or clear in text B1
diagram showing incident ray in glass/perspex (no arrow needed) **and** correct refraction out into air B1
adjustment of (angle of incidence of) ray until along surface/just no longer emerges B1
(measure) correct angle marked or described clearly or C marked on diagram B1

**(b) (i)** converging or convex B1

**(ii)** ray from top through middle of lens undeviated B1
other ray from top of object to same position on film M1
correct image labelled/drawn/marked A1

**(iii)** ratio of size/height/length/distance of image to size/height/length/distance of object B1

**(iv)** 0.4(0) (±0.05) no ecf **(iii)** B1

**(v)** upside down // inverted // real // other side of lens to object // nearer lens than object B1

**(vi)** (otherwise) not focussed // to/adjust focus // to produce a clear/sharp image // B1

(otherwise) rays do not converge on film // to converge rays onto film // image on film // object at different distance C1
(otherwise) image formed in front of film // object now further A1

**43.**

**(a) (i)** **one** correctly reflected ray (by eye) B1

**(ii)** **two** reflected rays traced back to an image B1
(labelled) image in correct position (by eye) B1

**(iii)** any **two** of:
virtual
full size/mag = 1 **or** same distance from mirror as C
**laterally** inverted (**ign** upright)
dimmer B2

**(iv)** more comfortable/no neck strain/no need to look up/reflects to eyes B1

**(b) (i)** ($c$ =) 3(.00) × $10^8$ (m/s) **or** 3(.00) × $10^5$ (km/s) **or** used in equation B1
($f$ =) $c/\lambda$ **or** (3.0 × $10^8$/their **stated** value/330)/4.0 × $10^{-7}$ C1
7.5 × $10^{14}$ Hz **or** correct answer from **stated** value (incl. unit) **or** 8.2/8.25/8.3 × $10^8$ Hz A1

**44.**

| | | | |
|---|---|---|---|
| **(a)** | **(i)** | horizontal ray from Q to pool edge **and** on to P from corner | B1 |
| | | critical angle marked *C* **or** obvious | B1 |
| | **(ii)** | for $i$ = 90° **or** horizontal ray | B1 |
| | | angle(in water) equals/cannot be less than critical/*C* | B1 |
| | **(iii)** | ($n$ =) sin $i$/sin $r$ **or** 1/sin $C$ **or** 1/$n$ = sin $C$ **or** sin 90°/sin 49° **or** 1/sin 49° | B1 |
| | | 1.3(2501) | B1 |
| | **(iv)** | decreases | B1 |
| **(b)** | **(i)** | any **two** of:<br>real<br>less bright<br>further from lens<br>beyond 2*f* | B2 |
| | **(ii)** | straight ray from R to **top** of image | B1 |
| | **(iii)** | where ray crosses principal axis, vertical line (L **or** drawn lens) | B1 |
| | **(iv)** | paraxial ray from R to lens refracted to top of image **or** paraxial ray from lens to top of image, traced back to R | M1 |
| | | F marked | A1 |
| | **(v)** | 1.6 – 1.9 cm **or** attempt to use 1/$u$ + 1/$v$ | C1 |
| | | 19 – 23 cm (2 sig. fig. only) | A1 |

**45.**

| | | | |
|---|---|---|---|
| **(a)** | **(i)** | smallest angle of incidence for total internal reflection<br>or greatest angle of incidence that allows refraction<br>or angle of incidence for (refracted) ray along surface/angle of refraction 90° | B1 |
| | **(ii)** | correct angle marked to normal (by eye) | B1 |
| | **(iii)** | ray along surface or reflected ray correct (by eye) or both rays | B1 |
| **(b)** | | ray in air refracted away from normal | B1 |
| **(c)** | | refractive index = sin $i$/sin $r$ algebraic or numerical e.g. 1.5 = sin 50/sin $r$ | C1 |
| | | 31° accept 30.71, 30.7 degree symbol required somewhere | A1 |

**46.**

| | | | |
|---|---|---|---|
| **(a)** | **(i)** | correct direction of refraction at **both** faces (**not** along normal)<br>blue below red **and** blue and red diverge | B1<br>B1 |
| | **(ii)** | any **two** from orange, yellow, green | B1 |
| **(b)** | **(i)** | **total internal reflection or** angle of incidence greater than critical angle | B1 |
| | **(ii)** | all colours reflected at same angle **or** all have i = r | B1 |

**47.**

| | | | |
|---|---|---|---|
| **(a)** | **(i)** | one ray from M correctly reflected – checked by eye<br>two rays from M correctly reflected – checked by eye – **and** traced back to image | C1<br>A1 |
| | **(ii)** | image point **clearly** marked at intersection/correct place checked by eye | B1 |
| **(b)** | | 0.34 m **cao** | B1 |

**48.**

| | | | |
|---|---|---|---|
| **(a)** | **(i)** | correct ray | B1 |
| | **(ii)** | correct angle marked to normal | B1 |
| | **(iii)** | (the angle) between the incident ray and the normal (at the point of contact) | B1 |
| **(b)** | | correct ray from hat to eye<br>0.85–1.15 m | B1<br>B1 |

**49.**

| | | | |
|---|---|---|---|
| **(a)** | | red | B1 |
| **(b)** | **(i)** | equal to | B1 |
| | **(ii)** | less than | B1 |
| **(c)** | | two correct refractions on Fig. 6.2<br>no dispersion **and** ray ends close to P | M1<br>A1 |

**50.**

**(a)** reflections correct by eye B1

**(b)** **all** the ray **reflects** back (into the denser medium/glass)
**or** reflection **and** no refraction/escape into air B1

**(c)** more calls **or** greater bandwidth **or** more/faster data(/sec)/information **or** better quality **or** less power loss/energy loss/attenuation **or** greater distance (between repeaters) **or** harder to tap **or** less noise/interference B1

**(d)** $f = v/\lambda$ in any form numerical or algebraic C1
$3.3 \times 10^{14}$ Hz A1

**51.**

**(a)** any ray from top of object correct through lens within 1mm of optical centre or F B1
other ray from same point correct through lens meeting 1st ray **and** none wrong B1

**(b)** **(i)** image size/object size (accept image distance/object distance **or** v/u) B1

**(ii)** 0.55–0.65 **ecf diagram in (a) sizes or distances** B1

**(c)** rays completed to retina but would meet behind retina B1

**52.**

**(a)** **(i)** reflected ray correct by eye **and** normal B1
**(ii)** 40 ° B1
40 ° **or** same as angle of incidence B1

**(b)** **diagram** with object, mirror, image in approx. correct position B1
at least 1 ray drawn from object/ray-box correctly reflecting from mirror B1
at least 2 rays extrapolated back to image position B1

**OR(b)** **diagram** with object, mirror, image in approx. correct position
**OR** B1
Use of search pin behind mirror shown/stated B1
no parallax used to locate image **or** described B1
(ignore arrows/do not insist on dotted lines)

**53.**

| | | |
|---|---|---|
| **(a)** | 46 (°) | B1 |
| **(b)** | angle of incidence | B1 |
| | when refracted ray is along surface<br>**or** minimum angle of incidence for Total Internal Reflection | B1 |
| **(c)** | sin *i*/sin *r* **or** 1/sin*C* | B1 |
| | sin 63/sin 40 **or** 1/sin 46 | C1 |
| | 1.39 (accept 1.3860 –1.3902) | A1 |
| **(d)** | correct reflected ray by eye | B1 |

**54.**

| | | | |
|---|---|---|---|
| **(a)** | **(i)** | correct normal (by eye to centre of circle) | **M1** |
| | | angle between normal and ray 1 marked | **A1** |
| **(b)** | | ray 1 sensibly reflected **and no** refracted ray | **B1** |
| | | ray 2 bends upwards (ignore reflection) | **B1** |
| | | ray 3 undeviated (ignore all rays leaving bubble) | **B1** |
| **(c)** | **(i)** | sin i/sin r **or** ratio of speed in air/vacuum to speed in medium | **B1** |
| | | (**ignore** real/apparent depth) | |
| | **(ii)** | 1.33 or 0.75 | |
| | | (accept 1.326, 1.3, 0.754, 0.8 **not** 1.325, 1, 0.76) | **B1** |